Random Signals and Noise

A receiver observes a waveform carrying information together with noise. Probability describes this uncertainty; correlation and spectra describe how the waveform behaves across time and frequency. Filtering and detection use these descriptions to help recover the information.



Probability, random signals, spectra and filtering.

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Symbols

Symbol	Meaning
X, x	Random variable and its value. $X(t)$ is a random process; $x(t)$ is a realization.
$f_X(x)$, $F_X(x)$	Probability density and cumulative distribution.
μ_X, σ_X^2	Mean and variance.
$R_X(\tau)$, $C_X(\tau)$	Autocorrelation and autocovariance.

Symbol	Meaning
$R_{XY}(\tau)$	$E\{X(t+\tau)\,Y(t)\}$: the first process carries the lag.
$S_X(f)$, $S_{XY}(f)$	Power spectrum and cross spectrum; f is in hertz.
$h(t)$, $H(f)$	Impulse response and frequency response.
$N_0/2$	Two-sided white-noise spectral density.
B, W	Bandwidths defined locally. An ideal lowpass edge B corresponds to full positive-lobe bandpass width $2B$ where stated.

$$\operatorname{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \qquad \operatorname{sinc}(0) = 1.$$

Random Variables and Probability Laws

Events and random variables

Even when a receiver repeats the same measurement, its voltage can vary because of noise. Probability describes this uncertainty and lets us predict how often a reading will cross a decision threshold. A random variable represents one such uncertain numerical reading; it does not mean that every possible value is equally likely.

A random experiment has an uncertain outcome. Its sample space S contains all possible outcomes; an event A is a subset of S.

$$P(S) = 1, \qquad 0 \le P(A) \le 1,$$

$$P(A \cup B) = P(A) + P(B), \qquad A \cap B = \varnothing.$$

If A occurs n_A times in n repeated trials, n_A/n is its observed relative frequency.

For a fair die, $S = \{1,2,3,4,5,6\}$ and the even event is $A = \{2,4,6\}$, so $P(A) = 3/6 = 1/2$.

A random variable assigns a real value to each outcome:

$$X : S \to \mathbb{R}, \qquad X(s_k) = x.$$

X denotes the variable; x denotes a possible value. A discrete variable has a finite or countable set of values. The continuous variables considered below possess probability densities.

Probability mass and cumulative distribution

The probability mass function (PMF) of a discrete variable is

$$p_X(x_k) = P(X = x_k), \qquad p_X(x_k) \ge 0, \qquad \sum_k p_X(x_k) = 1.$$

A Bernoulli variable has $P(X = 1) = p$ and $P(X = 0) = 1 - p$, where $0 \le p \le 1$.

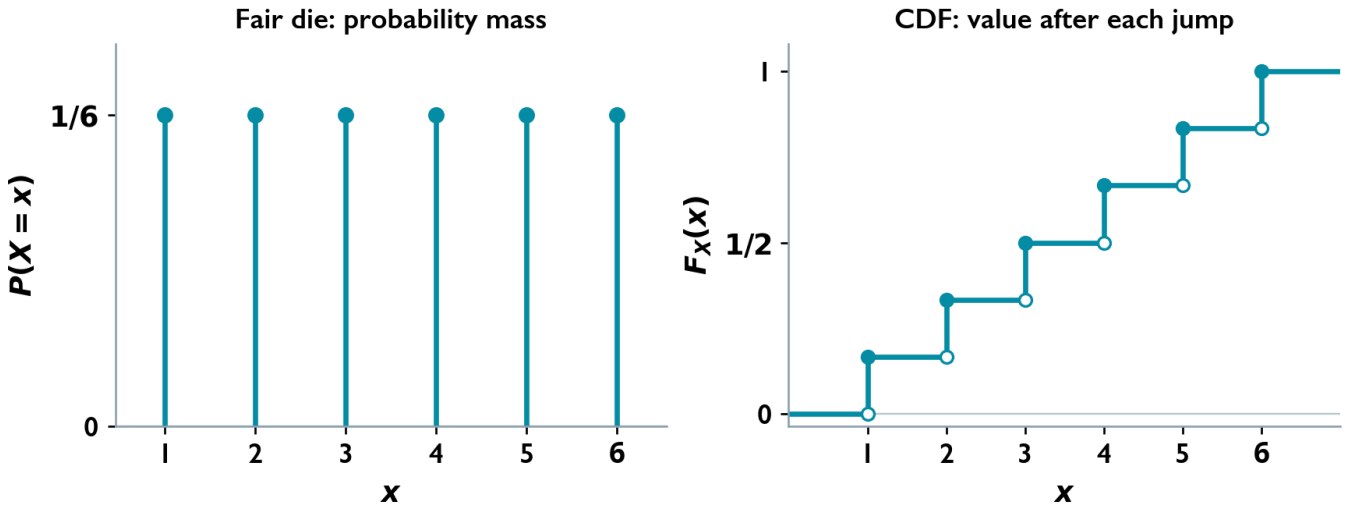
The cumulative distribution function (CDF) is

$$F_X(x) = P(X \le x).$$

It is nondecreasing, lies in $[0,1]$, and satisfies

$$F_X(-\infty) = 0, \qquad F_X(+\infty) = 1, \qquad P(a < X \le b) = F_X(b) - F_X(a).$$

For a discrete variable, $F_X(x) = \sum_{x_k \le x} p_X(x_k)$. A fair die has six masses of $1/6$, so its CDF increases by $1/6$ at each face value. A point mass is a CDF jump: $P(X = x_0) = F_X(x_0) - F_X(x_0^-)$, where $F_X(x_0^-)$ is the left limit. A mixed law retains both its point masses and its continuous density.



Probability masses and CDF of a fair die.

Probability density

A probability density function (PDF) satisfies

$$f_X(x) \geq 0, \qquad \int_{-\infty}^{\infty} f_X(x)\, dx = 1,$$

$$F_X(x) = \int_{-\infty}^{x} f_X(u)\, du, \qquad f_X(x) = \frac{dF_X(x)}{dx},$$

where the derivative exists. Thus

$$P(a < X \leq b) = \int_a^b f_X(x)\, dx.$$

A density is not a point probability: $P(X = x) = 0$. Its value may exceed one; its total area must equal one. A normalized histogram of repeated voltage measurements approximates this shape: intervals with more area occur more often. For a continuous reading, the probability of a finite voltage range matters, rather than the probability of one exact voltage.

Uniform distribution

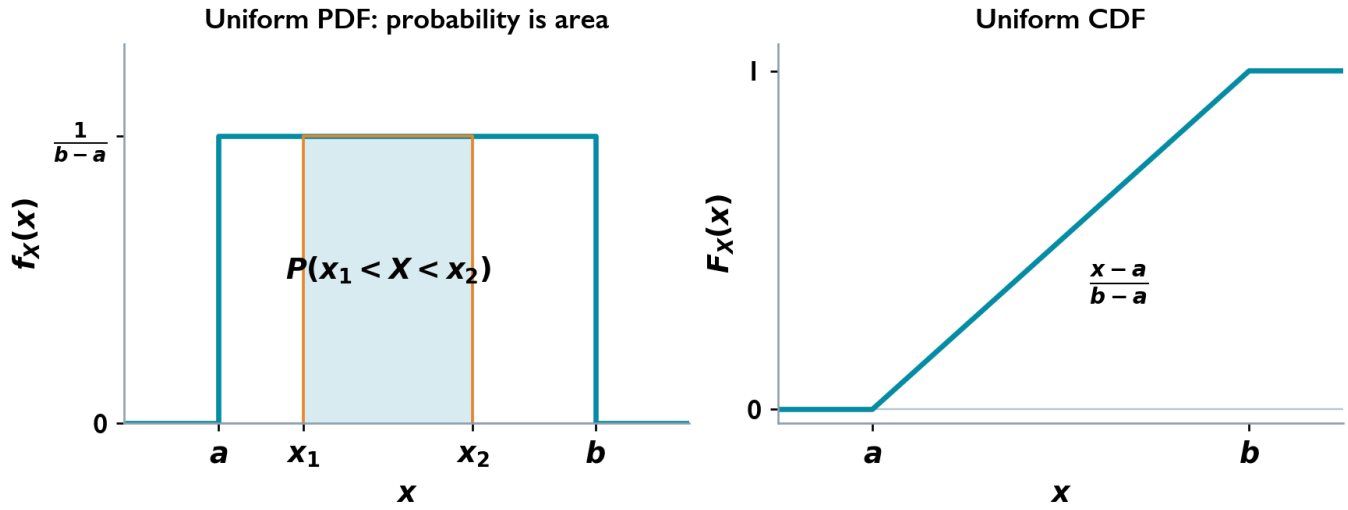
For X uniform on $[a, b]$, with $a < b$,

$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise}, \end{cases}$$

$$F_X(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b. \end{cases}$$

For $a \le c < d \le b$**,**

$$P(c < X \le d) = \frac{d-c}{b-a}.$$



Uniform PDF and CDF; the shaded area is an interval probability.

Gaussian distribution and Q function

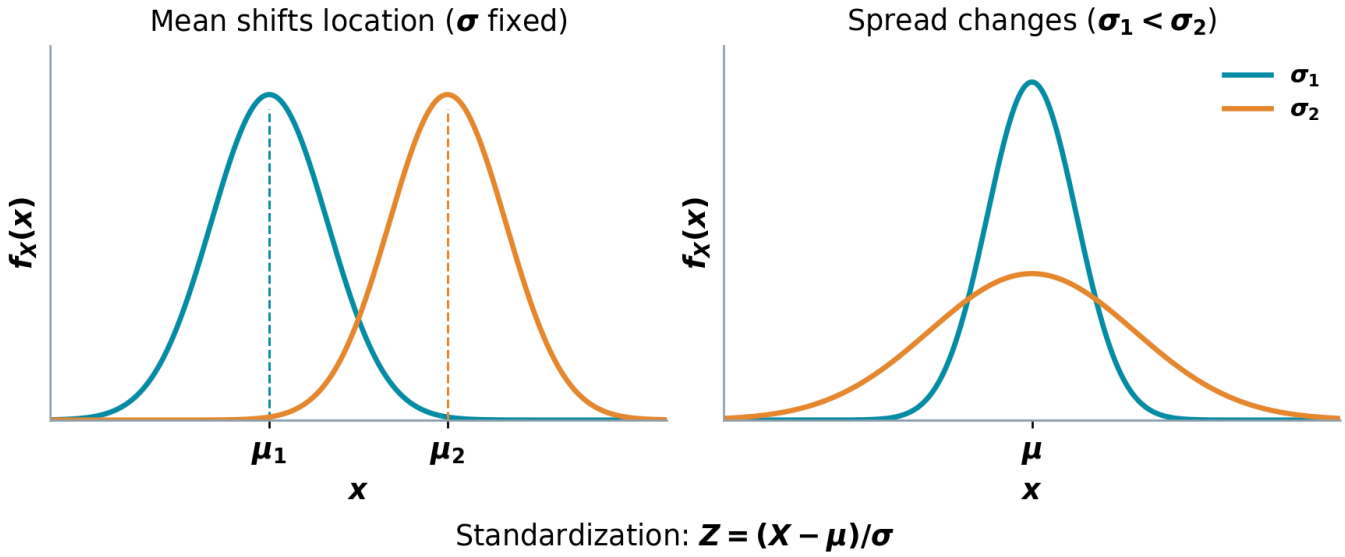
Gaussian noise is useful in receiver analysis because many small, approximately independent disturbances can combine into a nearly Gaussian voltage. The model must still suit the physical situation: knowing only a mean and a variance does not establish that a distribution is Gaussian.

A Gaussian variable $X \sim \mathcal{N}\left(\mu_X, \sigma_X^2\right)$ **has mean** μ_X**, variance** σ_X^2**, standard deviation** $\sigma_X > 0$**, and density**

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma_X} \exp\left[-\frac{\left(x-\mu_X\right)^2}{2\sigma_X^2}\right], \qquad x \in \mathbb{R}.$$

Standardization gives

$$Z = \frac{X-\mu_X}{\sigma_X} \sim \mathcal{N}(0,1).$$



Gaussian location, spread, and standardization.

A receiver makes an error when noise pushes its measured value beyond the wrong side of a threshold. The Q function gives the probability in a Gaussian tail, so it converts a threshold distance measured in standard deviations into an error probability.

Define the standard Gaussian upper tail by

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-u^2/2}\, du.$$

Then

$$P(X > c) = Q\left(\frac{c - \mu_X}{\sigma_X}\right), \qquad Q(-z) = 1 - Q(z), \qquad Q(0) = \frac{1}{2}.$$

Example: binary PAM decision error

Question

Binary symbols use levels $+A$ and $-A$, where $A > 0$. The received value is the transmitted level plus Gaussian noise $N \sim \mathcal{N}(0, \sigma_N^2)$, independent of the transmitted symbol. With decision threshold zero, find the error probability after transmitting $+A$.

Solution

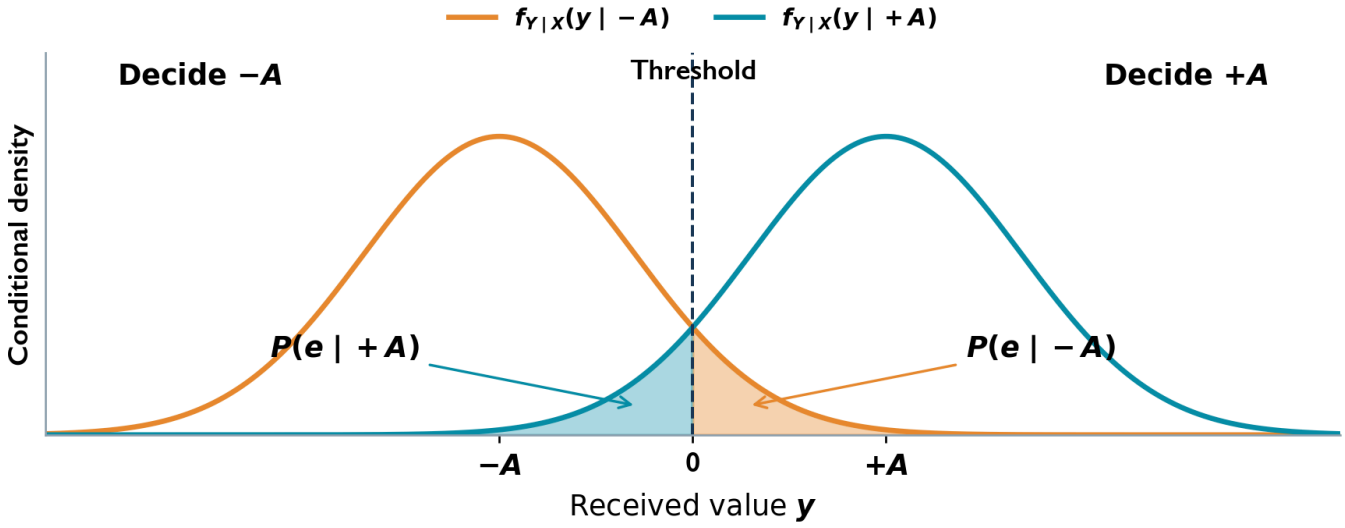
Given $+A$, $Y = A + N \sim \mathcal{N}(A, \sigma_N^2)$, and an error occurs when $Y < 0$:

$$P(\text{error} \mid +A) = \int_{-\infty}^{0} \frac{e^{-(y-A)^2/(2\sigma_N^2)}}{\sqrt{2\pi}\,\sigma_N}\, dy.$$

With $z = (y - A)/\sigma_N$ and Gaussian symmetry,

$$P(\text{error} \mid +A) = P\left(Z < -\frac{A}{\sigma_N}\right) = Q\left(\frac{A}{\sigma_N}\right).$$

The conditional error after sending $-A$ is identical, so $P_e = Q(A/\sigma_N)$ for the symmetric binary model. Increasing the signal level relative to the noise spread moves the decision threshold farther into the tail and reduces errors.



Zero-threshold PAM decision and Gaussian error tails.

Central limit theorem

Let $X_1, \ldots, X_n$ be independent identically distributed variables with common finite mean μ_X and finite positive variance σ_X^2. For their sum,

$$Y_n = \sum_{k=1}^{n} X_k, \qquad E\{Y_n\} = n\mu_X, \qquad \sigma_{Y_n}^2 = n\sigma_X^2.$$

As $n \to \infty$, the distribution of

$$Z_n = \frac{Y_n - n\mu_X}{\sigma_X\sqrt{n}}$$

approaches $\mathcal{N}(0,1)$. This explains the Gaussian model for a sum of many small independent noise contributions. For finite n it is an approximation; independent Gaussian summands give an exactly Gaussian sum.

Worked Problems

A cubic cumulative distribution

Question

A random variable X has cumulative distribution

$$F_X(x) = \begin{cases} 0, & x < 0, \\ Ax^3, & 0 \le x \le 10, \\ B, & x > 10. \end{cases}$$

Determine A and B, and obtain and sketch the density.

Solution

Normalization and right continuity

$$\lim_{x \to \infty} F_X(x) = B = 1, \qquad F_X(10) = 1000A = 1.$$

$$A = 10^{-3}, \qquad B = 1.$$

Density The CDF is continuous, so differentiating gives

$$f_X(x) = \begin{cases} 3 \times 10^{-3} x^2, & 0 < x < 10, \\ 0, & \text{otherwise}. \end{cases}$$

Check and sketch

$$\int_0^{10} 3 \times 10^{-3} x^2 \, dx = 10^{-3} \left[x^3 \right]_0^{10} = 1.$$

The density rises parabolically from zero to a limiting height of 0.3 on $(0, 10)$ and vanishes outside this interval.

Testing an exponential density

Question

Determine whether

$$g(x) = \begin{cases} e^{-x/2}, & x \ge 0, \\ 0, & x < 0 \end{cases}$$

is a valid probability density.

Solution

Nonnegativity $g(x) \geq 0$ for every x.

Total area

$$\int_{-\infty}^{\infty} g(x)\,dx = \int_{0}^{\infty} e^{-x/2}\,dx = \left[-2e^{-x/2}\right]_{0}^{\infty} = 2 \neq 1.$$

g is not a normalized probability density.

The corresponding normalized density is

$$f_X(x) = \tfrac{1}{2}e^{-x/2}u(x).$$

Normalizing a symmetric exponential law

Question

Find k such that

$$f_X(x) = ke^{-|x|}, \qquad -\infty < x < \infty,$$

is a probability density.

Solution

Normalization Since $e^{-|x|} > 0$, require $k \geq 0$. Split at the origin:

$$1 = k\int_{-\infty}^{0} e^{x}\,dx + k\int_{0}^{\infty} e^{-x}\,dx = k + k = 2k.$$

$$k = \frac{1}{2}, \qquad f_X(x) = \frac{1}{2}e^{-|x|}.$$

By symmetry, each half-line has probability $1/2$.

A trapezoidal probability density

Question

Let

$$f_X(x) = \begin{cases} ax, & 0 \leq x \leq 1, \\ a, & 1 < x \leq 2, \\ a(3-x), & 2 < x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Determine a, obtain and sketch the CDF, and calculate $P(X \leq 1.5)$.

Solution

Normalization

$$1 = a\int_0^1 x\,dx + a\int_1^2 dx + a\int_2^3 (\,3-x\,)\,dx = \frac{a}{2} + a + \frac{a}{2} = 2a.$$

$$a = \frac{1}{2}.$$

Accumulated probability

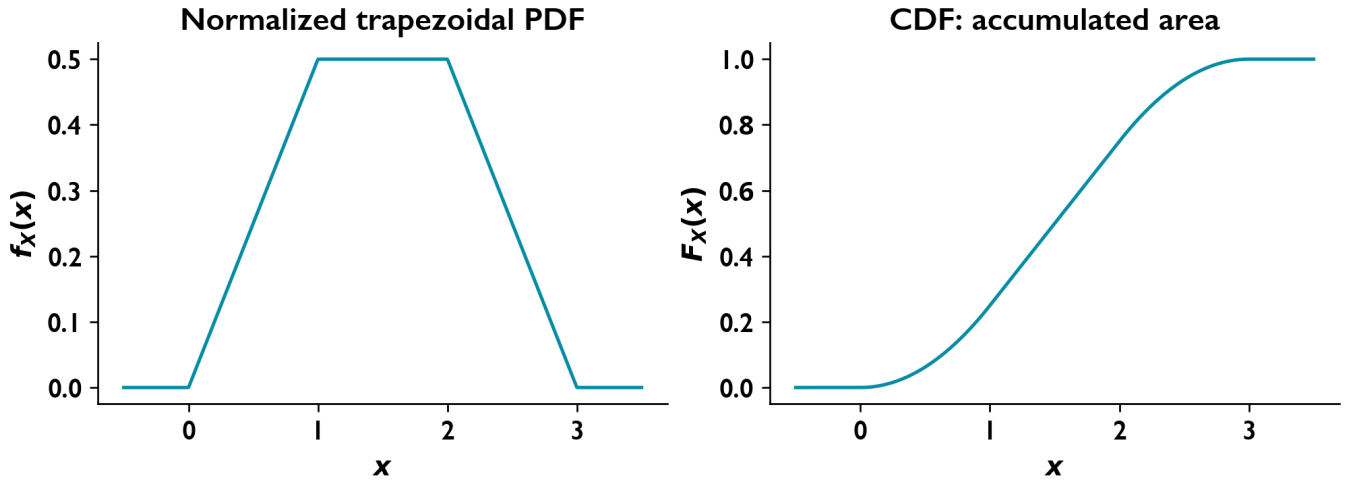
$$\begin{aligned}
F_X(\,x\,) &= \textstyle\int_0^x \frac{t}{2}\,dt = \frac{x^2}{4}, && 0 \le x < 1,\\
F_X(\,x\,) &= \tfrac{1}{4} + \textstyle\int_1^x \frac{1}{2}\,dt = \frac{x}{2} - \frac{1}{4}, && 1 \le x < 2,\\
F_X(\,x\,) &= 1 - \textstyle\int_x^3 \frac{3-t}{2}\,dt = 1 - \frac{(\,3-x\,)^{\,2}}{4}, && 2 \le x < 3.
\end{aligned}$$

$$F_X(\,x\,) = \begin{cases} 0, & x < 0,\\ x^2/4, & 0 \le x < 1,\\ x/2 - 1/4, & 1 \le x < 2,\\ 1 - (\,3-x\,)^{\,2}/4, & 2 \le x < 3,\\ 1, & x \ge 3. \end{cases}$$

The CDF is continuous and increases quadratically, linearly, then along a concave quadratic segment.

Requested probability

$$P(\,X \le 1.5\,) = F_X(\,1.5\,) = \frac{1.5}{2} - \frac{1}{4} = \frac{1}{2}.$$



The normalized trapezoidal density and its accumulated probability.

Normalizing a two-level density

Question

A density is specified by

$$f_X(x) = \begin{cases} k, & 0 < x < 1, \\ 1/3, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Determine k.

Solution

Total area

$$1 = \int_0^1 k\,dx + \int_1^2 \frac{1}{3}\,dx = k + \frac{1}{3}.$$

$$k = \frac{2}{3}.$$

Both density levels are nonnegative.

A shifted exponential law with a point mass

Question

A random voltage has generalized density

$$f_X(x) = k\,u(x+7)\,e^{-1.72(x+7)} + 0.35\delta(x-1),$$

where u is the unit step and δ is the Dirac impulse.

- Determine the normalizing constant.
- Obtain and sketch the CDF.
- Calculate $P(X \le 0.5)$, $P(X > 3)$, and $P(0.5 \le X \le 3)$.

Solution

Continuous and discrete probability The impulse assigns $P(X = 1) = 0.35$; the continuous part must integrate to 0.65.

$$1 = k\int_{-7}^{\infty} e^{-1.72(x+7)}\,dx + 0.35 = \frac{k}{1.72} + 0.35.$$

$$k = 1.72(0.65) = 1.118.$$

Cumulative distribution

$$F_X(x) = \begin{cases} 0, & x < -7, \\ 0.65\left[1 - e^{-1.72(x+7)}\right], & -7 \le x < 1, \\ 0.65\left[1 - e^{-1.72(x+7)}\right] + 0.35, & x \ge 1. \end{cases}$$

The CDF rises from zero at -7, jumps by 0.35 at 1, and approaches one asymptotically.

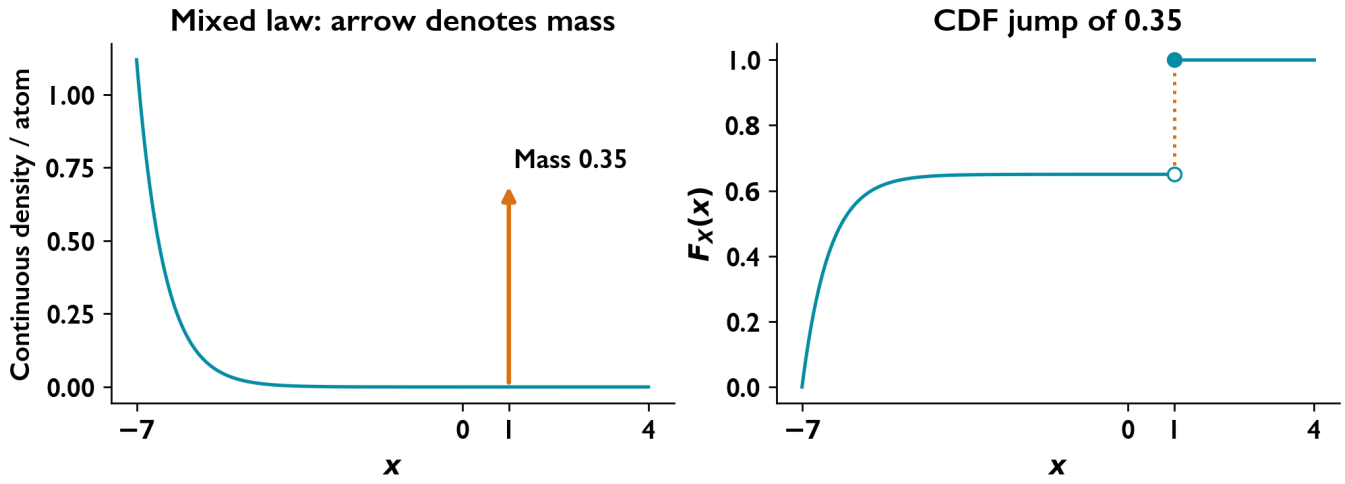
Requested probabilities There are no point masses at 0.5 or 3.

$$P(X \le 0.5) = 0.65\left(1 - e^{-12.9}\right) \approx 0.6499983763.$$

$$P(X > 3) = 0.65e^{-17.2} \approx 2.20317 \times 10^{-8}.$$

$$\begin{aligned} P(0.5 \le X \le 3) &= F_X(3) - F_X(0.5) \\ &= 0.35 + 0.65\left(e^{-12.9} - e^{-17.2}\right). \end{aligned}$$

$$P(0.5 \le X \le 3) \approx 0.3500016017.$$



The arrow represents a probability mass of 0.35; it is not an ordinary finite density height.

Cumulative probabilities of an Erlang density

Question

A continuous random variable has density

$$f_X(x) = xe^{-x}u(x).$$

Obtain its CDF and calculate $P(X \le 1)$, $P(1 \le X \le 2)$, and $P(X > 2)$.

Solution

Cumulative distribution For $x \ge 0$, integration by parts gives

$$F_X\left(x\right) = \int_0^x te^{-t}\,dt = \left[-\left(t+1\right)e^{-t}\right]_0^x = 1-\left(1+x\right)e^{-x}.$$

$$F_X\left(x\right) = \begin{cases} 0, & x<0, \\ 1-\left(1+x\right)e^{-x}, & x\geq 0. \end{cases}$$

Its limit at infinity is one, verifying normalization.

Requested probabilities

$$P\left(X\leq 1\right) = 1-\frac{2}{e} \approx 0.264241.$$

$$\begin{aligned} P\left(1\leq X\leq 2\right) &= F_X\left(2\right)-F_X\left(1\right) \\ &= \tfrac{2}{e}-\tfrac{3}{e^2}. \end{aligned}$$

$$P\left(1\leq X\leq 2\right) \approx 0.329753.$$

$$P\left(X>2\right) = 1-F_X\left(2\right) = \frac{3}{e^2} \approx 0.406006.$$

The cumulative distribution of a uniform variable

Question

For $a<b$, let

$$f_X\left(x\right) = \begin{cases} 1/\left(b-a\right), & a\leq x\leq b, \\ 0, & \text{otherwise}. \end{cases}$$

Obtain and sketch its CDF.

Solution

Integration over the support

$$F_X\left(x\right) = \int_a^x \frac{1}{b-a}\,dt = \frac{x-a}{b-a}, \qquad a\leq x\leq b.$$

No probability lies below a, and all probability has accumulated by b.

$$F_X\left(x\right) = \begin{cases} 0, & x<a, \\ \left(x-a\right)/\left(b-a\right), & a\leq x\leq b, \\ 1, & x>b. \end{cases}$$

The sketch is horizontal at zero, rises linearly from $\left(a,0\right)$ to $\left(b,1\right)$ with slope $1/\left(b-a\right)$, then remains at one.

Recovering a density from a sinusoidal CDF

Question

A random variable has CDF

$$F_X\left(x\right) = \begin{cases} 0, & x < 0, \\ k \sin\left(\pi x/40\right), & 0 \le x \le 10, \\ k \sin\left(\pi/4\right), & x > 10. \end{cases}$$

Determine k, obtain the density, and calculate $P\left(X \le 4\right)$.

Solution

Normalization

$$\lim_{x \to \infty} F_X\left(x\right) = k \sin\left(\pi/4\right) = 1.$$

$$k = \sqrt{2}\,.$$

Differentiation The CDF is continuous at both support boundaries, so no point masses occur there.

$$f_X\left(x\right) = \begin{cases} \dfrac{\sqrt{2}\,\pi}{40} \cos\left(\pi x/40\right), & 0 < x < 10, \\ 0, & \text{otherwise}. \end{cases}$$

The density is nonnegative on its support, and

$$\int_0^{10} f_X\left(x\right)\,dx = \sqrt{2}\,\sin\left(\pi/4\right) = 1.$$

Requested probability

$$P\left(X \le 4\right) = \sqrt{2}\,\sin\left(\pi/10\right) \approx 0.437016.$$

Probability between two thresholds

Question

A continuous random variable has

$$F_X\left(x\right) = \begin{cases} 0, & x < 0, \\ 10^{-3}x^3, & 0 \le x \le 10, \\ 1, & x > 10. \end{cases}$$

Find $P\left(3 \le X \le 7\right)$.

Solution

Use the CDF difference. The CDF is continuous, so the endpoints carry no probability mass:

$$P(\,3 \le X \le 7\,) = F_X(\,7\,) - F_X(\,3\,) = 10^{-3}\left(7^3 - 3^3\right) = 0.316.$$

A unit-area function that fails the density test

Question

Determine whether the function

$$g(\,x\,) = \begin{cases} \dfrac{3}{2} - x, & 0 \le x \le 2, \\ 0, & \text{otherwise} \end{cases}$$

is a valid probability density. Justify the answer.

Solution

Total area.

$$\int_{-\infty}^{\infty} g(\,x\,)\,dx = \int_0^2 \left(\frac{3}{2} - x\right) dx = \left[\frac{3}{2}x - \frac{x^2}{2}\right]_0^2 = 1.$$

Nonnegativity. For $3/2 < x \le 2$, the proposed function is negative; in particular $g(\,2\,) = -1/2$.

$$g \text{ is not a probability density.}$$

A density must satisfy both unit area and nonnegativity; satisfying the area condition alone is insufficient.

Counting independent errors in a block

Question

Each of n independently transmitted bits has error probability p. Let K count the errors. Derive $P(\,K \le k\,)$ for an integer $0 \le k \le n$. Evaluate the probability of at most three errors when $n = 1000$ and $p = 10^{-4}$.

Solution

Probability of an error pattern. A specified pattern with j errors and $n - j$ correct bits has probability $p^j\,(\,1-p\,)^{\,n-j}$. There are $\binom{n}{j}$ such patterns:

$$P(\,K = j\,) = \binom{n}{j} p^j\,(\,1-p\,)^{\,n-j}, \qquad 0 \le j \le n.$$

Accumulate the permitted counts.

$$P(\,K \le k\,) \;=\; \sum_{j\,=\,0}^{k} \binom{n}{j} p^{j}\,(\,1-p\,)^{\,n\,-\,j}.$$

The zero-error term is included because zero is also at most k. For the specified block,

$$P(\,K \le 3\,) \;=\; \sum_{j\,=\,0}^{3} \binom{1000}{j} \left(\,10^{-4}\,\right)^{\,j} (\,0.9999\,)^{\,1000\,-\,j} \approx 0.999996175.$$

The complementary probability of more than three errors is approximately 3.82500×10^{-6}.

Tail probabilities and shifted noise support

Question

An observation is $Y = N$ when no signal is present and $Y = S + N$ when a signal is present, where

$$f_N(\,n\,) \;=\; \begin{cases} \dfrac{1}{10}\left(1 - \dfrac{|n|}{10}\right), & |n| \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

For $S = 6$, declare a signal present when $Y > 4$. Find the false-alarm and detection probabilities. Find a threshold that gives zero false-alarm probability and the largest possible detection probability. Finally, find the smallest $S \ge 0$ that permits both perfect detection and zero false alarms.

Solution

Useful tail area. For $0 \le a \le 10$,

$$P(\,N > a\,) \;=\; \int_{a}^{10} \frac{1}{10}\left(1 - \frac{n}{10}\right) dn = \frac{(\,10-a\,)^{\,2}}{200}.$$

Symmetry gives $P(\,N > -a\,) = 1 - (\,10-a\,)^{\,2}/200$. Threshold at 4.

$$P_{\mathrm{FA}} = P(\,N > 4\,) \;=\; \frac{6^2}{200} = 0.18,$$

$$P_{\mathrm{D}} = P(\,6 + N > 4\,) \;=\; P(\,N > -2\,) \;=\; 1 - \frac{8^2}{200} = 0.68.$$

Zero false alarms. With no signal, Y lies in $[\,-10, 10\,]$. The smallest admissible threshold is 10; increasing it would reduce detection:

$$\text{Declare a signal when } Y > 10, \qquad P_{\mathrm{D}} = P(\,N > 4\,) \;=\; 0.18.$$

Separate the supports. Under a signal, the support is $[\,S-10, S+10\,]$. Complete separation requires

$$S - 10 \geq 10 \quad \Rightarrow \quad S_{\min} = 20.$$

At $S = 20$, the supports touch only at one point, which has zero probability for this continuous law.

A shifted Laplace observation and a prescribed tail probability

Question

An observation is $Y = W$ with no signal and $Y = 6 + W$ with a signal, where

$$f_W(w) = e^{-2|w|}, \qquad -\infty < w < \infty.$$

Declare a signal present when $Y > \gamma$. Find the threshold giving a false-alarm probability of 0.001 and the corresponding detection probability.

Solution

Positive tail. The density is normalized since $2\int_0^\infty e^{-2w}dw = 1$. For $\gamma \geq 0$,

$$P(W > \gamma) = \int_\gamma^\infty e^{-2w}dw = \frac{1}{2}e^{-2\gamma}.$$

Set this equal to 0.001:

$$\gamma = \frac{1}{2}\ln(500) \approx 3.10730.$$

Shifted event. With a signal, the event is $W > \gamma - 6$, whose threshold is negative. The left-tail probability is $F_W(w) = \frac{1}{2}e^{2w}$ for $w < 0$, so

$$\begin{aligned} P_D &= 1 - \tfrac{1}{2}e^{2(\gamma - 6)} \\ &= 1 - 250e^{-12} \approx 0.998464. \end{aligned}$$

The shift moves most of the signal-present distribution above the same threshold.

Recognizing a Gaussian density by completing the square

Question

A Gaussian variable has density

$$f_X(x) = c\,e^{-2x^2 + x}, \qquad -\infty < x < \infty.$$

Find the normalizing constant and determine the probability that X lies outside $[-1, 1]$.

Solution

Identify the parameters.

$$-2x^2 + x = -2\left(x - \frac{1}{4}\right)^2 + \frac{1}{8}.$$

Comparison with the Gaussian density gives

$$\mu_X = \frac{1}{4}, \qquad \sigma_X^2 = \frac{1}{4}, \qquad c = \sqrt{\frac{2}{\pi}}\, e^{-1/8}.$$

Standardize each tail. With $Z = \left(X - 1/4\right) / \left(1/2\right)$**,**

$$P\left(X < -1\right) = P\left(Z < -2.5\right) = Q\left(2.5\right),$$

$$P\left(X > 1\right) = P\left(Z > 1.5\right) = Q\left(1.5\right).$$

The two disjoint tails therefore give

$$P\left(X \notin [-1, 1]\right) = Q\left(2.5\right) + Q\left(1.5\right) \approx 0.0730169.$$

Probabilities for a nonzero-mean Gaussian variable

Question

Let $X \sim \mathcal{N}\left(4, 9\right)$**, where the second parameter is variance. Find** $P\left(X > 7\right)$ **and** $P\left(0 < X < 9\right)$**.**

Solution

Standardize with the standard deviation. Here $\sigma_X = 3$**, so** $Z = \left(X - 4\right)/3 \sim \mathcal{N}\left(0, 1\right)$**:**

$$P\left(X > 7\right) = Q\left(\frac{7 - 4}{3}\right) = Q\left(1\right) \approx 0.158655.$$

For the interval,

$$\begin{aligned} P\left(0 < X < 9\right) &= P\left(-\tfrac{4}{3} < Z < \tfrac{5}{3}\right) \\ &= Q\left(-4/3\right) - Q\left(5/3\right) \\ &= 1 - Q\left(4/3\right) - Q\left(5/3\right) \approx 0.860998. \end{aligned}$$

Threshold probabilities for Gaussian noise voltage

Question

A circuit noise voltage is $X \sim \mathcal{N}\left(0, 10^{-8}\right)$**, with variance in** V^2**. Find** $P\left(X > 10^{-4}\right)$**,** $P\left(X > 4 \times 10^{-4}\right)$**,** $P\left(-2 \times 10^{-4} < X < 10^{-4}\right)$**, and** $P\left(X > 10^{-4} \mid X > 0\right)$**.**

Solution

Convert the variance to voltage spread. The standard deviation is $\sigma = 10^{-4}\,\mathrm{V}$. Therefore

$$\begin{aligned} P\left(X > 10^{-4} \right) &= Q\left(1 \right) \approx 0.158655, \\ P\left(X > 4 \times 10^{-4} \right) &= Q\left(4 \right) \approx 3.16712 \times 10^{-5}, \\ P\left(-2 \times 10^{-4} < X < 10^{-4} \right) &= 1 - Q\left(2 \right) - Q\left(1 \right) \approx 0.818595. \end{aligned}$$

Condition on a positive voltage. Because the threshold event is contained in $\{X > 0\}$ and $P\left(X > 0 \right) = 1/2$,

$$P\left(X > 10^{-4} \mid X > 0 \right) = \frac{Q\left(1 \right)}{1/2} = 2Q\left(1 \right) \approx 0.317311.$$

Gaussian tails for two received voltage levels

Question

An equally likely binary symbol produces $2.5\,\mathrm{V}$ for bit zero or $0\,\mathrm{V}$ for bit one. The received voltage is that level plus independent noise $N \sim \mathcal{N}\left(0, 1.2\,\mathrm{V}^2 \right)$. A single threshold gives a symmetric binary channel. Find the threshold, all four conditional received-bit probabilities, and the average signal-to-noise ratio in decibels, using the mean-square transmitted voltage as signal power.

Solution

Symmetric threshold. Place it halfway between the two levels:

$$\gamma = 1.25\,\mathrm{V}.$$

Decide bit zero when the received voltage exceeds γ; otherwise decide bit one. Conditional probabilities. The noise standard deviation is $\sigma_N = \sqrt{1.2}\ \mathrm{V}$. The two error probabilities are equal:

$$q = Q\left(\frac{2.5 - 1.25}{\sqrt{1.2}} \right) = Q\left(\frac{1.25}{\sqrt{1.2}} \right) \approx 0.126917.$$

Writing the sent and received bits as S and R,

$$\begin{aligned} P\left(R = 1 \mid S = 0 \right) &= q, & P\left(R = 0 \mid S = 0 \right) &= 1 - q, \\ P\left(R = 0 \mid S = 1 \right) &= q, & P\left(R = 1 \mid S = 1 \right) &= 1 - q. \end{aligned}$$

Average SNR.

$$P_S = \frac{1}{2}\left(2.5 \right)^2 + \frac{1}{2}\left(0 \right)^2 = 3.125, \qquad P_N = 1.2.$$

$$\mathrm{SNR} = \frac{3.125}{1.2} \approx 2.60417, \qquad \mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(2.60417 \right) \approx 4.15669 \ \mathrm{dB}.$$

This specified power includes the nonzero DC component of the unipolar levels.

Gaussian decision probabilities with unequal symbol frequencies

Question

At a decision instant, the observed voltage is $R = X + N$, where X takes the values $+1$ V and -1 V. The independent noise sample satisfies

$$N \sim \mathcal{N} \left(0, 0.2 \ \mathrm{V}^2 \right).$$

Let $p = P(X = +1 \ \mathrm{V})$. A threshold rule selects the positive symbol when $R > \gamma$ and the negative symbol otherwise. Find the threshold that minimizes the probability of error for $p = 0.5$, $p = 0.3$, and $p = 0.75$. Evaluate the minimum error probability for $p = 0.75$.

Solution

Write the conditional Gaussian densities. Let $A = 1$ V and $\sigma_N = \sqrt{0.2}$ V. The two conditional means are $+A$ and $-A$, with the same variance:

$$f_{R|X} \left(r | \pm A \right) = \frac{1}{\sqrt{2\pi} \, \sigma_N} \exp \left[- \frac{\left(r \mp A \right)^2}{2\sigma_N^2} \right].$$

Express the two error tails. An error occurs above the threshold when the negative symbol was sent, or below it when the positive symbol was sent. Therefore

$$P_e \left(\gamma \right) = \left(1 - p \right) Q \left(\frac{\gamma + A}{\sigma_N} \right) + p Q \left(\frac{A - \gamma}{\sigma_N} \right).$$

Locate the boundary between the weighted densities. Differentiating the error probability gives

$$\frac{dP_e}{d\gamma} = p f_{R|X} \left(\gamma | A \right) - \left(1 - p \right) f_{R|X} \left(\gamma | -A \right).$$

Setting this derivative to zero is equivalent to equating the weighted densities:

$$p \exp \left[- \frac{\left(\gamma - A \right)^2}{2\sigma_N^2} \right] = \left(1 - p \right) \exp \left[- \frac{\left(\gamma + A \right)^2}{2\sigma_N^2} \right].$$

Taking the natural logarithm and collecting the terms in γ gives

$$\frac{2A\gamma}{\sigma_N^2} = \ln \left(\frac{1 - p}{p} \right), \qquad \gamma_{\mathrm{opt}} = \frac{\sigma_N^2}{2A} \ln \left(\frac{1 - p}{p} \right).$$

The weighted-density ratio increases with r, so the derivative changes from negative to positive at this threshold: it minimizes P_e.

Positive-symbol probability p	**Optimum threshold** γ_{opt}
0.5	$0\ \text{V}$
0.3	$0.1\ \ln\ (\,0.7/0.3\,)\ = 0.08473\ \text{V}$
0.75	$0.1\ \ln\ (\,0.25/0.75\,)\ = -0.10986\ \text{V}$

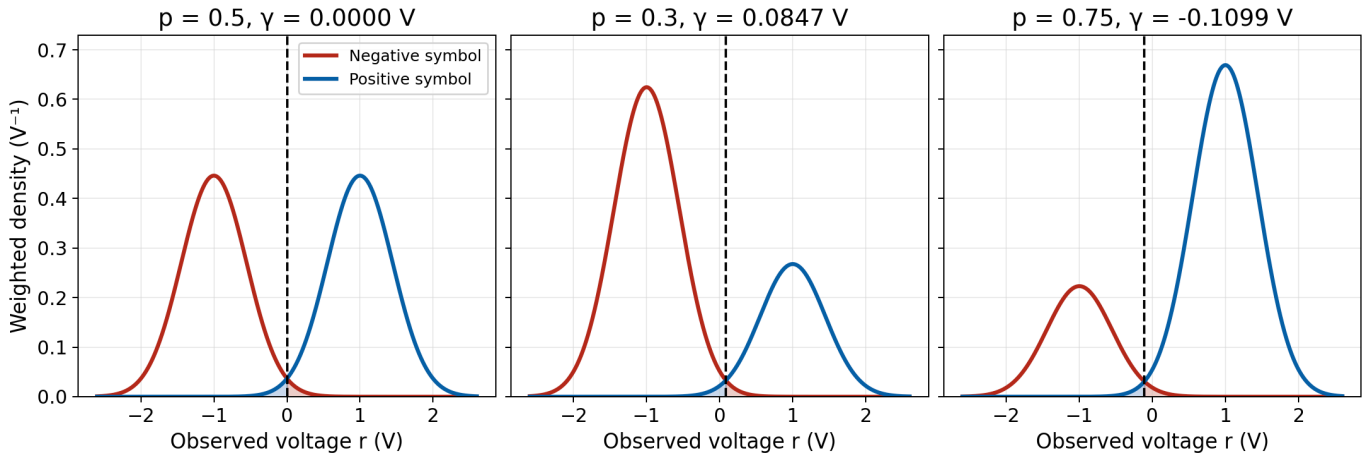
Evaluate the requested error probability. For $p = 0.75$,

$$\begin{aligned} P_{e,\ \min} &= 0.25Q\left(\tfrac{1-0.10986}{\sqrt{0.2}}\right) + 0.75Q\left(\tfrac{1+0.10986}{\sqrt{0.2}}\right) \\ &= 0.25Q\ (\,1.99041\,)\ + 0.75Q\ (\,2.48173\,) \\ &\approx 0.25\ (\,0.0232729\,)\ + 0.75\ (\,0.00653740\,)\ . \end{aligned}$$

Thus

$$P_{e,\ \min}\ \approx 0.0107213 \approx 1.072\%.$$

When positive symbols occur more often, the negative threshold makes the positive decision easier. The two tail probabilities still need their occurrence probabilities as weights; the total error is not either tail alone.



Weighted Gaussian densities for three symbol probabilities. The shaded tails contribute to the total error probability.

Prior probabilities and a Gaussian voltage threshold

Question

The received voltage is $Y = +1 + N$ under symbol s_1 and $Y = -1 + N$ under symbol s_2, where $N \sim \mathcal{N}\left(0, 0.1\ \mathrm{V}^2\right)$ is independent of the symbol. Let $p = P(s_1)$ and $P(s_2) = 1 - p$. Decide s_1 when $Y > \gamma$. Find the threshold that minimizes the average error probability for $p = 0.5$, $p = 0.7$, and $p = 0.2$.

Solution

Compare weighted densities. With $\sigma_N^2 = 0.1$ and levels $\pm A$, $A = 1$ V, choose s_1 where

$$p\, f_{Y|s_1}(y) > (1-p)\, f_{Y|s_2}(y).$$

The common Gaussian normalization cancels, and

$$\frac{f_{Y|s_1}(y)}{f_{Y|s_2}(y)} = \exp\left(\frac{(y+A)^2 - (y-A)^2}{2\sigma_N^2}\right) = \exp\left(\frac{2Ay}{\sigma_N^2}\right).$$

Therefore the optimum boundary is

$$\gamma = \frac{\sigma_N^2}{2A}\ln\left(\frac{1-p}{p}\right) = 0.05\ln\left(\frac{1-p}{p}\right)\ \mathrm{V}.$$

p	**Optimum γ**
0.5	0 V
0.7	$0.05\ln(3/7) \approx -0.0423649$ V
0.2	$0.05\ln 4 \approx 0.0693147$ V

A more probable positive symbol moves the threshold downward; a less probable positive symbol moves it upward.

Moments and Transformations

Mean, moments, and variance

A noise waveform changes from one reading to the next, but its average level and spread can remain stable. Moments summarize these features: the mean indicates an average bias, while the variance measures fluctuations about that bias. An expected value describes a probability-weighted average across measurements, rather than the value of each individual reading.

The mean is a probability-weighted average:

$$\mu_X = E\{X\} = \sum_k x_k p_X(x_k) \quad \text{or} \quad \mu_X = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

For a function g,

$$E\{g(X)\} = \sum_k g(x_k)\, p_X(x_k) \quad \text{or} \quad E\{g(X)\} = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx.$$

The raw moment of order n is $E\{X^n\}$. The first moment is the mean; the second is the mean-square value. For finite second moments,

$$\sigma_X^2 = E\{\left(X - \mu_X\right)^2\} = E\{X^2\} - \mu_X^2, \qquad \sigma_X = \sqrt{\sigma_X^2}.$$

The RMS value is $\sqrt{E\{X^2\}}$. Mean square equals variance only for zero mean. For voltage across resistance R, average power is $E\{X^2\}/R$. Positive and negative noise voltages can average to zero while still delivering nonzero power, because squaring removes their signs.

For constants c, d,

$$E\{cX + d\} = c\mu_X + d, \qquad \text{Var}\,(cX + d) = c^2\sigma_X^2.$$

For Bernoulli X, $X^2 = X$, giving $\mu_X = p$ and $\sigma_X^2 = p(1-p)$.

Estimates from measurements

For N independent identically distributed observations $x_1, \ldots, x_N$ with finite variance,

$$\widehat{\mu}_X = \frac{1}{N}\sum_{n=1}^{N} x_n,$$

$$\widehat{\sigma}_X^2 = \frac{1}{N-1}\sum_{n=1}^{N}\left(x_n - \widehat{\mu}_X\right)^2, \qquad N > 1.$$

These estimate the population mean and variance. The denominator $N-1$ makes the variance estimate unbiased under the stated conditions.

Transforming a variable

An amplifier changes the scale of a random voltage, and a DC offset shifts its average level. A transformation describes how the entire probability distribution changes when a device acts on its input; the output range changes along with the density.

For $Y=g(X)$, begin with

$$F_Y(y) = P(g(X) \le y).$$

Suppose X has a density and g is strictly monotone on its support, with inverse $h=g^{-1}$. Then

$$F_Y(y) = \begin{cases} F_X(h(y)), & g \text{ increasing}, \\ 1-F_X(h(y)), & g \text{ decreasing}. \end{cases}$$

Where the inverse is differentiable,

$$f_Y(y) = f_X(h(y)) \left| \frac{dh(y)}{dy} \right|.$$

Transform the support as well as the formula. If several inverse branches occur, each admissible branch contributes density. A constant transformation branch can create a point mass. With $A=\{x: g(x) = y_0\}$, its total mass is $P(Y=y_0) = P(X \in A) = F_Y(y_0) - F_Y(y_0^-)$, where the superscript minus denotes the left limit. Retain the remaining continuous density separately. A Gaussian input under $Y=cX+d$, $c \neq 0$, gives

$$Y \sim \mathcal{N}(c\mu_X + d, c^2\sigma_X^2).$$

Example: Gaussian voltage and averaging

Question

A zero-mean Gaussian voltage has standard deviation 10 V. Add 5 V. Find the probability of a positive reading and the probability that the average of two independent readings is positive.

Solution

Let $X \sim \mathcal{N}(0, 100)$ and $Y=X+5$. Then

$$\mu_Y = 5, \qquad \sigma_Y^2 = 100,$$

$$P(\,Y>0\,) = Q\left(\frac{0-5}{10}\right) = 1 - Q\,(\,0.5\,) \approx 0.6915.$$

For $M = (\,Y_1 + Y_2\,)/2$, independence gives

$$\mu_M = 5, \qquad \sigma_M^2 = \frac{100+100}{4} = 50,$$

$$P(\,M>0\,) = Q\left(-\frac{5}{\sqrt{50}}\right) = 1 - Q\left(\frac{5}{\sqrt{50}}\right) \approx 0.7602.$$

Averaging independent readings preserves the positive offset while reducing the random spread. The result illustrates why repeated measurements can make a weak DC level easier to distinguish from noise; correlated readings do not generally give the same reduction.

Example: double-exponential density

Question

For $f_X\,(\,x\,) = ae^{-b|x|}$ on $\mathbb{R}$, determine the constants required for a density, find its CDF, and evaluate $P(\,1<X<2\,)$.

Solution

Integrability requires $b>0$. Normalization gives

$$1 = 2a\int_0^\infty e^{-bx}\,dx = \frac{2a}{b}, \qquad a = \frac{b}{2}.$$

Integrating separately on either side of zero gives

$$F_X\,(\,x\,) = \begin{cases} \dfrac{1}{2}e^{bx}, & x<0, \\ 1-\dfrac{1}{2}e^{-bx}, & x\geq 0. \end{cases}$$

Thus

$$P(\,1<X<2\,) = F_X\,(\,2\,) - F_X\,(\,1\,) = \frac{1}{2}\,(\,e^{-b} - e^{-2b}\,).$$

Example: Rayleigh distribution

Unlike a signed voltage, a signal envelope is nonnegative. The Rayleigh distribution models the envelope of zero-mean narrowband Gaussian noise when its quadrature components are independent and have equal variance.

Question

For $b > 0$, a Rayleigh variable has

$$f_R(r) = \begin{cases} \dfrac{r}{b} e^{-r^2/(2b)}, & r > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find its CDF, mean, mean square, and variance.

Solution

Integration gives

$$F_R(r) = \begin{cases} 0, & r < 0, \\ 1 - e^{-r^2/(2b)}, & r \geq 0. \end{cases}$$

Integration by parts reduces the mean to a Gaussian integral:

$$E\{R\} = \frac{1}{b}\int_0^\infty r^2 e^{-r^2/(2b)}\, dr = \int_0^\infty e^{-r^2/(2b)}\, dr = \sqrt{\frac{b\pi}{2}}.$$

With $u = r^2/(2b)$,

$$E\{R^2\} = \frac{1}{b}\int_0^\infty r^3 e^{-r^2/(2b)}\, dr = 2b\int_0^\infty u e^{-u}\, du = 2b.$$

Therefore

$$\sigma_R^2 = 2b - \frac{b\pi}{2} = b\left(2 - \frac{\pi}{2}\right).$$

Worked Problems

A transformation producing both a discrete mass and a

continuous density

Question

Let $X \sim \mathcal{U}[-2,2]$ and $Y = g(X)$, where

$$g(x) = \begin{cases} -1, & -2 \le x < 0, \\ 2x/3, & 0 \le x < 3, \\ 2, & x \ge 3. \end{cases}$$

Find and sketch $F_X(x)$, $f_Y(y)$, and $F_Y(y)$.

Solution

Input distribution

$$f_X(x) = \tfrac{1}{4}\mathbf{1}_{(-2,2)}(x), \qquad F_X(x) = \begin{cases} 0, & x < -2, \\ (x+2)/4, & -2 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

Point mass and continuous density The input never reaches $x = 3$, so no mass occurs at $y = 2$.

$$\begin{aligned} P(Y = -1) &= P(X < 0) = \tfrac{1}{2}, \\ f_{Y,\text{cont}}(y) &= f_X(3y/2)\left|\tfrac{dx}{dy}\right| = \tfrac{1}{4}\tfrac{3}{2} = \tfrac{3}{8}, \qquad 0 < y < \tfrac{4}{3}. \end{aligned}$$

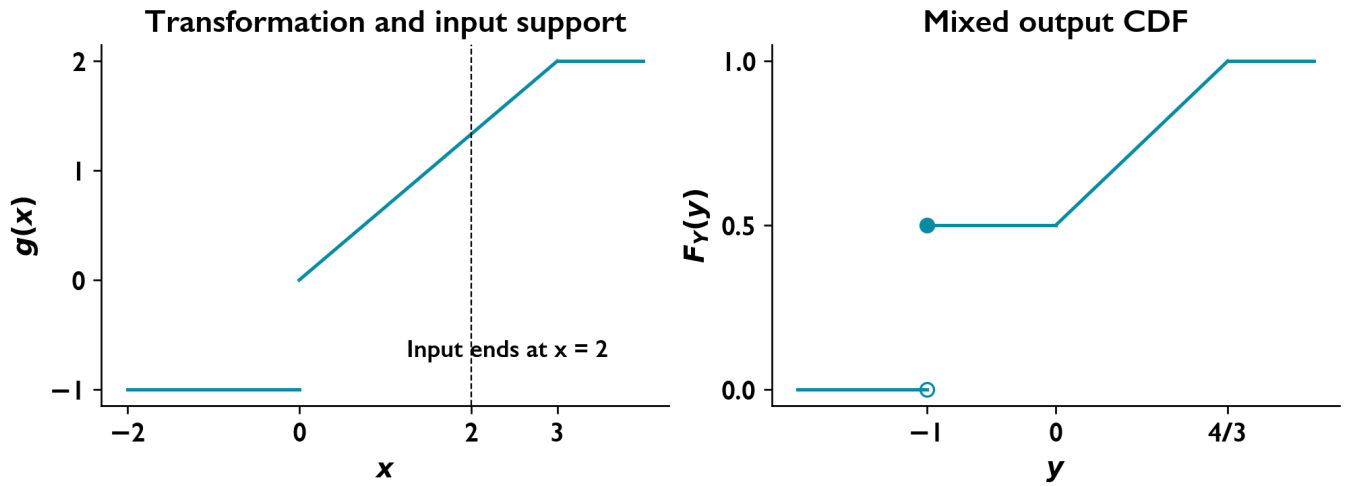
$$f_Y(y) = \frac{1}{2}\delta(y+1) + \frac{3}{8}\mathbf{1}_{(0,4/3)}(y)$$

The delta term denotes a probability mass, rather than a finite density height.

Output CDF

$$F_Y(y) = \begin{cases} 0, & y < -1, \\ \tfrac{1}{2}, & -1 \le y < 0, \\ \tfrac{1}{2} + \tfrac{3}{8}y, & 0 \le y < \tfrac{4}{3}, \\ 1, & y \ge \tfrac{4}{3}. \end{cases}$$

The CDF jumps by $1/2$ at -1, then increases linearly from $(0, 1/2)$ to $(4/3, 1)$.



The constant negative branch creates an atom at y = −1; the output never reaches the saturation branch at y = 2.

Moments of a quadratic density

Question

A continuous random variable has density

$$f_X(x) = \begin{cases} 3 \times 10^{-3} x^2, & 0 \le x \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

Determine its mean, second raw moment, and variance.

Solution

Mean

$$\mu_X = \int_0^{10} x \left(3 \times 10^{-3} x^2 \right) dx = 3 \times 10^{-3} \left[\frac{x^4}{4} \right]_0^{10}.$$

$$\mu_X = 7.5.$$

Second raw moment

$$E\{X^2\} = 3 \times 10^{-3} \int_0^{10} x^4 \, dx = 3 \times 10^{-3} \left[\frac{x^5}{5} \right]_0^{10}.$$

$$E\{X^2\} = 60.$$

Variance and standard deviation

$$\sigma_X^2 = E\{X^2\} - \mu_X^2 = 60 - (7.5)^2.$$

$$\sigma_X^2 = 3.75, \qquad \sigma_X = \sqrt{3.75} \approx 1.93649.$$

Moments of an exponential random variable

Question

For $a > 0$, let

$$f_X(x) = ae^{-ax}u(x).$$

Determine the mean, second raw moment, variance, and standard deviation.

Solution

Mean Integration by parts gives

$$\mu_X = a\int_0^\infty xe^{-ax}\,dx = \left[-xe^{-ax}\right]_0^\infty + \int_0^\infty e^{-ax}\,dx.$$

$$\mu_X = \frac{1}{a}.$$

Second raw moment

$$E\{X^2\} = a\int_0^\infty x^2e^{-ax}\,dx = \left[-x^2e^{-ax}\right]_0^\infty + 2\int_0^\infty xe^{-ax}\,dx.$$

$$E\{X^2\} = \frac{2}{a^2}.$$

Variance and standard deviation

$$\sigma_X^2 = \frac{2}{a^2} - \left(\frac{1}{a}\right)^2.$$

$$\sigma_X^2 = \frac{1}{a^2}, \qquad \sigma_X = \frac{1}{a}.$$

Moments and probabilities of a shifted parabolic density

Question

A continuous random variable X has

$$f_X(x) = \begin{cases} \frac{3}{32}\left(-x^2+8x-12\right), & 2<x<6, \\ 0, & \text{otherwise.} \end{cases}$$

Verify that this is a valid density. Find the mean, second raw moment, variance, and the probabilities $P(X>2)$, $P(X>4)$, and $P(X>6)$.

Solution

Check the density. On its support,

$$-x^2+8x-12 = (x-2)(6-x) \geq 0.$$

Center the variable using $u = x-4$, so $-2<u<2$ and $f_X(x) = \frac{3}{32}\left(4-u^2\right)$. Its total area is

$$\int_2^6 f_X(x)\,dx = \frac{3}{32}\int_{-2}^{2} (4-u^2)\,du = \frac{3}{32}\left(16-\frac{16}{3}\right) = 1.$$

Find the moments. Symmetry about $x=4$ gives $E\{X-4\}=0$, hence

$$\mu_X = 4.$$

The variance is

$$\begin{aligned} \sigma_X^2 &= \tfrac{3}{32}\int_{-2}^{2} u^2(4-u^2)\,du \\ &= \tfrac{3}{32}\left(\tfrac{64}{3}-\tfrac{64}{5}\right) = \tfrac{4}{5}. \end{aligned}$$

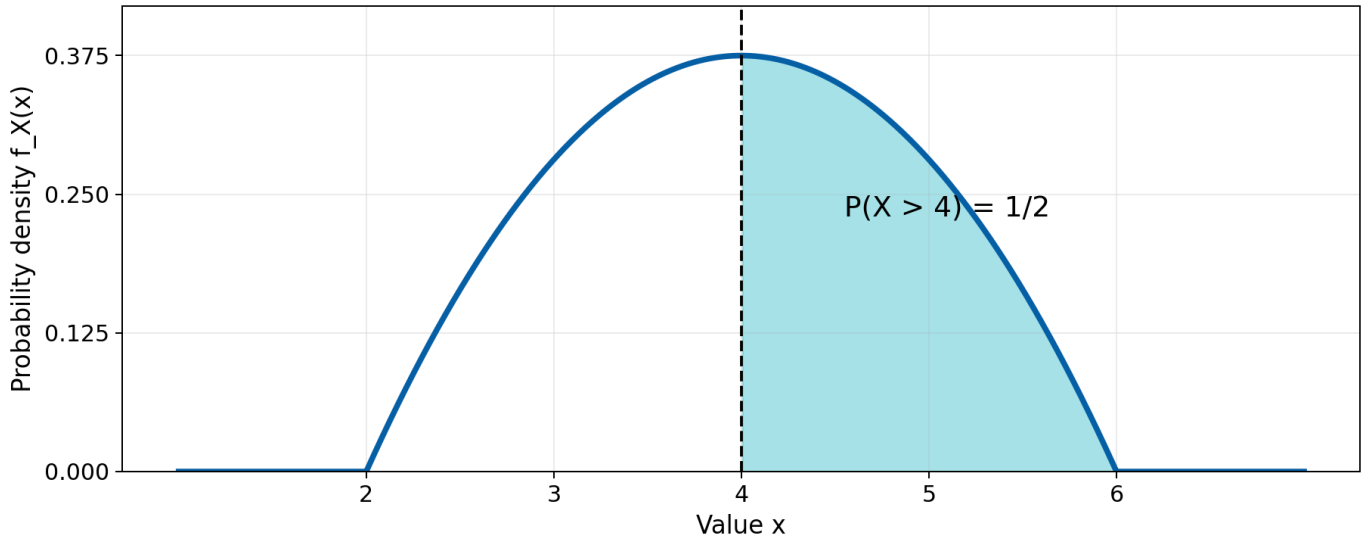
Therefore,

$$E\{X^2\} = \sigma_X^2+\mu_X^2 = \frac{4}{5}+16 = \frac{84}{5}.$$

Read the tail probabilities. The entire distribution lies between 2 and 6, and symmetry places half its probability above 4:

$$P(X>2) = 1, \qquad P(X>4) = \frac{1}{2}, \qquad P(X>6) = 0.$$

The support endpoints have zero probability because X is continuous.



The parabolic density is centered at four. Its right half has probability one-half.

Normalizing an asymmetric density and finding signal power

Question

The instantaneous amplitude X of a random signal has density

$$f_X(x) = \begin{cases} \frac{a}{2}(x+2), & -2 \le x < 0, \\ a, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Determine the normalizing constant and the signal's mean-square power $E\{X^2\}$.

Solution

Normalization. The rising part has triangular area a, and the constant part has rectangular area $2a$:

$$1 = \int_{-2}^{0} \frac{a}{2}(x+2)\,dx + \int_{0}^{2} a\,dx = a + 2a.$$

$$a = \frac{1}{3}.$$

Mean-square power. Integrate x^2 over both parts:

$$\begin{aligned} E\{X^2\} &= \tfrac{a}{2}\int_{-2}^{0} x^2(x+2)\,dx + a\int_{0}^{2} x^2\,dx \\ &= \tfrac{a}{2}\left[\tfrac{x^4}{4} + \tfrac{2x^3}{3}\right]_{-2}^{0} + a\left[\tfrac{x^3}{3}\right]_{0}^{2} \\ &= \tfrac{2a}{3} + \tfrac{8a}{3} = \tfrac{10}{9}. \end{aligned}$$

Moments of independent Gaussian combinations

Question

Let X and Y be independent Gaussian random variables, each with mean 3 and variance 9. Define $Z = 2X + Y$. Find $E\{XY\}$, the mean and variance of Z, and the density $f_Y(y)$.

Solution

Use independence.

$$E\{XY\} = E\{X\}E\{Y\} = 3 \times 3 = 9, \qquad \text{Cov}(X, Y) = 0.$$

Find the mean and variance. Linearity of expectation gives

$$\mu_Z = 2\mu_X + \mu_Y = 2(3) + 3 = 9.$$

For the variance, square each coefficient and include the covariance term:

$$\begin{aligned} \sigma_Z^2 &= 4\sigma_X^2 + \sigma_Y^2 + 4\,\text{Cov}(X, Y) \\ &= 4(9) + 9 = 45. \end{aligned}$$

Thus $Z \sim \mathcal{N}(9, 45)$, since it is a linear combination of independent Gaussian variables.

Write the requested density. With $\mu_Y = 3$ and $\sigma_Y = 3$,

$$f_Y(y) = \frac{1}{3\sqrt{2\pi}} \exp\left[-\frac{(y-3)^2}{18}\right], \qquad -\infty < y < \infty.$$

The variance is 9, while the standard deviation in the normalization factor is 3.

Probabilities after a logarithmic transformation

Question

Let $X \sim \mathcal{U}[0, 5]$ and

$$Y = \frac{\log(1 + \mu X)}{\log(1 + \mu)}, \qquad \mu > 0.$$

Determine $P(Y < 1)$, $P(1/2 \le X \le 1)$, and $P(1/2 \le Y \le 1)$. Evaluate the last result for $\mu = 100$.

Solution

Transform the event. The mapping is increasing, with inverse

$$x = \frac{(1 + \mu)^y - 1}{\mu}.$$

Consequently $Y < 1$ is equivalent to $X < 1$:

$$P(\,Y < 1\,) = \frac{1}{5}.$$

The untransformed interval has length $1/2$:

$$P(\,1/2 \le X \le 1\,) = \frac{1/2}{5} = 0.1.$$

Transform both interval endpoints.

$$\frac{1}{2} \le Y \le 1 \quad \Leftrightarrow \quad \frac{\sqrt{1+\mu}\,-1}{\mu} \le X \le 1.$$

Thus

$$P(\,1/2 \le Y \le 1\,) = \frac{1}{5}\left(1 - \frac{\sqrt{1+\mu}\,-1}{\mu}\right).$$

For $\mu = 100$, this is approximately 0.181900. The transformation changes which input interval corresponds to the same numerical output limits.

Half-wave and full-wave rectification of Gaussian noise

Question

Let $X \sim \mathcal{N}\left(0, 10^{-8}\right)$. A half-wave rectifier produces $Y = \max\,(X, 0)$. Find its CDF, its complete probability law, and its mean. Explain why an inverse-derivative density formula alone is insufficient. Then find the density and mean for a full-wave rectifier $V = |X|$.

Solution

Let $\sigma = 10^{-4}\,\mathrm{V}$ and $\Phi(\,z\,) = 1 - Q(\,z\,)$.

Half-wave CDF and point mass. If $y < 0$, $P(\,Y \le y\,) = 0$. If $y \ge 0$, $\{Y \le y\} = \{X \le y\}$, hence

$$F_Y(\,y\,) = \begin{cases} 0, & y < 0, \\ \Phi(\,y/\sigma\,), & y \ge 0. \end{cases}$$

There is a jump of $1/2$ at zero. The complete law consists of

$$P(\,Y = 0\,) = \frac{1}{2}, \qquad f_{Y,\mathrm{cont}}(\,y\,) = \frac{e^{-y^2/(\,2\sigma^2\,)}}{\sigma\sqrt{2\pi}} \text{ for } y > 0,$$

with continuous density zero elsewhere. Equivalently, its generalized density is $\frac{1}{2}\delta(\,y\,) + f_{Y,\mathrm{cont}}(\,y\,)$. The branch $x \le 0$ is constant and maps probability $1/2$ onto a single point; a Jacobian formula for invertible branches cannot recover that mass.

Half-wave mean.

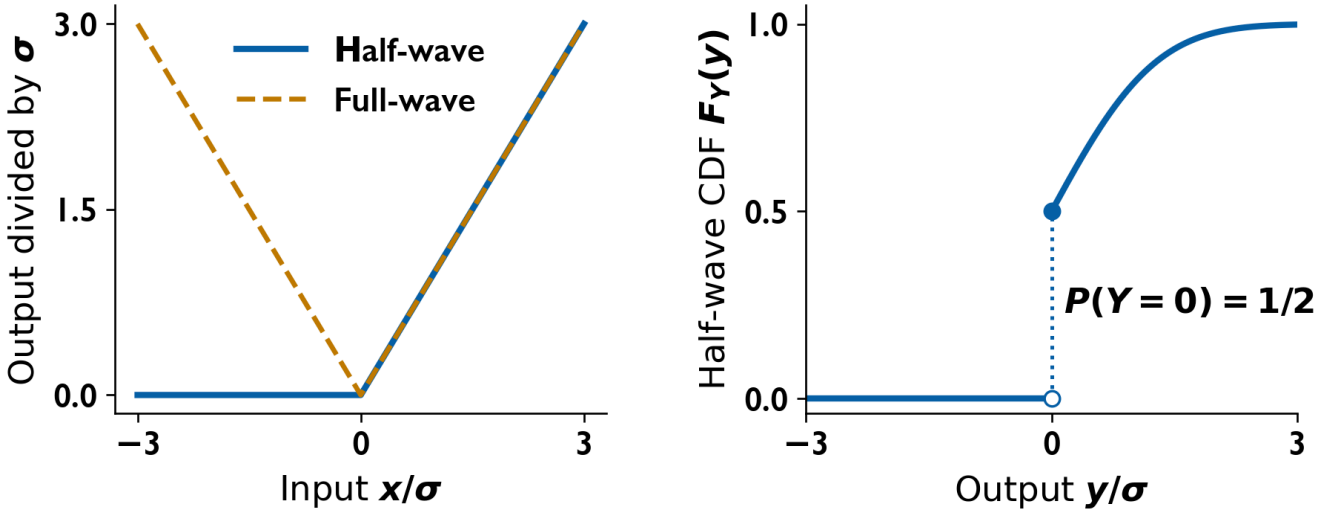
$$E\{Y\} = \frac{1}{\sigma\sqrt{2\pi}} \int_0^\infty y e^{-y^2/(2\sigma^2)}\, dy = \frac{\sigma}{\sqrt{2\pi}} \approx 3.98942 \times 10^{-5}\,\mathrm{V}.$$

Full-wave branches. For $v > 0$, the inverse values are $x = v$ and $x = -v$, each with unit Jacobian magnitude:

$$f_V(v) = \begin{cases} \dfrac{\sqrt{2/\pi}}{\sigma} e^{-v^2/(2\sigma^2)}, & v > 0, \\ 0, & v < 0. \end{cases}$$

There is no mass at zero because $P(X = 0) = 0$. Both halves now contribute to the mean:

$$E\{V\} = \sigma\sqrt{\frac{2}{\pi}} \approx 7.97885 \times 10^{-5}\,\mathrm{V}.$$



Half-wave rectification maps all negative inputs to zero, creating a CDF jump of one-half; full-wave rectification maps both signs to positive outputs.

Gaussian input through square-law and limiting devices

Question

Let $X \sim \mathcal{N}(0, \sigma^2)$ with $\sigma > 0$. Find the output probability law for each device: a square-law device $Y = aX^2$; a limiter $Y = -b$ for $X \le -b$, $Y = X$ for $|X| < b$, and $Y = b$ for $X \ge b$, where $b > 0$; and a hard limiter $Y = a$ for $X > 0$, $Y = 0$ for $X = 0$, and $Y = b$ for $X < 0$.

Solution

Square-law inverse branches. For $a \neq 0$, only outputs with $y/a > 0$ are possible. The two inverse values $x = \pm\sqrt{y/a}$ have equal Gaussian density, and $|g'(x)| = 2|a||x|$. Thus

$$f_Y(\,y\,) \;=\; \begin{cases} \dfrac{1}{\sigma\sqrt{2\pi|a||y|}}\,\exp\left(-\dfrac{y}{2a\sigma^2}\right), & y/a>0,\\[2ex] 0, & y/a<0. \end{cases}$$

There is no atom at zero for $a\neq 0$, despite the integrable density singularity there. For $a=0$, the whole law is $P(\,Y=0\,)\;=1$.

Soft limiter: retain both kinds of probability. Each saturated tail becomes a point mass:

$$P(\,Y=-b\,)\;=P(\,Y=b\,)\;=Q\left(\,b/\sigma\,\right).$$

The unsaturated branch retains the original density:

$$f_{Y,\,\mathrm{cont}}\,(\,y\,)\;=\frac{e^{-y^2/\,(\,2\sigma^2\,)}}{\sigma\sqrt{2\pi}},\qquad -b<y<b,$$

zero outside. The continuous probability is $1-2Q\left(\,b/\sigma\,\right)$, so the masses and continuous part sum to one.

Hard limiter: a discrete law. Since the input is continuous and symmetric,

$$P(\,X>0\,)\;=P(\,X<0\,)\;=\frac{1}{2},\qquad P(\,X=0\,)\;=0.$$

If $a\neq b$, the output has mass $1/2$ at each of a and b. If $a=b$, these contributions combine into mass one at that common value. The explicitly defined branch at $X=0$ creates no additional probability mass.

Joint and Conditional Distributions

Joint density and marginal densities

Two receiver measurements may share a disturbance, so their separate distributions do not describe their full relationship. A joint distribution describes which pairs of readings are likely together. This matters when one measurement can help us interpret or predict another.

For variables X, Y,

$$F_{X,Y}(x,y) = P(X \le x, Y \le y).$$

When a joint density exists,

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x\, \partial y}, \qquad f_{X,Y}(x,y) \ge 0,$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx\, dy = 1.$$

The support is the region where the density is nonzero. For a region D,

$$P((X,Y) \in D) = \iint_D f_{X,Y}(x,y)\, dx\, dy.$$

A marginal distribution describes one reading by itself, ignoring the value of the other. Integrate out the other variable to obtain a marginal density:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx.$$

Conditional density and moments

After observing one measurement, we can update what we expect from the other. The conditional density is this updated description; its mean gives the expected unobserved reading and its variance describes how uncertain that reading remains.

For $f_Y(y) > 0$,

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}.$$

This is a normalized density in x, with the support restricted to the fixed value y. Similarly, $f_{Y|X}(y \mid x) = f_{X,Y}(x,y)/f_X(x)$ when $f_X(x) > 0$.

The conditional mean and variance follow from that density:

$$\mu_{X\mid Y}(y) = E\{X \mid Y=y\} = \int_{-\infty}^{\infty} x f_{X\mid Y}(x \mid y)\, dx,$$

$$\sigma^2_{X\mid Y}(y) = \int_{-\infty}^{\infty} x^2 f_{X\mid Y}(x \mid y)\, dx - \mu_{X\mid Y}(y)^2.$$

The same integral with x^n gives the conditional moment of order n.

Independence and covariance

Independence means

$$P(X \in A, Y \in B) = P(X \in A)\, P(Y \in B).$$

For density-bearing pairs, equivalently,

$$f_{X,Y}(x,y) = f_X(x)\, f_Y(y), \qquad F_{X,Y}(x,y) = F_X(x)\, F_Y(y).$$

Under independence, the conditional density equals its marginal density.

For an integrable function g,

$$E\{g(X,Y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)\, f_{X,Y}(x,y)\, dx\, dy.$$

Covariance indicates whether deviations from the two means tend to occur in the same or opposite directions. It measures this particular type of relationship, rather than every possible dependence. In particular,

$$\operatorname{Cov}(X,Y) = E\{\left(X-\mu_X\right)\left(Y-\mu_Y\right)\} = E\{XY\} - \mu_X\mu_Y.$$

For $\sigma_X, \sigma_Y > 0$, the correlation coefficient is

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

Independent variables with finite second moments are uncorrelated. Zero covariance does not generally imply independence. It does imply independence for a jointly Gaussian pair; Gaussian marginal densities alone are insufficient.

Example: sine and cosine of a uniform angle

Question

Let Z be uniform on $[0, 2\pi]$, with $X = \sin Z$ and $Y = \cos Z$. Find their marginal densities. Are they uncorrelated? Are they independent?

Solution

For $-1 < x < 1$, the sine map has two inverse branches. Each has derivative magnitude $\sqrt{1-x^2}$ in the forward map, so

$$f_X(x) = \frac{2}{2\pi\sqrt{1-x^2}} = \frac{1}{\pi\sqrt{1-x^2}}.$$

The cosine map gives the same form:

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}, \qquad -1 < y < 1.$$

Both densities are zero outside their stated intervals. Over one full cycle,

$$\mu_X = \mu_Y = 0,$$

$$E\{XY\} = \frac{1}{2\pi}\int_0^{2\pi} \sin z \cos z \, dz = \frac{1}{4\pi}\int_0^{2\pi} \sin(2z)\, dz = 0.$$

Thus $\mathrm{Cov}(X, Y) = 0$. Nevertheless,

$$X^2 + Y^2 = 1,$$

so knowing X restricts Y: they are dependent. Their joint law lies on a circle and has no ordinary density over the plane. A shared random phase can therefore link two components even when their covariance is zero.

Example: a joint density on a triangular support

Question

Let

$$f_{X,Y}(x,y) = \begin{cases} Ke^{-x-y}, & x \geq y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find K, the marginal and conditional densities, independence, covariance, and correlation coefficient.

Solution

Normalization gives

$$1 = K\int_0^\infty \int_y^\infty e^{-x-y}\,dx\,dy = \frac{K}{2}, \qquad K = 2.$$

The marginal densities are

$$f_X(x) = \int_0^x 2e^{-x-y}\,dy = 2e^{-x}\left(1-e^{-x}\right), \qquad x \geq 0,$$

$$f_Y(y) = \int_y^\infty 2e^{-x-y}\,dx = 2e^{-2y}, \qquad y \geq 0.$$

Both are zero for negative arguments. The joint support prevents factorization, so the variables are dependent. Conditioning gives

$$f_{X\,|\,Y}(x\,|\,y) = e^{-(x-y)}, \qquad x \geq y, \quad y \geq 0,$$

and zero for $x < y$. With $u = x - y$,

$$\mu_{X\,|\,Y}(y) = \int_0^\infty (y+u)\,e^{-u}\,du = y+1,$$

$$\sigma^2_{X\,|\,Y}(y) = \int_0^\infty u^2 e^{-u}\,du - 1^2 = 1.$$

Integrating the marginal densities yields

$$\mu_X = \frac{3}{2}, \quad E\{X^2\} = \frac{7}{2}, \quad \sigma_X^2 = \frac{5}{4},$$

$$\mu_Y = \frac{1}{2}, \quad E\{Y^2\} = \frac{1}{2}, \quad \sigma_Y^2 = \frac{1}{4}.$$

Finally,

$$E\{XY\} = \int_0^\infty y\,(\,y+1\,)\;2e^{-2y}\,dy = 1,$$

$$\text{Cov}\;(\,X,Y\,) = 1 - \frac{3}{2}\cdot\frac{1}{2} = \frac{1}{4}, \qquad \rho_{X,Y} = \frac{1/4}{\sqrt{5/4}\sqrt{1/4}} = \frac{1}{\sqrt{5}}.$$

Worked Problems

Normalization and marginals on a curved support

Question

The joint density is

$$f_{XY}\,(\,x,y\,) = \begin{cases} k\,(\,1+2xy\,), & 0 < x \le 1, \quad 0 < x^2 < y \le x + 0.5 \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find:

- **The normalization constant k.**
- **The marginal densities $f_X\,(\,x\,)$ and $f_Y\,(\,y\,)$.**

Solution

Support

$$x + \tfrac{1}{2} \le 1 \quad \Rightarrow \quad \mathcal{D} = \{\,(\,x,y\,) : 0 < x \le \tfrac{1}{2},\; x^2 < y \le x + \tfrac{1}{2}\}.$$

Normalization

$$\begin{aligned} 1 &= k\int_0^{1/2}\int_{x^2}^{x+1/2}\,(\,1+2xy\,)\;dy\,dx \\ &= k\int_0^{1/2}\left(\tfrac{1}{2} + \tfrac{5}{4}x + x^3 - x^5\right)dx = k\tfrac{161}{384}. \end{aligned}$$

$$k = \frac{384}{161}$$

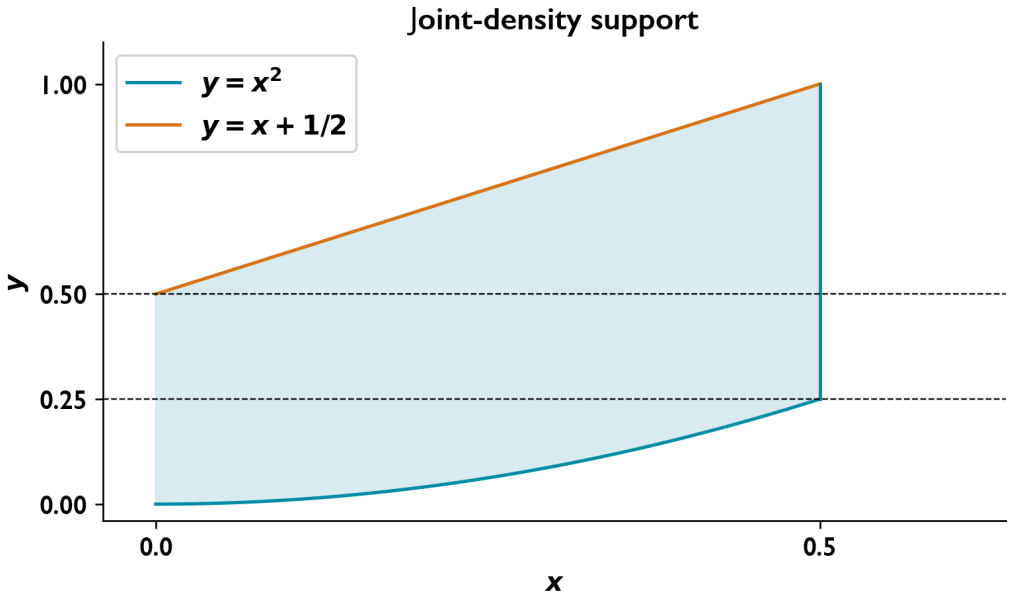
Marginal of X

$$f_X(x) = \begin{cases} \dfrac{384}{161}\left(\frac{1}{2} + \frac{5}{4}x + x^3 - x^5\right), & 0 < x < \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Marginal of Y For fixed y, let $\ell = \max(0, y - 1/2)$ and $u = \min\left(1/2, \sqrt{y}\right)$.

$$f_Y(y) = k\int_{\ell}^{u} (1 + 2xy)\, dx = k\left[x + yx^2\right]_{\ell}^{u}.$$

$$f_Y(y) = \frac{384}{161}\begin{cases} \sqrt{y} + y^2, & 0 < y < \frac{1}{4}, \\ \frac{1}{2} + \frac{1}{4}y, & \frac{1}{4} \le y < \frac{1}{2}, \\ 1 - y + y^2 - y^3, & \frac{1}{2} \le y < 1, \\ 0, & \text{otherwise.} \end{cases}$$



The curved support fixes the marginal integration limits. Horizontal boundary changes occur at y = 1/4 and y = 1/2.

A joint density on a triangular strip

Question

The joint density is

$$f_{XY}(x, y) = \begin{cases} kx, & 1 \le x + y \le 2, \quad x, y \ge 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find k, $f_X(x)$, and $f_Y(y)$.

Solution

Normalization Set $s = x + y$; the Jacobian is one.

$$1 = k \int_1^2 \int_0^s x \, dx \, ds = \frac{k}{2} \int_1^2 s^2 ds = \frac{7k}{6}, \qquad k = \frac{6}{7}.$$

Marginal of X

$$f_X(x) = kx \int_{\max(0,\,1-x)}^{2-x} dy.$$

$$f_X(x) = \begin{cases} \frac{6}{7}x, & 0 \le x \le 1, \\ \frac{6}{7}x(2-x), & 1 < x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Marginal of Y

$$f_Y(y) = k \int_{\max(0,\,1-y)}^{2-y} x \, dx.$$

$$f_Y(y) = \begin{cases} \frac{3}{7}(3-2y), & 0 \le y \le 1, \\ \frac{3}{7}(2-y)^2, & 1 < y \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

A joint exponential density on an ordered support

Question

Two continuous random variables have joint density

$$f_{X,Y}(x,y) = \begin{cases} ke^{-(x+y)}, & x > y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

- **Determine k and both marginal densities.**
- **Determine whether X and Y are independent.**
- **Obtain $f_{X \mid Y}(x \mid y)$.**

Solution

Normalization

$$1 = \int_0^\infty \int_y^\infty k e^{-(x+y)}\, dx\, dy = k \int_0^\infty e^{-2y}\, dy = \frac{k}{2}.$$

$$k = 2.$$

Marginal densities For fixed $x > 0$, the support requires $0 < y < x$:

$$f_X(x) = \int_0^x 2e^{-x-y}\, dy = 2e^{-x}\left(1 - e^{-x}\right).$$

For fixed $y > 0$, the support requires $x > y$:

$$f_Y(y) = \int_y^\infty 2e^{-x-y}\, dx = 2e^{-2y}.$$

$$f_X(x) = 2e^{-x}\left(1 - e^{-x}\right) u(x), \qquad f_Y(y) = 2e^{-2y} u(y).$$

Independence For $0 < x < y$, both marginals are positive but the joint density is zero. Therefore it cannot factor as $f_X(x)\, f_Y(y)$.

$$X \text{ and } Y \text{ are dependent.}$$

Conditional density For $y > 0$,

$$f_{X \mid Y}(x \mid y) = \frac{2e^{-x-y}}{2e^{-2y}} = e^{-(x-y)}, \qquad x > y.$$

$$f_{X \mid Y}(x \mid y) = \begin{cases} e^{-(x-y)}, & x > y, \\ 0, & x \le y. \end{cases} \qquad y > 0.$$

Its integral over (y, ∞) equals one.

An exponential conditional law with a random rate

Question

Let

$$f_{X,Y}(x, y) = y e^{-y(x+1)} u(x)\, u(y).$$

Obtain $f_Y(y)$, determine whether the variables are independent, and find $f_{X \mid Y}(x \mid y)$.

Solution

Marginal density of Y For $y > 0$,

$$f_Y(y) = \int_0^\infty y e^{-y(x+1)}\, dx = y e^{-y} \int_0^\infty e^{-yx}\, dx = e^{-y}.$$

$$f_Y(y) = e^{-y}u(y).$$

Independence For $x \geq 0$,

$$f_X(x) = \int_0^\infty ye^{-(x+1)y}\,dy = \frac{1}{(x+1)^2}.$$

On the positive quadrant,

$$f_X(x)\,f_Y(y) = \frac{e^{-y}}{(x+1)^2} \neq ye^{-y(x+1)}.$$

$$X \text{ and } Y \text{ are dependent.}$$

Conditional density For $y > 0$,

$$f_{X\mid Y}(x \mid y) = \frac{ye^{-y(x+1)}}{e^{-y}} = ye^{-yx}, \qquad x \geq 0.$$

$$f_{X\mid Y}(x \mid y) = \begin{cases} ye^{-yx}, & x \geq 0, \\ 0, & x < 0. \end{cases} \qquad y > 0.$$

Normalization follows from $\int_0^\infty ye^{-yx}\,dx = 1$. The conditional rate depends on y.

Conditional statistics on a curved joint support

Question

Let X and Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} 6xy, & 0 \leq x \leq 1, \quad 0 \leq y \leq \sqrt{x}, \\ 0, & \text{otherwise.} \end{cases}$$

Determine:

- Both marginal densities and whether X and Y are independent.
- The conditional density $f_{X\mid Y}(x \mid y)$.
- The conditional mean and variance of X given $Y = y$.
- The joint support in the (x,y) plane and the cross moment $E\{XY\}$.

Solution

Support and normalization

The same region can be written as

$$\mathcal{D} = \{(x,y) : 0 \leq y \leq 1,\ y^2 \leq x \leq 1\}.$$

The support lies between $y = 0$ and $y = \sqrt{x}$ for $0 \le x \le 1$:

$$\int_0^1 \int_0^{\sqrt{x}} 6xy\, dy\, dx = \int_0^1 3x^2\, dx = 1.$$

Marginal densities

$$\begin{aligned} f_X(x) &= \int_0^{\sqrt{x}} 6xy\, dy = 3x^2, & 0 \le x \le 1, \\ f_Y(y) &= \int_{y^2}^1 6xy\, dx = 3y\left(1 - y^4\right), & 0 \le y \le 1. \end{aligned}$$

Both densities are zero outside their stated supports. Thus

$$f_X(x) = 3x^2 \mathbf{1}_{[0,1]}(x), \qquad f_Y(y) = 3y\left(1 - y^4\right)\mathbf{1}_{[0,1]}(y).$$

The joint density is zero for $x < y^2$, whereas $f_X(x)\, f_Y(y)$ is positive at interior points in that region. Therefore X and Y are dependent.

Conditional density

For $0 < y < 1$,

$$f_{X \mid Y}(x \mid y) = \begin{cases} \dfrac{2x}{1 - y^4}, & y^2 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Conditional moments

$$\begin{aligned} E\{X \mid Y = y\} &= \tfrac{2}{1 - y^4} \int_{y^2}^1 x^2\, dx = \tfrac{2\left(1 - y^6\right)}{3\left(1 - y^4\right)}, \\ E\{X^2 \mid Y = y\} &= \tfrac{2}{1 - y^4} \int_{y^2}^1 x^3\, dx = \tfrac{1 + y^4}{2}. \end{aligned}$$

Hence

$$E\{X \mid Y = y\} = \frac{2\left(1 + y^2 + y^4\right)}{3\left(1 + y^2\right)},$$

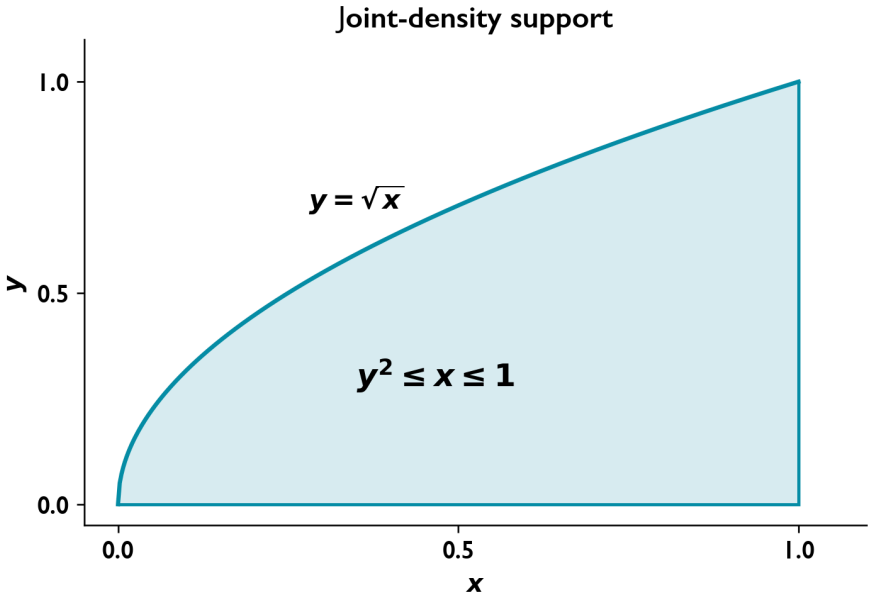
$$\operatorname{Var}(X \mid Y = y) = \frac{1 + y^4}{2} - \left[\frac{2\left(1 + y^2 + y^4\right)}{3\left(1 + y^2\right)}\right]^2.$$

The density ratio is defined for $0 < y < 1$. Its continuous limits give mean $2/3$ and variance $1/18$ as $y \to 0$, and mean 1 and variance 0 as $y \to 1$.

Cross moment

$$
\begin{aligned}
E\{XY\} &= \int_0^1 \int_0^{\sqrt{x}} 6x^2y^2 \, dy \, dx \\
&= 2\int_0^1 x^{7/2} \, dx = \tfrac{4}{9}.
\end{aligned}
$$

These are scalar random variables; their joint density determines $E\{XY\}$, not a temporal correlation at an unspecified lag.



The joint support is bounded by y = sqrt(x), y = 0 and x = 1.

Conditional probabilities in a four-symbol channel

Question

A channel has input symbol $S \in \{A, B, C, D\}$ with probabilities

$$P(S = A) = \frac{3}{16}, \quad P(S = B) = \frac{1}{4}, \quad P(S = C) = \frac{5}{16}, \quad P(S = D) = \frac{1}{4}.$$

Its received symbol R has the following conditional probabilities; rows denote the sent symbol and columns the received symbol.

$P(R = r \mid S = s)$	A	B	C	D
A	0.97	0.03	0	0
B	0.05	0.95	0	0
C	0	0	0.97	0.03
D	0	0	0.05	0.95

Find $P(R = A)$, the probability $P(R \neq S)$, and all four probabilities $P(S = s \mid R = D)$.

Solution

Received-symbol probability. Sum the joint probabilities over all possible inputs:

$$P(R = A) = \frac{3}{16}(0.97) + \frac{1}{4}(0.05) = 0.194375.$$

Error probability. Each input's error probability is one minus its diagonal entry:

$$\begin{aligned} P(R \neq S) &= \left(\tfrac{3}{16} + \tfrac{5}{16}\right)(0.03) + \left(\tfrac{1}{4} + \tfrac{1}{4}\right)(0.05) \\ &= 0.04. \end{aligned}$$

Condition on the received symbol.

$$P(R = D) = \frac{5}{16}(0.03) + \frac{1}{4}(0.95) = \frac{79}{320}.$$

Bayes' rule uses $P(S = s, R = D) = P(S = s)\,P(R = D \mid S = s)$:

$$\begin{aligned} P(S = A \mid R = D) &= 0, \\ P(S = B \mid R = D) &= 0, \\ P(S = C \mid R = D) &= \tfrac{3}{79}, \\ P(S = D \mid R = D) &= \tfrac{76}{79}. \end{aligned}$$

These posterior probabilities sum to one.

Gaussian marginals with a non-Gaussian joint law

Question

The joint density is

$$f_{X,Y}(x,y) = \begin{cases} \frac{K}{\pi} e^{-(x^2+y^2)/2}, & xy \geq 0, \\ 0, & xy < 0. \end{cases}$$

Find K and the marginal laws. Determine whether the variables are jointly Gaussian, independent, or uncorrelated. Find $f_{X|Y}(x \mid y)$ and decide whether it is Gaussian.

Solution

Normalize the two occupied quadrants. The first and third quadrants give identical integrals:

$$1 = \frac{2K}{\pi}\left(\int_0^\infty e^{-x^2/2}\,dx\right)^2 = \frac{2K}{\pi}\frac{\pi}{2} = K.$$

Thus $K = 1$.

Obtain the marginals. For $x > 0$, integrate over $y > 0$; for $x < 0$, integrate over $y < 0$:

$$f_X(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}, \qquad f_Y(y) = \frac{e^{-y^2/2}}{\sqrt{2\pi}}.$$

Both variables are individually $\mathcal{N}(0,1)$.

Test dependence and covariance. Opposite signs never occur, although independent standard-normal variables would have opposite signs with probability $1/2$. Hence they are dependent. Moreover

$$E\{XY\} = \frac{2}{\pi}\left(\int_0^\infty x e^{-x^2/2}\,dx\right)^2 = \frac{2}{\pi}.$$

Since both means vanish and both variances equal one,

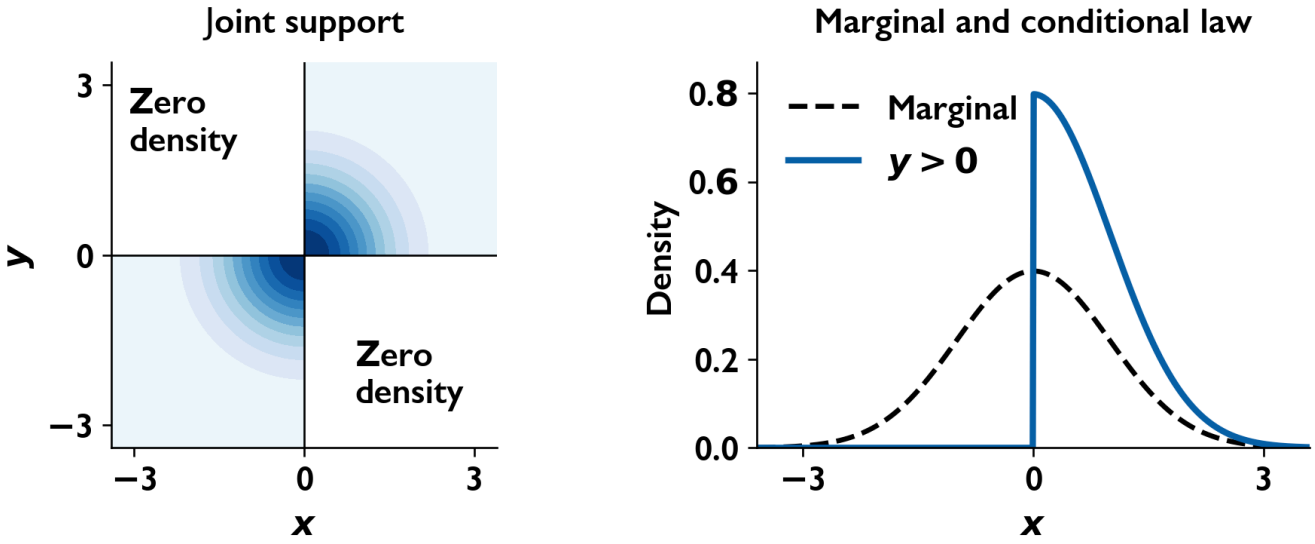
$$\operatorname{Cov}(X,Y) = \rho_{X,Y} = \frac{2}{\pi} \neq 0.$$

They are therefore correlated. The covariance matrix is nonsingular, but this joint density vanishes in two open quadrants. A nonsingular jointly Gaussian density is positive everywhere, so these Gaussian marginals are not jointly Gaussian.

Conditional law. For $y \neq 0$,

$$f_{X\,|\,Y}(\,x\,|\,y\,) = \begin{cases} \sqrt{\dfrac{2}{\pi}}e^{-x^2/2}, & xy > 0, \\ 0, & xy < 0. \end{cases}$$

It is a positive half-normal law when $y > 0$ and a reflected half-normal law when $y < 0$, rather than a Gaussian law on the entire real line. A conditional density at the single event $Y = 0$ is not uniquely determined by the joint density, since $P(\,Y = 0\,) = 0$.



Matching-sign support couples two Gaussian marginals. Conditioning on a positive Y leaves a positive half-normal density.

Sum of independent exponential variables

Question

Independent variables have densities $f_X(\,x\,) = \alpha e^{-\alpha x}$ for $x > 0$ and $f_Y(\,y\,) = \beta e^{-\beta y}$ for $y > 0$, zero elsewhere, where $\alpha, \beta > 0$. Find the density of $Z = X + Y$ for unequal rates and for equal rates.

Solution

Respect the support in the convolution. The output is positive. For $z > 0$, both $x > 0$ and $z - x > 0$ require $0 < x < z$:

$$f_Z(\,z\,) = \int_0^z \alpha e^{-\alpha x}\beta e^{-\beta(\,z-x\,)}\,dx.$$

For $\alpha \neq \beta$, integration gives

$$f_Z(\,z\,) = \begin{cases} \dfrac{\alpha\beta}{\beta-\alpha}(e^{-\alpha z} - e^{-\beta z}), & z > 0, \\ 0, & z \leq 0. \end{cases}$$

The expression is nonnegative for either ordering of the positive rates and integrates to one.

Equal rates. If $\alpha = \beta = \lambda$, the integrand is constant with respect to x:

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z > 0, \\ 0, & z \le 0. \end{cases}$$

As a check, the means add: $E\{Z\} = 1/\alpha + 1/\beta$, giving $2/\lambda$ in the equal-rate case.

Weighted error probabilities from conditional densities

Question

An observation has conditional densities

$$f_{Y|0}(y) = \begin{cases} 1/2, & -1.8 \le y \le 0.2, \\ 0, & \text{otherwise}, \end{cases}$$

$$f_{Y|1}(y) = \begin{cases} y + 0.2, & -0.2 \le y \le 0.8, \\ 1.8 - y, & 0.8 < y \le 1.8, \\ 0, & \text{otherwise}. \end{cases}$$

Let the prior probabilities of the rectangular and triangular laws be p_0 and p_1, respectively, where $p_0 + p_1 = 1$ and both are positive. Decide state one when $Y > \lambda$. Find the conditional and average error probabilities, the best threshold and its error probability. Plot the average error for $-0.2 \le \lambda \le 0.2$ when $(p_0, p_1) = (0.6, 0.4)$. Compare the result with $(0.4, 0.6)$ and with equal priors.

Solution

Conditional probabilities. An error under state zero is the right-tail event; under state one it is the left-tail event:

$$e_0(\lambda) = P(Y > \lambda \mid 0), \qquad e_1(\lambda) = P(Y \le \lambda \mid 1).$$

Integrating the specified densities gives

$$e_0(\lambda) = \begin{cases} 1, & \lambda < -1.8, \\ (0.2 - \lambda)/2, & -1.8 \le \lambda \le 0.2, \\ 0, & \lambda > 0.2, \end{cases}$$

$$e_1(\lambda) = \begin{cases} 0, & \lambda < -0.2, \\ (\lambda + 0.2)^2/2, & -0.2 \le \lambda \le 0.8, \\ 1 - (1.8 - \lambda)^2/2, & 0.8 < \lambda \le 1.8, \\ 1, & \lambda > 1.8. \end{cases}$$

Therefore

$$P_e(\lambda) = p_0 e_0(\lambda) + p_1 e_1(\lambda).$$

Best threshold. The conditional supports overlap only on $[-0.2, 0.2]$. In that interval,

$$P_e(\lambda) = \frac{p_0}{2}(0.2 - \lambda) + \frac{p_1}{2}(\lambda + 0.2)^2,$$

$$\frac{dP_e}{d\lambda} = -\frac{p_0}{2} + p_1(\lambda + 0.2), \qquad \frac{d^2 P_e}{d\lambda^2} = p_1 > 0.$$

The unconstrained minimum is $p_0/(2p_1) - 0.2$; restrict it to the overlap:

$$\lambda_* = \min\left(0.2, \frac{p_0}{2p_1} - 0.2\right).$$

Its value is

$$P_{e,\,\min} = \begin{cases} 0.2p_0 - \dfrac{p_0^2}{8p_1}, & p_0 \le 0.8p_1, \\ 0.08p_1, & p_0 > 0.8p_1. \end{cases}$$

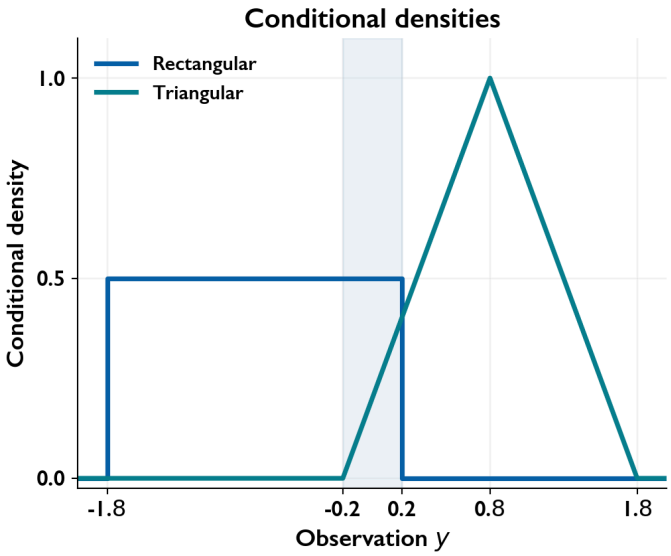
Numerical comparisons.

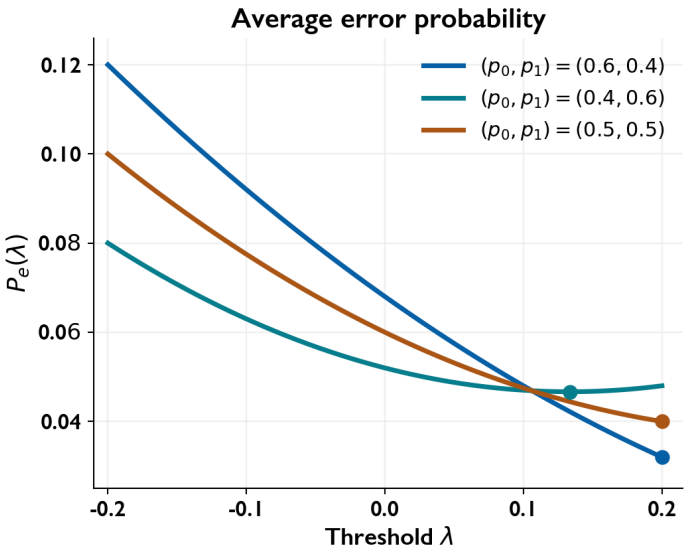
(p_0, p_1)	λ_*	$P_{e,\,\min}$
$(0.6, 0.4)$	0.2	0.032
$(0.4, 0.6)$	$2/15 \approx 0.133333$	$7/150 \approx 0.0466667$
$(0.5, 0.5)$	0.2	0.04

For $(0.6, 0.4)$, the plotted curve is

$$P_e(\lambda) = 0.2\lambda^2 - 0.22\lambda + 0.068, \qquad -0.2 \le \lambda \le 0.2.$$

It decreases from 0.12 to 0.032 over this interval. The reverse prior assignment moves the minimum inside the overlap because the triangular-law state is then more probable.


Conditional densities
Rectangular
Triangular
Conditional density
1.0
0.5
0.0
-1.8
-0.2
0.2
0.8
1.8
Observation y


Average error probability
$(p_0, p_1) = (0.6, 0.4)$
$(p_0, p_1) = (0.4, 0.6)$
$(p_0, p_1) = (0.5, 0.5)$
$P_e(\lambda)$
0.12
0.10
0.08
0.06
0.04
-0.2
-0.1
0.0
0.1
0.2
Threshold λ

Random processes, stationarity and ergodicity

The process and its ensemble

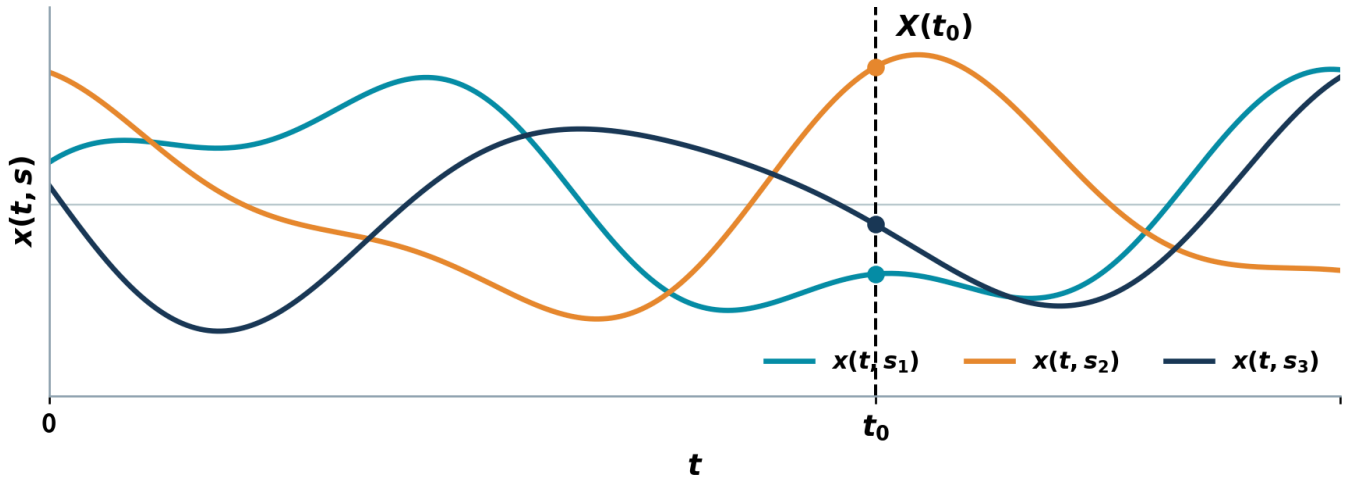
A communication receiver observes a waveform determined by transmitted symbols and noise. Before reception, those symbols and the exact noise waveform are uncertain, so we model the received signal as a random process. This model describes the possible waveforms and lets us calculate their average power and statistical behavior. A fully specified test tone can instead be a deterministic signal.

A random process assigns a waveform to each outcome s of an experiment:

$$X(t, s), \qquad x_k(t) = X(t, s_k).$$

Here t is time, and $x_k(t)$ is one sample function, or realization: the particular trace observed during one transmission. The collection of possible realizations is the ensemble. Fixing the outcome gives a waveform; fixing the time gives a random variable $X(t_0)$, whose value can differ across repeated transmissions.

Three sample functions; dots form a fixed-time ensemble slice



Realizations of a process. The vertical slice selects the random variable at one fixed time.

At a fixed time, the ensemble mean averages the amplitudes across possible realizations. It is a deterministic function of time. If a density exists,

$$\mu_X(t) = E\{X(t)\} = \int_{-\infty}^{\infty} x f_{X(t)}(x)\, dx.$$

Different realizations need not be independent merely because they belong to the same ensemble. Independence is an additional assumption about how observations or experiments are obtained.

What stationarity requires

Stationarity lets us use the same statistical description at different observation times. In a receiver, it is often an approximation over an interval during which operating conditions remain stable; switching a transmitter on or changing its power can invalidate that approximation. The mathematical stationarity conditions below are exact conditions on the model.

First-order stationarity means that the one-time probability law is unchanged by a time shift h:

$$F_{X(t+h)}(x) = F_{X(t)}(x) \qquad \text{for every } t, h, x.$$

When the mean exists, it is therefore constant. A constant mean alone does not establish this equality.

Strict-sense stationarity requires the joint law of every finite collection $X(t_1), \ldots, X(t_n)$ to remain unchanged when all times are shifted by the same h. It concerns all joint probability laws, rather than only a one-time density.

For a real process with finite second moments, wide-sense stationarity (WSS) requires

$$E\{X(t)\} = \mu_X = \text{constant}, \qquad R_X(t_1, t_2) = E\{X(t_1)\,X(t_2)\} = R_X(t_2 - t_1).$$

Writing $\tau = t_2 - t_1$ gives $R_X(\tau) = E\{X(t)\,X(t+\tau)\}$. Strict-sense stationarity with finite second moments implies WSS. WSS alone does not establish strict-sense stationarity.

Ensemble averages and time averages

An ensemble average compares realizations at fixed times. In practice, a receiver or measuring instrument often has one long waveform record rather than access to all possible realizations. Ergodicity explains when averages from that record can recover the ensemble statistics. A time average follows one realization over an observation duration L:

$$\bar{x_k} = \lim_{L \to \infty} \frac{1}{L} \int_{-L/2}^{L/2} x_k(t)\, dt.$$

In the almost-sure interpretation used here, mean ergodicity means that this limit equals μ_X for almost every realization. Correlation ergodicity means, for the lag under consideration,

$$R_X(\tau) = \lim_{L \to \infty} \frac{1}{L} \int_{-L/2}^{L/2} x_k(t)\, x_k(t+\tau)\, dt.$$

The limits must exist. Ergodicity is specified for the statistic being recovered; WSS by itself does not justify replacing an ensemble expectation by a time average. A finite observation gives an estimate, so a short record need not accurately reproduce the ensemble result.

With a normalized resistive load, mean square represents average power:

$$P_X = E\{X^2\,(\,t\,)\} = \mu_X^2 + \sigma_X^2.$$

The deterministic mean contributes power μ_X^2, while the centered fluctuation has power σ_X^2. Total RMS is $\sqrt{\mu_X^2 + \sigma_X^2}$; fluctuation RMS is σ_X. These powers coincide with DC and AC power, respectively, when the centered process has no random constant component. For a voltage across resistance r, divide the mean-square voltage by r to obtain power in watts.

Example: Random phase

Question

Is $X\,(\,t\,) = A\cos\,(\,2\pi f_c t + \Theta\,)$ stationary when A and $f_c \neq 0$ are constants and $\Theta \sim \mathcal{U}[0, 2\pi)$?

Solution

A time shift replaces Θ by $\Theta + 2\pi f_c h$, taken modulo 2π. This phase is still uniform. Because the same phase shift applies to every observed time, all finite-dimensional joint laws remain unchanged: the process is strictly stationary.

$$\mu_X\,(\,t\,) = \frac{A}{2\pi}\int_0^{2\pi} \cos\,(\,2\pi f_c t + \theta\,)\,d\theta = 0.$$

Its correlation, derived below, is $R_X\,(\,\tau\,) = \frac{A^2}{2}\cos\,(\,2\pi f_c \tau\,)$, so it is also WSS. Integrating one sinusoidal realization over a growing interval gives time mean zero and the same time correlation; it is ergodic for these two statistics.

Example: Random amplitude

Question

Is $X\,(\,t\,) = A\cos\,(\,2\pi f_c t\,)$ strictly stationary if $A \sim \mathcal{U}\,[\,0, 1\,]$ and $f_c \neq 0$?

Solution

Since $E\{A\} = 1/2$,

$$\mu_X\,(\,t\,) = \frac{1}{2}\cos\,(\,2\pi f_c t\,).$$

The mean varies with absolute time. The process is neither strictly stationary nor WSS. A random amplitude with fixed phase therefore behaves differently from a fixed amplitude with uniform random phase.

Example: Random frequency

Question

Show that $X(t) = \sin(2\pi W t)$ is nonstationary when $W \sim \mathcal{U}[0, B]$ and $B > 0$ hertz.

Solution

For $t \neq 0$,

$$\mu_X(t) = \frac{1}{B}\int_0^B \sin(2\pi w t)\, dw = \frac{1 - \cos(2\pi B t)}{2\pi B t}.$$

At $t = 0$, the mean is zero, also the limit of this expression. The mean is not constant, which rules out both forms of stationarity.

Example: A Gaussian amplitude through an integrator

Question

Let $X(t) = A\cos(2\pi f_c t)$, where $A \sim \mathcal{N}(0, \sigma_A^2)$, $\sigma_A^2 > 0$ and $f_c \neq 0$. Find the law of $Y(t_k)$ for

$$Y(t) = \int_0^t X(u)\, du,$$

and determine whether $Y(t)$ is stationary.

Solution

Direct integration gives

$$Y(t) = \frac{A\sin(2\pi f_c t)}{2\pi f_c}.$$

At a fixed time t_k, define $K_k = \sin(2\pi f_c t_k)/(2\pi f_c)$. Then

$$E\{Y(t_k)\} = 0, \qquad \sigma_Y^2(t_k) = K_k^2\sigma_A^2.$$

For $K_k \neq 0$, the output is Gaussian with density

$$f_{Y(t_k)}(y) = \frac{1}{\sqrt{2\pi}\,|K_k|\sigma_A}\exp\left(-\frac{y^2}{2K_k^2\sigma_A^2}\right).$$

If $K_k = 0$, the output is identically zero and its law is a point mass at zero. The variance changes with time, so the process is neither WSS nor strictly stationary, despite its constant zero mean.

Stationarity and ergodicity conditions

Question

State the conditions for a random process $X(t)$ to be stationary and ergodic. Distinguish wide-sense stationarity from strict stationarity.

Solution

Wide-sense stationarity. The second moment must be finite, the mean must be constant, and the autocorrelation must depend only on the time separation:

$$E\{X(t)\} = \mu_X, \qquad E\{X^2(t)\} < \infty, \qquad E\{X(t+\tau)\,X(t)\} = R_X(\tau).$$

A constant mean alone does not establish stationarity.

Strict stationarity. Every joint distribution is unchanged by a common shift of all observation times:

$$F_{X(t_1),\ldots,X(t_n)}(x_1,\ldots,x_n) = F_{X(t_1+a),\ldots,X(t_n+a)}(x_1,\ldots,x_n)$$

for all n, times, thresholds, and shifts a. With finite second moments, strict stationarity implies wide-sense stationarity.

Ergodicity. For a stationary process, a time average from almost every sufficiently long realization must equal the corresponding ensemble statistic. Mean and correlation ergodicity require, respectively,

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} X(t)\,dt = \mu_X,$$

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} X(t+\tau)\,X(t)\,dt = R_X(\tau)$$

almost surely, for each fixed τ. Ergodicity allows one long recording to represent the ensemble. Stationarity by itself does not guarantee either equality.

An exponential process driven by one uniform random variable

Question

Let $X \sim \mathcal{U}[0,2]$ and define

$$V(t) = 6e^{Xt}.$$

Find $E\{V(t)\}$, the autocorrelation $R_V(t_1,t_2)$, and the mean-square value $E\{V^2(t)\}$. Determine whether the process is stationary.

Solution

Average over the random parameter. A realization keeps the same value of X at every time. Since $f_X(x) = 1/2$ on $[0,2]$,

$$E\{V(t)\} = \frac{1}{2}\int_0^2 6e^{xt}\,dx = \begin{cases} \dfrac{3\left(e^{2t}-1\right)}{t}, & t \neq 0, \\ 6, & t = 0. \end{cases}$$

The value at $t = 0$ follows directly from $V(0) = 6$, or from the limit of the expression above.

Compute the two-time autocorrelation. The two samples share X, so their exponents add before taking the expectation:

$$R_V(t_1,t_2) = E\{V(t_1)\,V(t_2)\} = 18\int_0^2 e^{x(t_1+t_2)}\,dx.$$

Thus

$$R_V(t_1,t_2) = \begin{cases} \dfrac{18(e^{2(t_1+t_2)}-1)}{t_1+t_2}, & t_1+t_2 \neq 0, \\ 36, & t_1+t_2 = 0. \end{cases}$$

Set the two times equal.

$$E\{V^2(t)\} = R_V(t,t) = \begin{cases} \dfrac{9\left(e^{4t}-1\right)}{t}, & t \neq 0, \\ 36, & t = 0. \end{cases}$$

Check stationarity. The mean changes with time, and the autocorrelation depends on t_1+t_2 rather than only on t_1-t_2. Therefore $V(t)$ is not wide-sense stationary. It is also not strictly stationary: $V(0)$ is deterministic, whereas $V(t)$ is nondegenerate for $t \neq 0$. The random growth rate makes the waveform's statistics change as time passes.

Mean, AC power, and RMS of a waveform

Question

For $x\left(t\right) = 1 + \cos\left(2\pi f_0 t\right)$ with $f_0 > 0$, determine the time-average value, normalized AC power, and RMS value.

Solution

Separate the constant and varying parts. Over $T_0 = 1/f_0$, the cosine averages to zero:

$$\bar{x} = \frac{1}{T_0}\int_0^{T_0} x\left(t\right)\,dt = 1.$$

The AC component is $x_{\text{ac}}\left(t\right) = x\left(t\right) - \bar{x} = \cos\left(2\pi f_0 t\right)$, so

$$P_{\text{ac}} = \frac{1}{2}, \qquad P_{\text{dc}} = 1.$$

The cross term averages to zero. Consequently

$$\bar{x^2} = 1 + \frac{1}{2} = \frac{3}{2}, \qquad x_{\text{rms}} = \sqrt{\frac{3}{2}} \approx 1.2247.$$

RMS includes both the DC and AC contributions; removing the DC gives AC RMS $1/\sqrt{2}$.

Comparing time and ensemble moments

Question

A random process is $X\left(t\right) = A\cos\left(2\pi f_0 t + \Theta\right)$, where A and $f_0 > 0$ are constants and $\Theta \sim \mathcal{U}[0, 2\pi)$. Calculate the first and second moments by averaging one realization over an integer number of periods and by ensemble averaging over phase. Compare the results.

Solution

Time averages of a realization. For any fixed phase θ and positive integer M, let $T_0 = 1/f_0$:

$$\begin{aligned}\frac{1}{MT_0}\int_0^{MT_0} A\cos\left(2\pi f_0 t + \theta\right)\,dt &= 0,\\ \frac{1}{MT_0}\int_0^{MT_0} A^2\cos^2\left(2\pi f_0 t + \theta\right)\,dt &= \frac{A^2}{2}.\end{aligned}$$

These are exact for integer periods, rather than merely large-sample approximations.

Ensemble averages at a fixed time.

$$\begin{aligned}E\{X\left(t\right)\} &= \frac{A}{2\pi}\int_0^{2\pi}\cos\left(2\pi f_0 t + \theta\right)\,d\theta = 0,\\ E\{X^2\left(t\right)\} &= \frac{A^2}{2\pi}\int_0^{2\pi}\cos^2\left(2\pi f_0 t + \theta\right)\,d\theta = \frac{A^2}{2}.\end{aligned}$$

Thus time and ensemble averages agree for both requested moments. In addition,

$$X\left(t+\bar{\tau}\right)X\left(t\right)=\frac{A^2}{2}\cos\left(2\pi f_0\tau\right)=R_X\left(\tau\right),$$

so a long observation of any realization also yields the ensemble autocorrelation in this fixed-amplitude model.

Autocorrelation, autocovariance and crosscorrelation

Energy signals and random processes

A waveform has a time structure: nearby samples may carry related values, and that relationship can weaken or reverse as their separation grows. Autocorrelation measures this structure by comparing a signal with a delayed copy. For a centered random process, it describes how fluctuations at one time relate to fluctuations later, helping us assess memory and the usefulness of earlier samples in prediction.

For a real deterministic signal $x(t)$ of finite energy,

$$\mathcal{E}_x = \int_{-\infty}^{\infty} x^2(t)\,dt < \infty, \qquad R_x(\tau) = \int_{-\infty}^{\infty} x(t)\,x(t+\tau)\,dt.$$

The correlation integrates the overlap between the signal and a shifted copy. At zero lag, $R_x(0) = \mathcal{E}_x$: this is energy, not time-averaged power.

For a real random process, use an ensemble expectation:

$$R_X(t_1, t_2) = E\{X(t_1)\,X(t_2)\}.$$

Only after verifying WSS may it be reduced to

$$R_X(\tau) = E\{X(t)\,X(t+\tau)\}, \qquad R_X(0) = E\{X^2(t)\} = P_X.$$

For a correlation-ergodic process, the corresponding time integral must be divided by its observation duration before taking the infinite-duration limit. The deterministic energy integral has no such normalization.

Removing the mean

A constant signal offset can create correlation even when the fluctuations themselves have no time dependence. Autocovariance removes this mean contribution and measures the correlation of deviations from the mean:

$$C_X(t_1, t_2) = E\{\left[X(t_1) - \mu_X(t_1)\right]\left[X(t_2) - \mu_X(t_2)\right]\}$$

$$= R_X(t_1, t_2) - \mu_X(t_1)\,\mu_X(t_2).$$

For WSS,

$$C_X(\tau) = R_X(\tau) - \mu_X^2, \qquad C_X(0) = \sigma_X^2.$$

Thus correlation and covariance coincide for a zero-mean process. Values at two times are uncorrelated when their covariance is zero. Independence implies zero covariance when the moments exist; the converse requires extra assumptions, such as joint Gaussianity.

Conditions every autocorrelation must satisfy

For real WSS processes with finite average power,

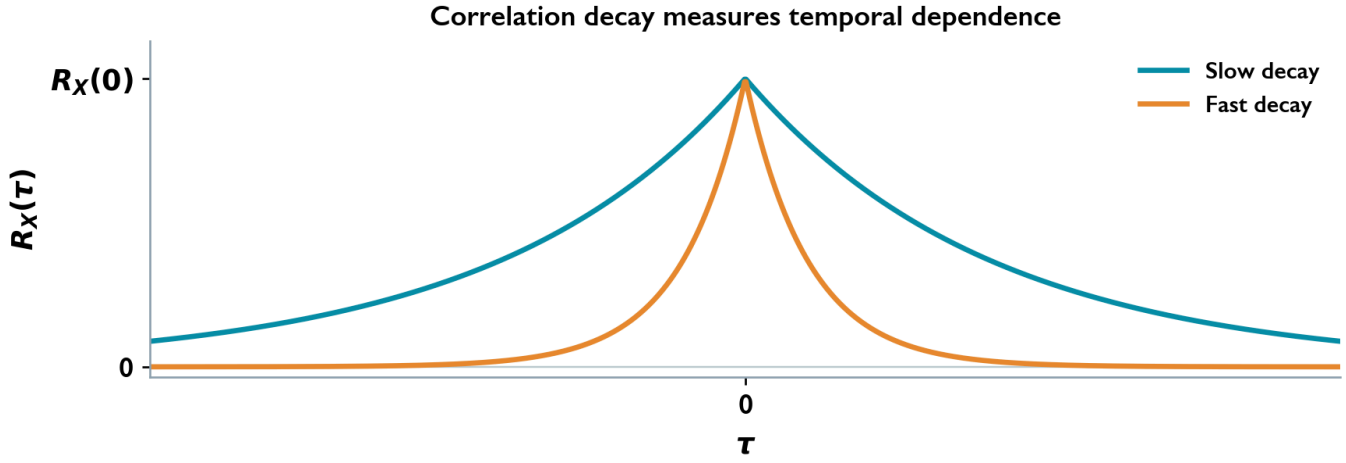
$$R_X(0) \geq 0, \qquad R_X(-\tau) = R_X(\tau), \qquad |R_X(\tau)| \leq R_X(0).$$

The magnitude bound follows without assuming a positive correlation. Expanding a nonnegative mean square gives

$$0 \leq E\{[X(t+\tau) \pm X(t)]^2\} = 2R_X(0) \pm 2R_X(\tau).$$

Hence $-R_X(0) \leq R_X(\tau) \leq R_X(0)$. The same symmetry and magnitude bound hold for finite-energy R_x, with energy at the origin.

These are necessary checks, not a complete validity test. A proposed WSS autocorrelation must also have a nonnegative power spectrum. A negative value of $R_X(\tau)$ at a nonzero lag is allowed; a negative spectral power density is not.



Normalized broad and narrow autocorrelation peaks. The broader peak retains correlation over larger time separations.

Slowly varying fluctuations, such as the output of a low-pass noise filter with a small cutoff frequency, often have a broad correlation peak: samples separated by a small delay still tend to vary together. Rapid fluctuations can have a narrow peak, so their correlation can weaken over shorter delays. This does not imply that every correlation decays monotonically or tends to zero: a sinusoidal component produces a persistent oscillation.

Example: Correlation of a random-phase sinusoid

Question

Find $R_X(\tau)$ for $X(t) = A\cos(2\pi f_c t + \Theta)$ with fixed A, f_c and $\Theta \sim \mathcal{U}[0, 2\pi)$.

Solution

Apply $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$:

$$R_X(\tau) = \frac{A^2}{2}\cos(2\pi f_c\tau) + \frac{A^2}{2}E\{\cos(4\pi f_c t + 2\pi f_c\tau + 2\Theta)\}.$$

The second term averages to zero over the uniform phase. Therefore

$$R_X(\tau) = \frac{A^2}{2}\cos(2\pi f_c\tau), \qquad P_X = \frac{A^2}{2}.$$

It is even, depends only on lag, and attains its greatest magnitude at zero and at other integer half-period lags.

Example: DC and sinusoidal components in noise

Question

How do a deterministic DC offset and an independent random-phase sinusoid affect correlation?

Solution

If $X(t) = a + N(t)$, where a is constant and N is zero-mean WSS noise, expansion gives

$$R_X(\tau) = a^2 + aE\{N(t)\} + aE\{N(t+\tau)\} + R_N(\tau) = a^2 + R_N(\tau).$$

No independence assumption is needed for the constant a. If instead $X(t) = A\cos(2\pi f_0 t + \Theta) + N(t)$ and the uniform phase is independent of the noise, the cross terms vanish and

$$R_X(\tau) = \frac{A^2}{2}\cos(2\pi f_0\tau) + R_N(\tau).$$

Example: A finite pulse and a random time shift

Question

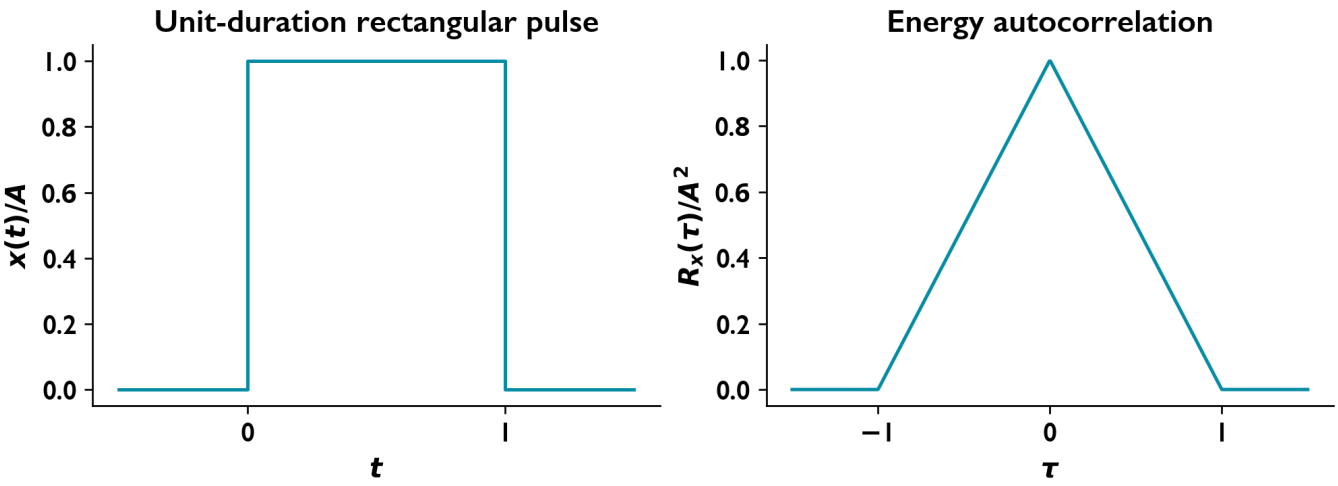
Find the energy correlation of $x(t) = Ap_T(t)$, where $p_T(t) = 1$ on $0 \le t < T$ and zero elsewhere. Does a single uniformly shifted pulse automatically become WSS?

Solution

The overlap length is $T - |\tau|$ for $|\tau| \le T$, and zero otherwise:

$$R_x(\tau) = \begin{cases} A^2(T - |\tau|), & |\tau| \le T, \\ 0, & |\tau| > T. \end{cases}$$

The zero-lag value is A^2T, the pulse energy.



The pulse and its energy correlation for unit duration. The peak is the signal energy.

For a random shift $U \sim \mathcal{U}[0, T)$, take $X(t) = Ap_T(t + U)$ with $A \neq 0$. Then $R_X(0, 0) = A^2$ but $R_X(T, T) = 0$. The mean square depends on absolute time, so this single finite pulse is not WSS. Its energy correlation must not be relabeled as a stationary process correlation.

Crosscorrelation and the lag convention

Crosscorrelation compares two waveforms as one is shifted relative to the other. A receiver can compare a known transmitted pulse with its delayed received copy to estimate the propagation delay: their overlap is greatest when the shift aligns them. The sign of that shift depends on which waveform carries the lag, so the convention must be fixed first.

For two real jointly WSS processes, put the lag on the first process:

$$R_{XY}(\tau) = E\{X(t + \tau)\, Y(t)\}, \qquad R_{YX}(\tau) = E\{Y(t + \tau)\, X(t)\}.$$

Joint WSS means that the individual means are constant and all auto- and crosscorrelations depend only on time differences. Individual WSS alone does not establish this joint property. Reversing the order reflects the lag axis:

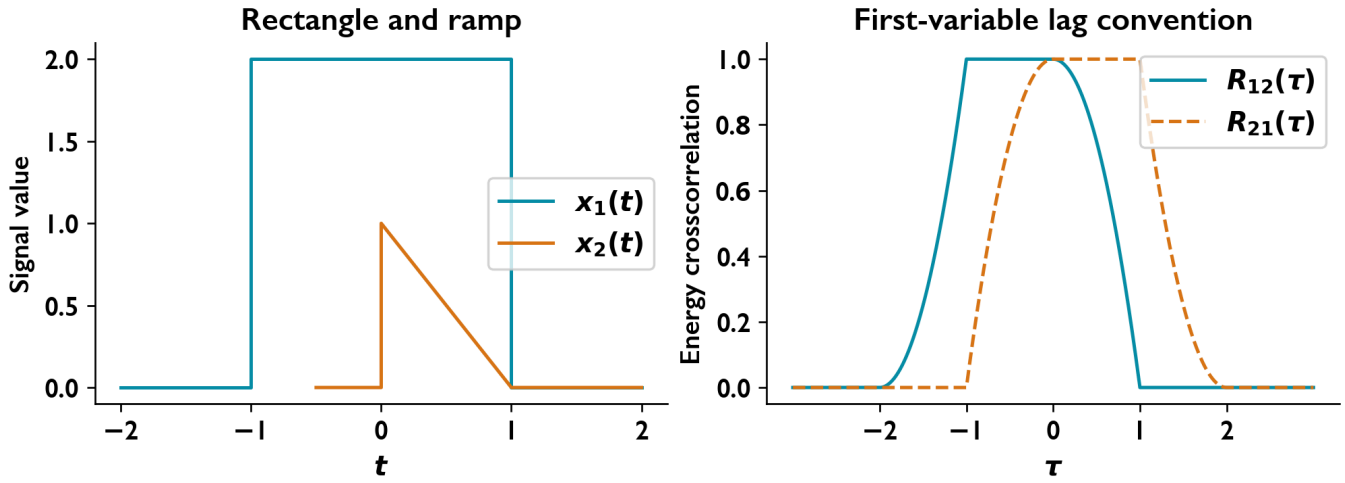
$$R_{XY}(\tau) = R_{YX}(-\tau).$$

Crosscorrelation need not be even or peak at zero. Crosscovariance is $C_{XY}(\tau) = R_{XY}(\tau) - \mu_X \mu_Y$.

The matching deterministic energy definition is

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t+\tau)\, y(t)\, dt,$$

with $R_{xy}(\tau) = R_{yx}(-\tau)$. The order must be fixed before identifying the sign of a delay.



Energy crosscorrelations of a rectangle and ramp. Exchanging the two signals reflects the lag axis.

Worked Problems

Energy autocorrelation of a rectangular pulse

Question

A real deterministic pulse is

$$x(t) = \begin{cases} A, & 0 \le t \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Obtain and sketch its energy autocorrelation,

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t+\tau)\, x(t)\, dt.$$

Solution

Overlap intervals The factors are simultaneously nonzero on the intersection of $[0,1]$ and $[-\tau, 1-\tau]$.

For $0 \le \tau \le 1$,

$$R_x(\tau) = \int_0^{1-\tau} A^2\, dt = A^2\,(1-\tau).$$

For $-1 \le \tau \le 0$,

$$R_x(\tau) = \int_{-\tau}^{1} A^2\, dt = A^2\,(1+\tau).$$

For $|\tau| > 1$, there is no overlap.

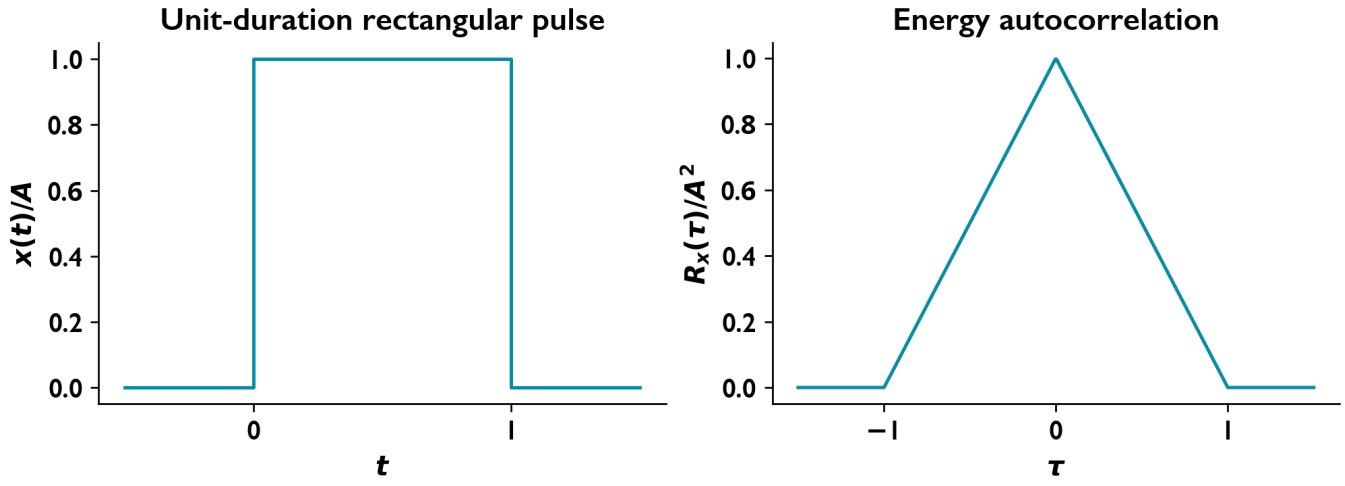
Autocorrelation

$$R_x(\tau) = \begin{cases} A^2\,(1-|\tau|), & |\tau| \le 1, \\ 0, & |\tau| > 1. \end{cases}$$

The graph is triangular and even, with peak A^2 and zeros at $\tau = \pm 1$.

Zero-lag energy

$$R_x(0) = E_x = \int_0^1 A^2\, dt = A^2.$$



Overlap of a unit-duration pulse produces its triangular energy autocorrelation.

Crosscorrelation of a rectangular pulse and a ramp

Question

Let

$$x_1(t) = \begin{cases} 2, & -1 \le t \le 1, \\ 0, & \text{otherwise}, \end{cases} \qquad x_2(t) = \begin{cases} 1-t, & 0 \le t \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Obtain and sketch the energy crosscorrelations

$$R_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t+\tau)\, x_2(t)\, dt,$$

$$R_{21}(\tau) = \int_{-\infty}^{\infty} x_2(t+\tau)\, x_1(t)\, dt.$$

Solution

Overlap geometry For R_{12}, intersect the ramp interval $[0,1]$ with the shifted rectangle interval $[-1-\tau, 1-\tau]$. The integrand is $2(1-t)$ on the intersection.

Partial and full overlaps

$$\begin{aligned} R_{12}(\tau) &= 2\int_{-1-\tau}^{1} (1-t)\, dt = (\tau+2)^2, && -2 \le \tau < -1, \\ R_{12}(\tau) &= 2\int_{0}^{1} (1-t)\, dt = 1, && -1 \le \tau \le 0, \\ R_{12}(\tau) &= 2\int_{0}^{1-\tau} (1-t)\, dt = 1-\tau^2, && 0 < \tau \le 1. \end{aligned}$$

Outside $[-2,1]$, there is no overlap.

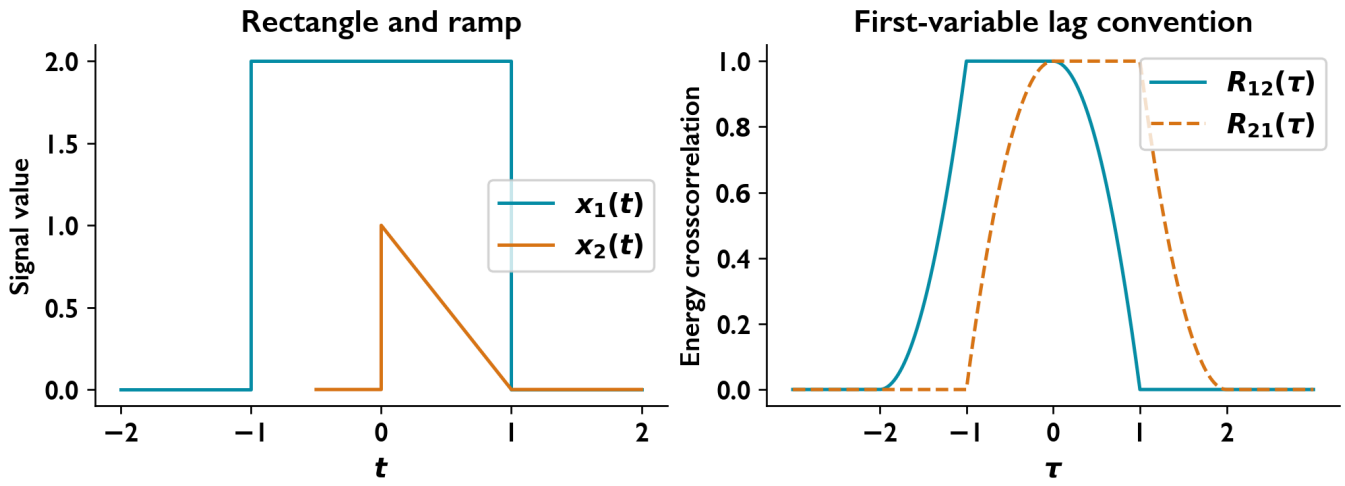
Forward-order crosscorrelation

$$R_{12}(\tau) = \begin{cases} (\tau+2)^2, & -2 \le \tau < -1, \\ 1, & -1 \le \tau \le 0, \\ 1-\tau^2, & 0 < \tau \le 1, \\ 0, & \text{otherwise}. \end{cases}$$

Reverse-order crosscorrelation For real signals, $R_{21}(\tau) = R_{12}(-\tau)$.

$$R_{21}(\tau) = \begin{cases} 1-\tau^2, & -1 \le \tau < 0, \\ 1, & 0 \le \tau \le 1, \\ (2-\tau)^2, & 1 < \tau \le 2, \\ 0, & \text{otherwise}. \end{cases}$$

The two graphs are reflections about zero lag. Each has maximum one and continuous joins between adjacent pieces.



Crosscorrelation order is defined by shifting the first signal. Reversal reflects the lag axis.

Testing proposed autocorrelation functions

Question

Determine whether each function can be the autocorrelation of a real WSS process with finite second moment.

- $r_1(\tau) = \sin(2\pi f_0 \tau)$, where $f_0 \neq 0$.
- $r_2(\tau) = \tau^2$.
- The piecewise function

$$r_3(\tau) = \begin{cases} 1 - |\tau|, & |\tau| \leq 1, \\ 1 + |\tau|, & |\tau| > 1. \end{cases}$$

Solution

Necessary conditions A real WSS autocorrelation must satisfy

$$R(-\tau) = R(\tau), \qquad |R(\tau)| \leq R(0).$$

Sine candidate r_1 is odd and nonzero, so it is not even. Also $r_1(0) = 0$ while r_1 is nonzero at some lags.

$$r_1 \text{ is invalid.}$$

Quadratic candidate Although r_2 is even,

$$r_2(0) = 0, \qquad r_2(1) = 1 > r_2(0).$$

$$r_2 \text{ is invalid.}$$

Piecewise candidate The function is even but violates the zero-lag bound:

$$r_3(0) = 1, \qquad r_3(2) = 3 > r_3(0).$$

r_3 is invalid.

Time-averaged correlation of a sinusoid

Question

For the deterministic waveform $x(t) = A\cos(2\pi f_0 t + \phi)$, with $f_0 > 0$, calculate its time-averaged autocorrelation and average normalized power. Express the correlation in terms of the period $T_0 = 1/f_0$.

Solution

Average over one period.

$$R_x(\tau) = \frac{1}{T_0}\int_0^{T_0} x(t+\tau)\,x(t)\,dt.$$

Using $\cos u \cos v = \frac{1}{2}\cos(u-v) + \frac{1}{2}\cos(u+v)$, the second cosine averages to zero:

$$R_x(\tau) = \frac{A^2}{2}\cos\left(\frac{2\pi\tau}{T_0}\right), \qquad P_x = R_x(0) = \frac{A^2}{2}.$$

This is a time average of a periodic power signal. The same expression is the ensemble autocorrelation of a fixed-amplitude sinusoid with a uniformly random phase.

Power and autocorrelation of a two-tone waveform

Question

For the deterministic waveform

$$x(t) = 10\cos(10t) + 20\cos(20t),$$

find its average power, verify that power using its complex Fourier-series coefficients, and obtain its time-average autocorrelation

$$\bar{R}_x(\tau) = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} x(t+\tau)\,x(t)\,dt.$$

Solution

Average power. Distinct harmonics have zero time-average cross product, and each squared cosine averages to $1/2$:

$$P_x = \frac{10^2}{2} + \frac{20^2}{2} = 250.$$

Fourier-series check. The fundamental angular frequency is 10 rad/s, and

$$x\left(t\right) = 5e^{j10t} + 5e^{-j10t} + 10e^{j20t} + 10e^{-j20t}.$$

Only $c_{\pm 1} = 5$ and $c_{\pm 2} = 10$ are nonzero. Thus

$$\sum_k |c_k|^2 = 2\left(5^2\right) + 2\left(10^2\right) = 250.$$

Time autocorrelation. Averaging the cosine-product identity removes cross terms and double-frequency terms:

$$\bar{R}_x\left(\tau\right) = 50\cos\left(10\tau\right) + 200\cos\left(20\tau\right).$$

At zero lag, $\bar{R}_x\left(0\right) = 250 = P_x$. This persistent waveform has finite average power; its unrestricted energy integral is infinite.

Testing correlation functions through their spectra

Question

For a real random process, determine which candidate functions can be autocorrelations and justify each decision:

$$\begin{aligned}
r_a\left(\tau\right) &= \begin{cases} 1, & |\tau| \le 1, \\ 0, & |\tau| > 1, \end{cases} \\
r_b\left(\tau\right) &= \delta\left(\tau\right) + \sin\left(2\pi f_0 \tau\right), \quad f_0 \neq 0, \\
r_c\left(\tau\right) &= e^{|\tau|}, \\
r_d\left(\tau\right) &= \begin{cases} 1 - |\tau|, & |\tau| \le 1, \\ 0, & |\tau| > 1. \end{cases}
\end{aligned}$$

Solution

A real autocorrelation is even; an ordinary finite-power autocorrelation also satisfies $|R_X\left(\tau\right)| \le R_X\left(0\right)$. Its Fourier transform must be nonnegative.

Rectangular candidate.

$$\mathcal{F}\{r_a\}\left(f\right) = \int_{-1}^{1} e^{-j2\pi f\tau}\, d\tau = 2\,\mathrm{sinc}\left(2f\right).$$

This spectrum has negative intervals; for example, its value at $f = 3/4$ is negative. Hence r_a is invalid even though it is even and bounded by its zero-lag value.

Impulse plus sine. The impulse is even, but the nonzero sine term is odd. Thus $r_b\left(-\tau\right) \neq r_b\left(\tau\right)$, so it cannot be a real autocorrelation, even under a generalized white-noise interpretation.

Growing exponential. For every nonzero lag, $r_c\left(\tau\right) > 1 = r_c\left(0\right)$; it is invalid.

Triangular candidate.

$$\mathcal{F}\{r_d\}\,(\,f\,) \;=\; \mathrm{sinc}^2\;(\,f\,) \;\geq 0.$$

It is valid. For example, passing unit-strength white noise through a unit-height pulse of duration one produces this correlation and unit output power.

An even bounded function that fails the PSD test

Question

A candidate autocorrelation is zero for $|\tau| > 2$ and joins the points $(\,-2, 0\,)$, $(\,-1, -1\,)$, $(\,0, 1.2\,)$, $(\,1, -1\,)$, and $(\,2, 0\,)$ by straight lines. Can it be the autocorrelation of a real WSS process? Justify the answer.

Solution

Write the piecewise function.

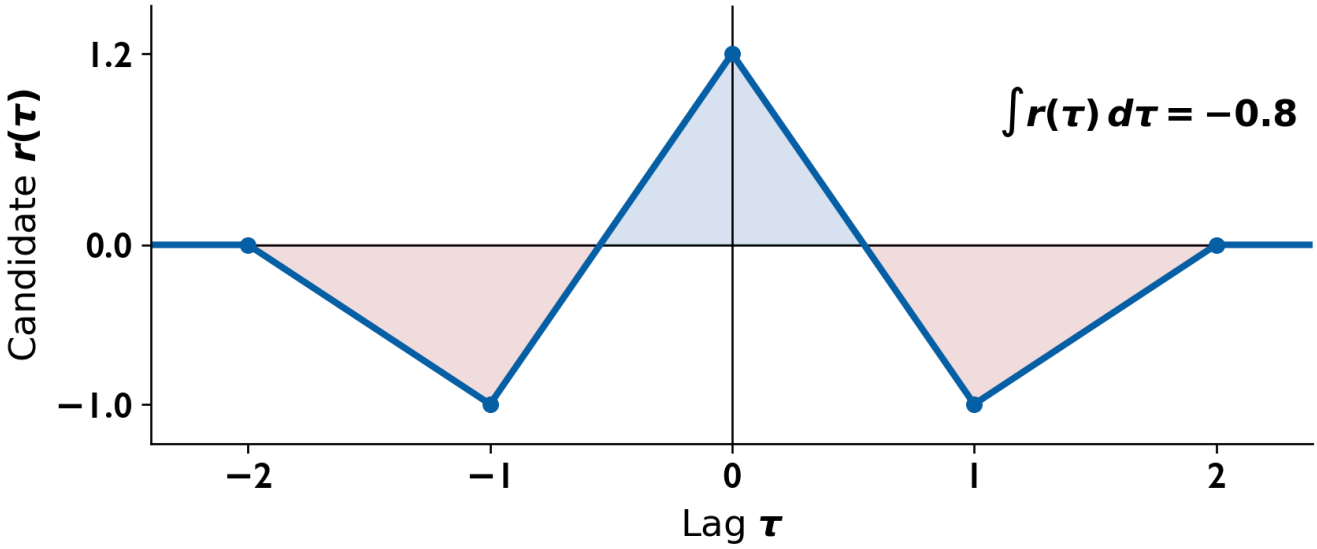
$$r\,(\,\tau\,) \;=\; \begin{cases} 1.2 - 2.2|\tau|, & |\tau| \leq 1, \\ |\tau| - 2, & 1 < |\tau| \leq 2, \\ 0, & |\tau| > 2. \end{cases}$$

It is even and satisfies $|r\,(\,\tau\,)\,| \leq r\,(\,0\,) \;= 1.2$, so those checks alone do not reject it.

Check the spectral value at zero.

$$\begin{aligned} S\,(\,0\,) \quad &= \int_{-\infty}^{\infty} r\,(\,\tau\,)\; d\tau \\ &= 2\left[\,\int_0^1 \,(\,1.2 - 2.2\tau\,)\; d\tau + \int_1^2 \,(\,\tau - 2\,)\; d\tau\right] \\ &= 2\,(\,0.1 - 0.5\,) \;= -0.8. \end{aligned}$$

A PSD cannot be negative. Therefore this candidate is invalid, despite its symmetry and maximum at zero.



The negative area exceeds the positive area, so the Fourier transform at zero is negative and the candidate cannot be an autocorrelation.

Power spectra, linear systems and random binary signals

Finite observation and power spectrum

A time-domain waveform shows how a signal changes during one observation. A power spectral density (PSD) shows how its average power is distributed over frequency, even when the individual waveform is unpredictable. This helps us choose a transmission bandwidth and determine which parts of a signal or noise a receiver filter will pass. Autocorrelation and PSD describe the same second-order behavior in the time-lag and frequency domains, respectively.

Observe a process over $[-T, T]$ and define its finite-window transform:

$$\Xi_T(f) = \int_{-T}^{T} X(t)\, e^{-j2\pi f t}\, dt.$$

Its power spectrum is

$$S_X(f) = \lim_{T \to \infty} \frac{E\{|\Xi_T(f)|^2\}}{2T},$$

when the limit exists. Spectral delta lines are interpreted by their weights.

Correlation and spectrum

For a real WSS process, autocorrelation and the two-sided power spectral density form a Fourier pair:

$$S_X(f) = \int_{-\infty}^{\infty} R_X(\tau)\, e^{-j2\pi f \tau}\, d\tau,$$

$$R_X(\tau) = \int_{-\infty}^{\infty} S_X(f)\, e^{j2\pi f \tau}\, df.$$

Here f is frequency in hertz, τ is time lag, and $j = \sqrt{-1}$. The spectrum distributes mean-square power over frequency:

$$P_X = R_X(0) = \int_{-\infty}^{\infty} S_X(f)\, df, \qquad S_X(f) \geq 0, \qquad S_X(-f) = S_X(f).$$

The area under the PSD over a frequency interval gives the mean-square power in that interval. Thus a filter can reduce noise by rejecting frequencies outside the useful signal band. With amplitude in volts, the spectral units are V^2/Hz; for a normalized load they can be expressed as watts per hertz. When the integral exists, $S_X(0) = \int R_X(\tau)\, d\tau$.

For a deterministic finite-energy signal, the corresponding Fourier transform of $R_x(\tau)$ is the energy spectral density $\psi_x(f) = \left|\xi_x(f)\right|^2$, where $\xi_x(f) = \int x(t)\, e^{-j2\pi ft}\, dt$. Its spectral area is energy. It is not the PSD of a WSS random process.

DC and sinusoidal lines

A constant offset concentrates its power at zero frequency, whereas a sinusoidal component concentrates power at its oscillation frequency and the matching negative frequency in a two-sided spectrum. These concentrated components appear as spectral lines.

Writing $X(t) = \mu_X + \tilde{X}(t)$ with a zero-mean WSS fluctuation gives

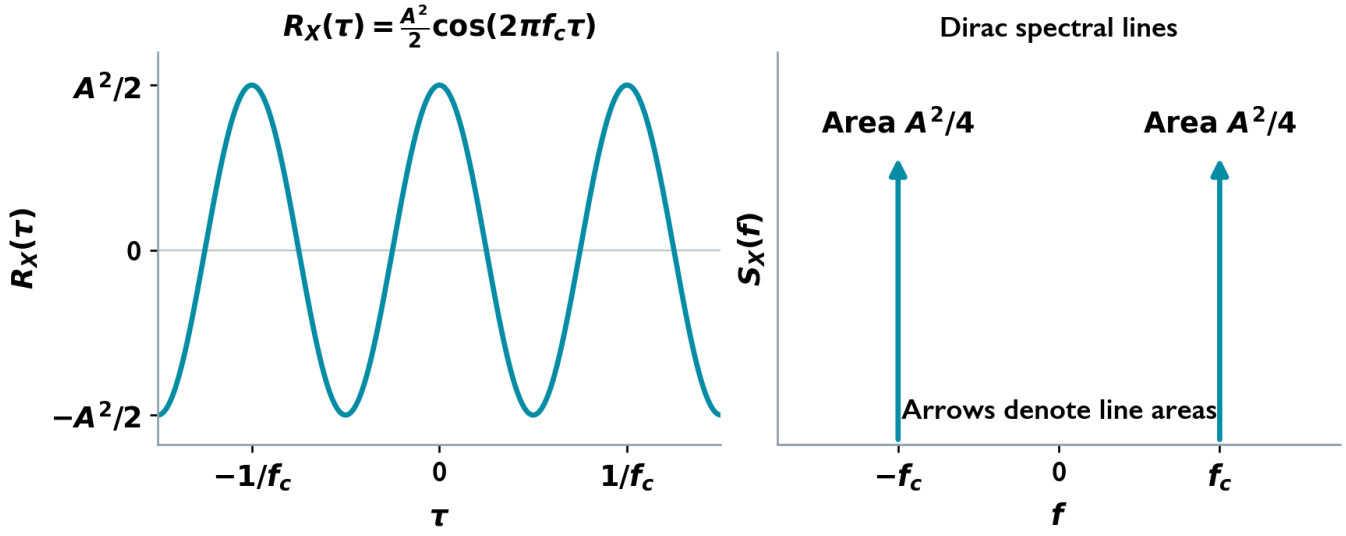
$$R_X(\tau) = \mu_X^2 + C_X(\tau), \qquad S_X(f) = \mu_X^2 \delta(f) + S_{\tilde{X}}(f).$$

The deterministic mean contributes a spectral line at zero. A delta line is characterized by its area or weight, not by an ordinary finite height. A DC line can also contain power from a random constant component; its weight alone does not determine the mean.

For the fixed-amplitude random-phase sinusoid,

$$R_X(\tau) = \frac{A^2}{2}\cos(2\pi f_c \tau), \qquad S_X(f) = \frac{A^2}{4}\left[\delta(f - f_c) + \delta(f + f_c)\right].$$

Each line carries power $A^2/4$, giving total power $A^2/2$.



Cosine correlation and the two spectral lines. Arrow labels indicate delta weights.

Example: Triangular correlation and triangular spectrum

Question

Transform a triangular correlation, and then find the correlation corresponding to a triangular PSD.

Solution

Define the dimensionless functions

$$\Lambda\left(u\right)=\begin{cases}1-|u|, & |u|\leq 1,\\ 0, & |u|>1,\end{cases}\qquad \operatorname{sinc}\left(u\right)=\frac{\sin\left(\pi u\right)}{\pi u},\quad \operatorname{sinc}\left(0\right)=1.$$

For a zero-mean process and time width $T>0$,

$$R_X\left(\tau\right)=\sigma^2\Lambda\left(\tau/T\right)\quad\Leftrightarrow\quad S_X\left(f\right)=\sigma^2T\operatorname{sinc}^2\left(fT\right).$$

For a frequency width $f_0>0$ and peak spectral density s_0,

$$S_X\left(f\right)=s_0\Lambda\left(f/f_0\right)\quad\Leftrightarrow\quad R_X\left(\tau\right)=s_0f_0\operatorname{sinc}^2\left(f_0\tau\right).$$

The zero-lag values agree with the spectral areas: σ^2 for the first pair and s_0f_0 for the second. A triangle in one domain becomes a squared sinc in the other; the scale factor follows the integration variable.

Example: A triangular PSD with a DC line

Question

For $S_X\left(f\right)=s_0\Lambda\left(f/f_0\right)+p_0\delta\left(f\right)$, find correlation, DC and AC powers, and sampling intervals giving uncorrelated samples. Interpret the line as a deterministic mean contribution.

Solution

The inverse transform gives

$$R_X\left(\tau\right)=s_0f_0\operatorname{sinc}^2\left(f_0\tau\right)+p_0.$$

Thus $\mu_X^2=p_0$, $P_{\mathrm{DC}}=p_0$, $P_{\mathrm{AC}}=s_0f_0$, and $P_X=p_0+s_0f_0$. The stated numerical unit-height, unit-line case is obtained with $s_0=1$ and $p_0=1$ in the relevant units.

After subtracting the mean contribution,

$$C_X(\tau) = s_0 f_0 \operatorname{sinc}^2(f_0\tau).$$

Consequently $C_X(kT_s) = 0$ for every nonzero integer k if

$$T_s = \frac{m}{f_0}, \qquad f_s = \frac{f_0}{m}, \qquad m = 1, 2, \ldots.$$

These samples are uncorrelated, although $R_X(kT_s) = p_0$ rather than zero. They need not be independent without joint Gaussianity. If the line were a random constant component instead, this covariance conclusion would require its mean and variance to be specified.

Cross spectra

With $R_{XY}(\tau) = E\{X(t+\tau)\,Y(t)\}$, define

$$S_{XY}(f) = \int_{-\infty}^{\infty} R_{XY}(\tau)\, e^{-j2\pi f\tau}\, d\tau,$$

$$R_{XY}(\tau) = \int_{-\infty}^{\infty} S_{XY}(f)\, e^{j2\pi f\tau}\, df.$$

For real jointly WSS processes,

$$S_{XY}(f) = S_{YX}(-f) = S_{YX}^*(f).$$

A cross spectrum describes how two processes are related at each frequency. Comparing a filter input with its output, for example, reveals a frequency-dependent relationship that can include a phase shift. It may therefore be complex and is not required to be nonnegative. Its integral gives the zero-lag crosscorrelation, $R_{XY}(0) = E\{X(t)\,Y(t)\}$.

A random bipolar binary sequence

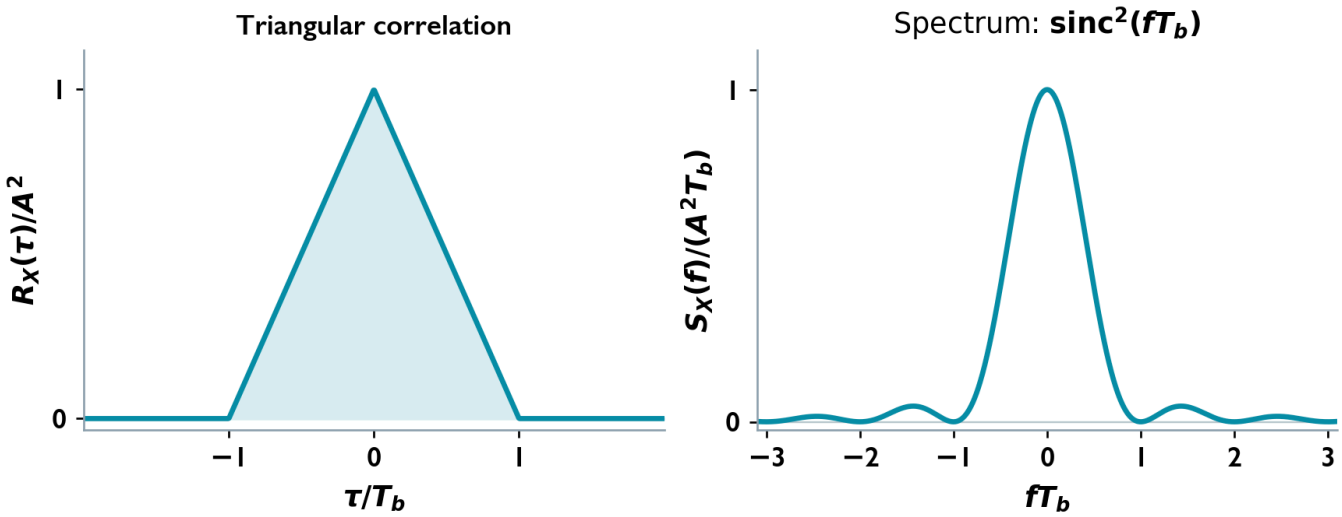
A digital transmitter converts successive data symbols into voltage pulses. Before the data are known, the possible pulse sequences form an ensemble of random waveforms. Holding each symbol for a longer time changes both how long nearby samples remain related and how widely the signal power spreads in frequency.

Let independent equiprobable symbols $A_k \in \{-1, +1\}$ be held for duration T. Randomize the timing origin uniformly over $[0, T)$, independently of the symbols, so no absolute time is privileged. Then $E\{X(t)\} = 0$ and $P_X = 1$.

For $|\tau| \le T$, the probability that two observations belong to the same symbol is $1 - |\tau|/T$. The same-symbol product is one; different symbols have expected product zero. Hence

$$R_X(\tau) = \begin{cases} 1 - |\tau|/T, & |\tau| \le T, \\ 0, & |\tau| > T, \end{cases} \qquad S_X(f) = T \operatorname{sinc}^2(fT).$$

For symbols $\pm A$, both expressions are multiplied by A^2.



Binary correlation and PSD for two symbol durations, using shared axis scales.

The bit rate is $R_b = 1/T$. First spectral zeros occur at $f = \pm 1/T = \pm R_b$, so the null-to-null main-lobe width is $2R_b$. A longer symbol produces broader correlation and a narrower main lobe; a shorter symbol reverses this relation. This connects faster data transmission to a wider signal spectrum and helps explain why a channel bandwidth limits the symbol rates it can carry without substantial pulse distortion. The squared-sinc sidelobes continue outside the main lobe, so this width is not a strict finite spectral support.

Worked Problems

Power spectrum of the sum of two processes

Question

Let $X(t)$ and $Y(t)$ be zero-mean WSS processes, and define $Z(t) = X(t) + Y(t)$. Assume joint WSS for the general spectral calculation.

- Determine $S_Z(f)$ in terms of the auto and cross spectra.
- Simplify the result when $X(t)$ and $Y(t)$ are independent.

Solution

Expand the autocorrelation

Using $R_{XY}(\tau) = E\{X(t+\tau)\,Y(t)\}$,

$$\begin{aligned} R_Z(\tau) &= E\{[X(t+\tau) + Y(t+\tau)]\,[X(t) + Y(t)]\} \\ &= R_X(\tau) + R_Y(\tau) + R_{XY}(\tau) + R_{YX}(\tau). \end{aligned}$$

Take the Fourier transform

Since $S_{YX}(f) = S_{XY}^{*}(f)$,

$$S_Z(f) = S_X(f) + S_Y(f) + 2\,\mathrm{Re}\,S_{XY}(f).$$

Independent processes

Independence and zero means give $R_{XY}(\tau) = 0$ at every lag. Therefore,

$$S_Z(f) = S_X(f) + S_Y(f).$$

The same simplification holds for processes with zero cross-correlation at every lag.

A spectral null produced by delayed-symbol subtraction

Question

An i.i.d. sequence $\{\alpha_n\}$ takes values -1 and $+1$ with equal probability. For real K, define

$$\beta_n = \alpha_n + K\alpha_{n-3}, \qquad X(t) = \sum_{n=-\infty}^{\infty} \beta_n g(t - nT).$$

Determine K if the power spectrum has a null at $f = 1/(3T)$. Assume $G(1/(3T)) \neq 0$, where $G(f)$ is the Fourier transform of $g(t)$.

Solution

Symbol correlation

The original symbols have zero mean and unit variance, so

$$R_\alpha[k] = \delta[k],$$

where $\delta[k]$ is the Kronecker delta. Expanding the filtered-symbol product gives

$$R_\beta[k] = \left(1 + K^2\right)\delta[k] + K\delta[k-3] + K\delta[k+3].$$

Spectral factor

$$\begin{aligned}\Phi_\beta(f) &= \textstyle\sum_{k=-\infty}^{\infty} R_\beta[k]\, e^{-j2\pi fkT} \\ &= 1 + K^2 + 2K\cos(6\pi fT).\end{aligned}$$

The fixed-clock pulse train has the period-averaged PSD

$$\bar{S}_X(f) = \frac{|G(f)|^2}{T}\Phi_\beta(f).$$

The same expression is the WSS PSD when an independent uniform timing offset over $[0, T)$ is included.

Apply the null condition

At $f = 1/(3T)$,

$$\Phi_\beta\left(\frac{1}{3T}\right) = 1 + K^2 + 2K = (1+K)^2.$$

Because the pulse spectrum is nonzero there,

$$(1+K)^2 = 0 \quad \Rightarrow \quad K = -1.$$

If $G(1/(3T)) = 0$, that null alone would impose no restriction on K.

The spectrum of a sinusoid with random amplitude and phase

Question

Consider

$$X(t) = Y\cos(\omega_c t + \Theta),$$

where $Y \sim \mathcal{U}[-3, 3]$ and $\Theta \sim \mathcal{U}[0, 2\pi)$ are independent. Let $f_c = \omega_c/(2\pi) > 0$. Determine the autocorrelation and power spectral density of $X(t)$.

Solution

Find the amplitude's second moment.

$$E\{Y\} = 0, \qquad E\{Y^2\} = \frac{1}{6}\int_{-3}^{3} y^2\, dy = 3.$$

The independent uniform phase gives $E\{X(t)\} = 0$.

Average the product over the phase. Using $\cos a \cos b = [\cos(a-b) + \cos(a+b)]/2$,

$$\begin{aligned} R_X(\tau) &= E\{X(t+\tau)\,X(t)\} \\ &= E\{Y^2\}\,E\{\cos(\omega_c(t+\tau)+\Theta)\cos(\omega_c t+\Theta)\} \\ &= \tfrac{E\{Y^2\}}{2}\cos(\omega_c\tau) + \tfrac{E\{Y^2\}}{2}E\{\cos(2\omega_c t+\omega_c\tau+2\Theta)\}. \end{aligned}$$

The last expectation is zero, so

$$R_X(\tau) = \frac{3}{2}\cos(\omega_c\tau) = \frac{3}{2}\cos(2\pi f_c\tau).$$

The constant mean and dependence on lag alone establish wide-sense stationarity.

Transform the correlation into its spectrum. With the Fourier convention in hertz,

$$\cos\ (\ 2\pi f_c \tau\ )\ \leftrightarrow\ \frac{1}{2}[\delta\ (\ f-f_c\ )\ +\delta\ (\ f+f_c\ )\ ].$$

Consequently,

$$S_X\ (\ f\ )\ =\frac{3}{4}[\delta\ (\ f-f_c\ )\ +\delta\ (\ f+f_c\ )\ ].$$

The area of each spectral line is $3/4$, giving total average power

$$\int_{-\infty}^{\infty}\ S_X\ (\ f\ )\ df=\frac{3}{2}=R_X\ (\ 0\ )\ .$$

Random amplitude and phase change the sinusoid between realizations, while its frequency stays fixed; its power is therefore concentrated at the two frequencies $\pm f_c$.

Correlation and line spectra of quadrature sinusoids

Question

Consider the deterministic periodic signals

$$m_1\ (\ t\ )\ =10\ \cos\ (\ 200\pi t\ )\ ,\qquad m_2\ (\ t\ )\ =20\ \cos\ \left(200\pi t+\frac{\pi}{2}\right),\qquad m_3\ (\ t\ )\ =m_1\ (\ t\ )\ +m_2\ (\ t\ )\ .$$

Using time-averaged correlations, find $R_2\ (\ \tau\ )$ and $R_3\ (\ \tau\ )$, the crosscorrelations $R_{21}\ (\ \tau\ )$ and $R_{12}\ (\ \tau\ )$, the power spectra of m_1 and m_3, and the average normalized powers of all three signals. Describe the spectral sketches.

Solution

Set the lag convention. For these periodic power signals, use

$$R_{ab}\ (\ \tau\ )\ =\ \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\ m_a\ (\ t+\tau\ )\ m_b\ (\ t\ )\ dt.$$

The carrier frequency is $f_0=100$ Hz. The time average of two equal-frequency cosines is

$$\langle\ A\ \cos\ (\ \omega_0 t+\alpha\ )\ B\ \cos\ (\ \omega_0 t+\beta\ )\ \rangle\ =\frac{AB}{2}\cos\ (\ \alpha-\beta\ )\ .$$

Calculate the autocorrelations.

$$R_1\ (\ \tau\ )\ =50\ \cos\ (\ 200\pi\tau\ )\ ,\qquad R_2\ (\ \tau\ )\ =200\ \cos\ (\ 200\pi\tau\ )\ .$$

For the cross terms,

$$R_{21}\ (\ \tau\ )\ =100\ \cos\ \left(200\pi\tau+\frac{\pi}{2}\right)\ =-100\ \sin\ (\ 200\pi\tau\ )\ ,$$

$$R_{12}\ (\ \tau\ )\ =100\ \cos\ \left(200\pi\tau-\frac{\pi}{2}\right)\ =100\ \sin\ (\ 200\pi\tau\ )\ .$$

They obey $R_{21}\ (\ \tau\ )\ =R_{12}\ (\ -\tau\ )$. Consequently,

$$\begin{aligned} R_3(\tau) &= R_1(\tau) + R_2(\tau) + R_{12}(\tau) + R_{21}(\tau) \\ &= 250\cos(200\pi\tau). \end{aligned}$$

Transform to power spectra. With frequency expressed in hertz,

$$\cos(2\pi f_0\tau) \leftrightarrow \frac{1}{2}[\delta(f-f_0) + \delta(f+f_0)].$$

Hence,

$$S_1(f) = 25[\delta(f-100) + \delta(f+100)],$$

$$S_3(f) = 125[\delta(f-100) + \delta(f+100)].$$

Each sketch has lines at $f = \pm 100$ Hz. The weights are 25 per line for m_1 and 125 per line for m_3; these are impulse areas, not finite density heights.

Read the normalized powers.

$$P_1 = R_1(0) = 50, \qquad P_2 = R_2(0) = 200, \qquad P_3 = R_3(0) = 250.$$

The two sinusoids are orthogonal over a period, so their powers add. A zero-lag crosscorrelation does not mean that crosscorrelation vanishes at every lag.

Testing proposed power spectral densities

Question

Determine which expressions can be the PSD of a real WSS process with finite mean-square power:

$$\begin{aligned} S_a(f) &= \delta(f) + \cos^2(2\pi f), \\ S_b(f) &= 10 + \delta(f-10), \\ S_c(f) &= e^{-2\pi|f-10|}, \\ S_d(f) &= e^{-2\pi(f^2-10)}. \end{aligned}$$

Explain any ideal-white-noise qualification.

Solution

A PSD for a real finite-power process must be a nonnegative even spectral measure with finite total area.

Constant-plus-periodic spectrum. S_a is nonnegative and even, but

$$\int_{-\infty}^{\infty} \cos^2(2\pi f)\, df = \infty.$$

It cannot describe an ordinary finite-power process. It can be interpreted as a generalized infinite-power noise spectrum plus a DC line, just as ideal white noise requires a generalized interpretation.

Unpaired spectral line. S_b is not even: its line at 10 has no matching line at -10. Its constant part also has infinite area. It is invalid for the stated real finite-power process.

Shifted exponential. S_c has finite area and is nonnegative, but is not even about $f = 0$. It is invalid for a real process; such an asymmetric spectrum can belong to a complex process.

Gaussian-shaped spectrum. $S_d = e^{20\pi} e^{-2\pi f^2}$ is nonnegative, even, and integrable. Thus it is valid, with

$$R_X(\tau) = \frac{e^{20\pi}}{\sqrt{2}} e^{-\pi\tau^2/2}, \qquad P_X = R_X(0) = \frac{e^{20\pi}}{\sqrt{2}}.$$

Power contained in selected frequency intervals

Question

A real signal has two-sided PSD

$$S_X(f) = \begin{cases} 10^{-6} f^2, & |f| \le 10^4 \text{ Hz}, \\ 0, & \text{otherwise}. \end{cases}$$

Find the spectral area over 0 to $10\,\text{kHz}$ and over 5 to $6\,\text{kHz}$. Also state the power in the corresponding real low-pass and band-pass frequency regions.

Solution

Integrate using frequencies in hertz.

$$\begin{aligned} \int_0^{10^4} S_X(f)\, df &= 10^{-6} \left[\tfrac{f^3}{3} \right]_0^{10^4} = \tfrac{10^6}{3}, \\ \int_{5000}^{6000} S_X(f)\, df &= \tfrac{10^{-6}}{3} \left(6000^3 - 5000^3 \right) = \tfrac{91000}{3}. \end{aligned}$$

These are the areas in the stated positive-frequency intervals. A real filter passes the matching negative-frequency intervals as well:

$$P_{|f| \le 10\,\text{kHz}} = \frac{2 \times 10^6}{3}, \qquad P_{5\,\text{kHz} \le |f| \le 6\,\text{kHz}} = \frac{182000}{3}.$$

The second passband contains 0.091, or 9.1%, of the total power. Distinguishing a positive-frequency PSD area from a physical two-sided passband avoids an incorrect factor of two.

Power spectrum of a random bipolar line waveform

Question

A bipolar nonreturn-to-zero waveform takes levels $+12$ V and -12 V, with bit rate 2400 bit/s. Model successive bits as independent and equally likely, and choose a timing origin uniformly over one bit interval so that the waveform is stationary. Find its PSD and verify its total mean-square power.

Solution

Bit statistics. Let $T = 1/2400$ s and let $A_n = \pm 12$ V with equal probabilities. Then

$$E\{A_n\} = 0, \qquad E\{A_n^2\} = 144, \qquad E\{A_n A_m\} = 0 \quad (\, n \neq m \,) \,.$$

Correlation. For $|\tau| < T$, two observations lie in the same bit interval with probability $1 - |\tau|/T$. Distinct independent bits contribute zero correlation:

$$R_X(\,\tau\,) = \begin{cases} 144\,(\,1 - |\tau|/T\,), & |\tau| \le T, \\ 0, & \text{otherwise.} \end{cases}$$

Spectrum. Transforming the triangular correlation gives

$$S_X(\,f\,) = 144T\,\mathrm{sinc}^2\,(\,fT\,) = 0.06\,\mathrm{sinc}^2\left(\frac{f}{2400}\right)\ \mathrm{V}^2/\mathrm{Hz}.$$

The first spectral zeros occur at $f = \pm 2400$ Hz. Its area is

$$\int_{-\infty}^{\infty} S_X(\,f\,)\,df = 144\ \mathrm{V}^2 = R_X(\,0\,) \,.$$

With zero mean, there is no DC spectral impulse.

Random Processes Through Linear Filters

A receiver filter acts on both the useful signal and the noise accompanying it. Although the input waveform is random, the same filter acts on every possible realization, so its output statistics can be predicted. A low-pass filter suppresses high-frequency fluctuations; a bandpass filter selects frequencies around a carrier. Reducing the passband can lower noise power, but a passband that is too narrow also removes useful signal content and distorts the message.

Convolution, mean, and correlation

Let a real WSS process $X(t)$ pass through a real LTI filter with impulse response $h(t)$ and frequency response $H(f)$. For an output with finite second moment,

$$Y(t) = h(t) * X(t) = \int_{-\infty}^{\infty} h(u)\, X(t-u)\, du.$$

Taking the expectation inside the integral gives

$$\mu_Y = \int_{-\infty}^{\infty} h(u)\, \mu_X\, du = \mu_X H(0).$$

The output mean depends on the filter's DC gain. A zero-mean input therefore gives a zero-mean output.

Expanding the expected product of two output values gives

$$\begin{aligned} R_Y(\tau) &= E\{Y(t+\tau)\, Y(t)\} \\ &= \iint h(u)\, h(v)\, R_X(\tau - u + v)\, du\, dv \\ &= R_X(\tau) * h(\tau) * h(-\tau). \end{aligned}$$

The mean is constant and the correlation depends only on lag, so the output is WSS under the stated existence condition. For Gaussian input, the output is also Gaussian.

Power spectra and input–output correlation

The Fourier transform converts the correlation convolution into

$$S_Y(f) = |H(f)|^2 S_X(f).$$

The squared magnitude is the power gain. At each frequency, the filter multiplies the input power density by this gain; integrating the result gives the total power reaching the next receiver stage. The output mean square and variance are

$$P_Y = R_Y(0) = \int_{-\infty}^{\infty} |H(f)|^2 S_X(f)\, df, \qquad \sigma_Y^2 = P_Y - \mu_Y^2.$$

Use the cross-correlation convention

$$R_{XY}(\tau) = E\{X(t+\tau)\,Y(t)\}.$$

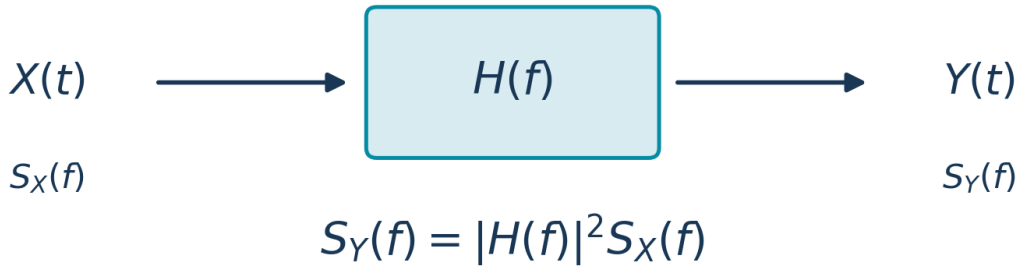
Then

$$\begin{aligned} R_{XY}(\tau) &= \int h(u)\,R_X(\tau+u)\,du = R_X(\tau) * h(-\tau), \\ R_{YX}(\tau) &= R_{XY}(-\tau) = R_X(\tau) * h(\tau). \end{aligned}$$

Their cross spectra are

$$S_{XY}(f) = S_X(f)\,H^*(f), \qquad S_{YX}(f) = S_X(f)\,H(f).$$

The output PSD retains the magnitude of the filter response; the cross spectrum also retains its phase. Output power alone therefore cannot describe how the filter shifts a waveform in time or phase.



Input and output power spectra of an LTI filter.

Example: Cascaded filters

> ### Question
>
> **A WSS process passes through two real filters: $X \to H_1 \to V \to H_2 \to Y$. Find the intermediate and final spectra and correlations.**

Solution

The equivalent response is

$$H_{\text{eq}}(f) = H_1(f)\,H_2(f), \qquad h_{\text{eq}} = h_1 * h_2.$$

Consequently,

$$S_V(f) = |H_1(f)|^2 S_X(f),$$

$$S_Y(f) = |H_2(f)|^2 S_V(f) = |H_1(f)\,H_2(f)|^2 S_X(f).$$

In the time domain,

$$R_V = R_X * h_1(\tau) * h_1(-\tau),$$

$$R_Y = R_X * h_{\text{eq}}(\tau) * h_{\text{eq}}(-\tau).$$

The intermediate process remains the output of the first filter; it is not a new independent input.

Example: A random-phase sinusoid through an RC filter

Question

Let $X(t) = A\cos(2\pi f_0 t + \Theta)$, where Θ is uniform on $[0, 2\pi)$, and let

$$H(f) = \frac{1}{1 + j2\pi RCf}.$$

Find the output autocorrelation and power.

Solution

The input has zero mean and

$$S_X(f) = \frac{A^2}{4}[\delta(f - f_0) + \delta(f + f_0)].$$

Evaluate the power gain at the two spectral lines:

$$S_Y(f) = \frac{A^2}{4\left[1 + (2\pi RCf_0)^2\right]}[\delta(f - f_0) + \delta(f + f_0)].$$

Inverse transformation gives

$$R_Y(\tau) = \frac{A^2}{2\left[1 + (2\pi RCf_0)^2\right]}\cos(2\pi f_0 \tau).$$

Thus

$$P_Y = R_Y(0) = \frac{A^2}{2\left[1 + (2\pi RCf_0)^2\right]}.$$

The filter changes amplitude and phase, while the autocorrelation reflects the amplitude change.

Worked Problems

Independent Gaussian noise branches and an estimation-error network

Question

Independent Gaussian white-noise processes $W_1(t)$ and $W_2(t)$ satisfy

$$R_{W_i}(\tau) = \frac{\eta}{2}\delta(\tau).$$

Identical real LTI filters $h(t)$ produce $N_1(t)$ and $N_2(t)$. Define

$$N = N_1 - N_2, \qquad \varepsilon = N_1 - h_o * N.$$

Find:

- The density of N and the joint density of N_1, N_2 at a fixed time.
- $P(N > -1)$.
- A simplified expression for ε.

Solution

Branch statistics Assume $\eta > 0$ and $0 < E_h < \infty$.

$$E_h = \int_{-\infty}^{\infty} |h(u)|^2 du, \qquad \sigma_b^2 = \frac{\eta E_h}{2}, \qquad S_{N_i}(f) = \frac{\eta}{2}|H(f)|^2.$$

The branch outputs are independent, zero-mean Gaussian processes.

Densities and probability

$$N \sim \mathcal{N}(0, \eta E_h).$$

$$f_N(n) = \frac{e^{-n^2/(2\eta E_h)}}{\sqrt{2\pi\eta E_h}}$$

$$f_{N_1N_2}(n_1, n_2) = \frac{1}{2\pi\sigma_b^2}\exp\left[-\frac{n_1^2 + n_2^2}{2\sigma_b^2}\right]$$

$$P(N > -1) = 1 - Q\left(\frac{1}{\sqrt{\eta E_h}}\right)$$

A numerical probability requires ηE_h.

Error network

$$\varepsilon = (\delta - h_o) * N_1 + h_o * N_2$$

$$S_\varepsilon(f) = \frac{\eta}{2}|H(f)|^2\left(|1-H_o(f)|^2 + |H_o(f)|^2\right).$$

Without a specified H_o, this is the general result. For minimum mean-square estimation,

$$H_o = \frac{S_{N_1 N}}{S_N} = \frac{1}{2}, \qquad h_o = \tfrac{1}{2}\delta,$$

$$\varepsilon = \frac{N_1 + N_2}{2}, \qquad \operatorname{Var}(\varepsilon) = \frac{\eta E_h}{4}$$

If $\eta E_h = 0$, the outputs vanish in mean square, the joint law is a point mass at the origin, and $P(N > -1) = 1$.

Output autocorrelation of a rectangular LTI filter

Question

A WSS input has

$$R_X(\tau) = \begin{cases} 1-|\tau|, & |\tau| \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

For $h(t) = u(t) - u(t-1)$, use convolution to find $R_Y(\tau)$.

Solution

Correlation convolution

$$\begin{aligned} R_Y &= R_X * h * \tilde{h}, \qquad \tilde{h}(t) = h(-t), \\ h * \tilde{h} &= \Lambda, \qquad \Lambda(\tau) = \max(1-|\tau|, 0), \\ R_Y &= \Lambda * \Lambda. \end{aligned}$$

Overlap integral

$$R_Y(\tau) = \int_{-1}^{1} (1-|v|)\,\max(1-|\tau - v|, 0)\,dv.$$

Set $a = |\tau|$ and split at the changes of sign inside the overlap:

$$R_Y(\tau) = \begin{cases} \frac{2}{3} - a^2 + \frac{1}{2}a^3, & 0 \le a \le 1, \\ \frac{1}{6}(2-a)^3, & 1 < a \le 2, \\ 0, & a > 2, \end{cases} \qquad a = |\tau|.$$

$$P_Y = R_Y(0) = \frac{2}{3}$$

At $|\tau| = 1$, both branches equal $1/6$.

Autocorrelation and spectrum of a shifted difference

Question

A real WSS process $X(t)$ is transformed according to

$$Y(t) = X(t+a) - X(t-a).$$

Determine:

- **The autocorrelation $R_Y(\tau)$ in terms of $R_X(\tau)$.**
- **The PSD $S_Y(f)$ in terms of $S_X(f)$.**

Solution

Expand the autocorrelation

The output mean is zero. With $R_X(\tau) = E\{X(t+\tau)X(t)\}$,

$$\begin{aligned} R_Y(\tau) &= E\{[X(t+\tau+a) - X(t+\tau-a)][X(t+a) - X(t-a)]\} \\ &= 2R_X(\tau) - R_X(\tau-2a) - R_X(\tau+2a). \end{aligned}$$

Thus

$$R_Y(\tau) = 2R_X(\tau) - R_X(\tau-2a) - R_X(\tau+2a).$$

Find the transfer function

$$H(f) = e^{j2\pi fa} - e^{-j2\pi fa} = 2j\sin(2\pi fa).$$

Consequently,

$$S_Y(f) = 4\sin^2(2\pi fa)\, S_X(f).$$

Zero-lag check

Evenness of R_X gives

$$P_Y = R_Y(0) = 2[R_X(0) - R_X(2a)].$$

A triangular Gaussian spectrum and low-pass filtering

Question

A zero-mean WSS Gaussian process has

$$S_X(f) = 4\times 10^{-5}\Lambda\left(\frac{f}{10^5}\right), \qquad \Lambda(u) = \begin{cases} 1-|u|, & |u|\le 1,\\ 0, & |u|>1. \end{cases}$$

Find its power and one-sided support bandwidth. An ideal unity-gain low-pass filter passes $|f|\le 50\,\text{kHz}$; find the output PSD and power. Determine the density of $X(0)$ and the smallest positive t_0 for which $X(0)$ and $X(t_0)$ are independent.

Solution

Integrate the triangular spectrum. Its height is 4×10^{-5} and its total base width is 2×10^5 Hz:

$$P_X = \frac{1}{2}\left(2\times 10^5\right)\left(4\times 10^{-5}\right) = 4, \qquad B_X = 100\,\text{kHz}.$$

Here B_X is the positive-frequency support edge; the two-sided support spans $200\,\text{kHz}$.

Retain the central part through the filter.

$$S_Y(f) = \begin{cases} 4\times 10^{-5}\left(1-\dfrac{|f|}{10^5}\right), & |f|\le 5\times 10^4,\\ 0, & \text{otherwise}. \end{cases}$$

Its area is

$$P_Y = 2\left(4\times 10^{-5}\right)\int_0^{5\times 10^4}\left(1-\frac{f}{10^5}\right)df = 3.$$

Use Gaussianity for the amplitude law. Since the mean is zero and the variance is four,

$$X(0)\sim\mathcal{N}(0,4), \qquad f_{X(0)}(x) = \frac{1}{\sqrt{8\pi}}e^{-x^2/8}.$$

Find the first independent sample. Fourier inversion of the triangle gives

$$R_X(\tau) = 4\,\text{sinc}^2\left(10^5\tau\right).$$

Samples from a Gaussian process are jointly Gaussian, so zero covariance implies independence. The first positive correlation zero occurs at

$$t_0 = 10^{-5}\,\text{s} = 10\,\mu\text{s}.$$

Gaussian samples at zero and nonzero correlation lags

Question

A zero-mean WSS Gaussian process has $R_X(\tau) = 4\,\mathrm{sinc}^2\left(10^4\tau\right)$, where $\mathrm{sinc}(u) = \sin(\pi u)/(\pi u)$. Determine its power, PSD, and one-sided support bandwidth. For an ideal unity-gain low-pass filter passing $|f| \le 5\,\mathrm{kHz}$, find the output PSD and power. Find the marginal densities of $X(0)$, $X\left(10^{-4}\right)$, and $X\left(1.5\times10^{-4}\right)$. Compare the independence of each later sample with $X(0)$.

Solution

Obtain the PSD and total power.

$$P_X = R_X(0) = 4, \qquad S_X(f) = 4\times10^{-4}\Lambda\left(\frac{f}{10^4}\right), \qquad B_X = 10\,\mathrm{kHz},$$

where $\Lambda(u) = \max(1-|u|,0)$. The bandwidth is the positive-frequency support edge.

Filter the spectrum.

$$S_Y(f) = \begin{cases} 4\times10^{-4}\left(1-\dfrac{|f|}{10^4}\right), & |f|\le 5000, \\ 0, & \text{otherwise.} \end{cases}$$

Thus

$$P_Y = 8\times10^{-4}\int_0^{5000}\left(1-\frac{f}{10^4}\right)df = 3.$$

Use the same marginal law at every time. Stationarity and Gaussianity give

$$f_{X(t)}(x) = \frac{1}{\sqrt{8\pi}}e^{-x^2/8}$$

for all three requested times. Identical marginals do not imply independence between different times.

Compare the two covariance values.

$$\begin{aligned} \mathrm{Cov}\left(X(0), X\left(10^{-4}\right)\right) &= 4\,\mathrm{sinc}^2(1) = 0, \\ \mathrm{Cov}\left(X(0), X\left(1.5\times10^{-4}\right)\right) &= 4\,\mathrm{sinc}^2(1.5) = \tfrac{16}{9\pi^2} \neq 0. \end{aligned}$$

The first pair is independent because it is jointly Gaussian with zero covariance. The second pair is dependent; its correlation coefficient is $4/(9\pi^2) \approx 0.04503$.

A Gaussian amplitude after a wide channel filter

Question

A zero-mean Gaussian process has

$$S_X(f) = \frac{10^{-2}}{2B}\,\mathrm{rect}\left(\frac{f}{2B}\right), \qquad B = 10^6\ \mathrm{Hz}.$$

It passes through

$$H(f) = 0.3\,\mathrm{rect}\left(\frac{f}{2B_{\mathrm{ch}}}\right), \qquad B_{\mathrm{ch}} = 10^9\ \mathrm{Hz}.$$

Determine $P(-0.6 \le Y(t) \le 0.6)$ at the filter output.

Solution

Output variance. The filter passes the entire input band because $B_{\mathrm{ch}} > B$. Hence

$$\sigma_Y^2 = \int_{-\infty}^{\infty} |H(f)|^2 S_X(f)\,df = 0.3^2\left(10^{-2}\right) = 9\times 10^{-4}.$$

Linear filtering preserves Gaussianity, so

$$Y(t) \sim \mathcal{N}\left(0, 9\times 10^{-4}\right), \qquad \sigma_Y = 0.03.$$

Gaussian interval. Both limits are 20 standard deviations from the mean:

$$P(-0.6 \le Y(t) \le 0.6) = 1 - 2Q\left(\frac{0.6}{0.03}\right) = 1 - 2Q(20).$$

Since $Q(20) \approx 2.75362\times 10^{-89}$, this probability is extremely close to one.

A Gaussian process with a DC mean through a derivative-delay network

Question

A WSS Gaussian input has mean $\mu_X = 2$ and a rectangular continuous spectrum $S_{X,\mathrm{cont}}(f) = 10^{-3}$ for $|f| \le 200\,\mathrm{Hz}$, zero elsewhere. A direct branch is added to one-half of the derivative branch; the sum is delayed by one second, producing $Y(t)$. Find the output mean, complete PSD, total power, stationarity, Gaussianity, and the density of $Y(1)$.

Solution

Include the mean in the complete PSD. A nonzero constant mean requires a DC spectral line. Thus the physically consistent complete input spectrum is

$$S_X(f) = 4\delta(f) + S_{X,\text{cont}}(f).$$

If the reported rectangle were interpreted as the entire PSD, its total power 0.4 would be smaller than $\mu_X^2 = 4$, which is impossible. The rectangle therefore specifies the continuous, centered-process spectrum.

Find the network gain and mean.

$$H(f) = e^{-j2\pi f}(1 + j\pi f), \qquad |H(f)|^2 = 1 + \pi^2 f^2.$$

The derivative contributes no constant mean and the delay leaves a constant unchanged:

$$\mu_Y = H(0)\,\mu_X = 2.$$

Find the output spectrum and power.

$$S_Y(f) = 4\delta(f) + \begin{cases} 10^{-3}\left(1 + \pi^2 f^2\right), & |f| \le 200, \\ 0, & |f| > 200. \end{cases}$$

The continuous area gives the variance:

$$\begin{aligned} \sigma_Y^2 &= 10^{-3}\int_{-200}^{200}\left(1 + \pi^2 f^2\right) df \\ &= 0.4 + \tfrac{16000\pi^2}{3} \approx 52638.2901. \end{aligned}$$

Therefore $P_Y = 4 + \sigma_Y^2 \approx 52642.2901$.

Identify the output law. The bandlimited input has a mean-square derivative. An LTI transformation preserves WSS and joint Gaussianity, so the output is WSS Gaussian. At $t = 1$,

$$Y(1) \sim \mathcal{N}(2, \sigma_Y^2), \qquad f_{Y(1)}(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left[-\frac{(y-2)^2}{2\sigma_Y^2}\right].$$

A delayed-sum differentiator and its spectral nulls

Question

A real stationary process with PSD $S_X(f)$ enters two branches: a direct branch and a branch delayed by $T > 0$. Their sum is differentiated to form $Y(t)$. Determine whether the output is stationary, derive its PSD, and identify the frequencies that cannot occur at the output. Assume the output has finite mean-square power.

Solution

Combine the branches.

$$Y(t) = \frac{d}{dt}[X(t) + X(t-T)], \qquad H(f) = j2\pi f(1 + e^{-j2\pi fT}).$$

The system is LTI. If the input is strictly stationary, a well-defined time-invariant output is also strictly stationary; for a WSS input, the finite-power output is WSS. Its mean is zero because differentiation removes a constant input mean.

Apply the squared gain. Using $\left|1+e^{-j2\pi fT}\right|^2 = 4\cos^2(\pi fT)$,

$$S_Y(f) = 16\pi^2 f^2 \cos^2(\pi fT)\, S_X(f).$$

The finite-power condition is

$$\int_{-\infty}^{\infty} 16\pi^2 f^2 \cos^2(\pi fT)\, S_X(f)\, df < \infty.$$

An ideal differentiator does not produce finite power for every possible WSS input.

Locate the rejected frequencies.

$$f = 0, \qquad f = \frac{2k+1}{2T}, \quad k \in \mathbb{Z}.$$

DC is removed by differentiation. At the other nulls, the delayed and direct components differ by an odd multiple of π and cancel before the differentiator.

Signal and noise powers through a channel and equalizer

Question

A zero-mean input signal and independent additive noise have spectra

$$S_M(f) = \begin{cases} 2(1-|f|/2), & |f| \le 2, \\ 0, & \text{otherwise}, \end{cases} \qquad S_N(f) = \begin{cases} 1/4, & |f| \le 3, \\ 0, & \text{otherwise}. \end{cases}$$

The channel magnitude is 2 on $|f| \le 1$ and 1 on $1 < |f| \le 2$, and zero elsewhere. Noise is added after the channel. The subsequent equalizer has magnitude 1 on $|f| \le 1$ and 2 on $1 < |f| \le 2$, and zero elsewhere. Its phase compensates the channel phase. Find the signal powers at the channel input and output, and the signal-to-noise ratios at the equalizer input and output.

Solution

Input signal power.

$$P_{M,\text{in}} = 2\int_0^2 (2-f)\, df = 4.$$

Channel output. Square each channel gain before integrating:

$$\begin{aligned} P_{M,\text{ch}} &= 2\left[4\int_0^1 (2-f)\, df + \int_1^2 (2-f)\, df\right] \\ &= 13. \end{aligned}$$

The added noise power is

$$P_{N,\,\text{in}} = \int_{-3}^{3} \frac{1}{4}\,df = \frac{3}{2}.$$

Consequently

$$\text{SNR}_{\text{in}} = \frac{13}{3/2} = \frac{26}{3} \approx 8.66667 \quad (\,9.37852\,\text{dB}\,).$$

Equalizer output. The channel–equalizer cascade has constant magnitude 2 across the signal support. It restores the spectral shape with an amplitude gain of 2:

$$P_{M,\,\text{out}} = 2^2\,(\,4\,) = 16.$$

The equalizer acts on the added noise alone with its own squared magnitude:

$$P_{N,\,\text{out}} = 2\left[\int_0^1 \frac{1}{4}\,df + \int_1^2 \frac{1}{4}\left(2^2\right)\,df\right] = \frac{5}{2}.$$

Therefore

$$\text{SNR}_{\text{out}} = \frac{16}{5/2} = 6.4 \quad (\,8.06180\,\text{dB}\,).$$

Restoring the signal shape also amplifies noise in the weaker channel band.

Noise enhancement in an inverse channel filter

Question

A signal of power 10^{-3} W is confined to $|f| \le 4000$ Hz. It passes through a channel with

$$H_{\text{ch}}\,(\,f\,) = \frac{4000}{j4000 + f}.$$

Independent white noise is added after the channel, with two-sided PSD $S_N\,(\,f\,) = q = 10^{-8}$ W/Hz. An equalizer has response $H_{\text{eq}}\,(\,f\,) = 1/H_{\text{ch}}\,(\,f\,)$ for $|f| \le 4000$ Hz and zero elsewhere. Find the output SNR.

Solution

Signal restoration. The cascade equals one throughout the signal band, so

$$P_{S,\,\text{out}} = 10^{-3}\ \text{W}.$$

Noise enhancement. The equalizer's squared magnitude is

$$\left|H_{\text{eq}}\,(\,f\,)\right|^2 = 1 + \left(\frac{f}{4000}\right)^2, \qquad |f| \le 4000.$$

Consequently

$$\begin{aligned} P_{N,\,\text{out}} &= 2q \int_0^{4000} \left[1 + \left(\tfrac{f}{4000} \right)^2 \right] df \\ &= 2q \left(4000 + \tfrac{4000}{3} \right) = \tfrac{8}{3} q \left(4000 \right) \\ &= 1.06667 \times 10^{-4} \text{ W}. \end{aligned}$$

Thus

$$\text{SNR}_{\text{out}} = 9.375, \qquad \text{SNR}_{\text{out, dB}} \approx 9.71971 \text{ dB}.$$

The inverse response boosts frequencies attenuated by the channel, and boosts noise at those frequencies as well.

White Noise, Filtered Noise, and Equivalent Bandwidth

Noise in a receiver comes partly from thermal agitation in its electronic components. Its exact waveform cannot be predicted, but its average power within the receiver's bandwidth can be estimated. A flat spectrum is a useful approximation when the noise density changes little over the operating band. This makes it possible to calculate how much noise a filter admits without knowing a particular noise waveform.

The white-noise model

Zero-mean ideal white noise has a constant two-sided power spectral density:

$$S_W(f) = \frac{N_0}{2}, \qquad -\infty < f < \infty.$$

Its autocorrelation is the inverse Fourier transform:

$$R_W(\tau) = \frac{N_0}{2}\delta(\tau).$$

The model has no correlation at nonzero lag. Its total ideal power is infinite, so finite noise power is evaluated after bandwidth limitation rather than by assigning a finite variance to an ideal unfiltered point value. Whiteness specifies the spectrum: equal frequency intervals contain equal average noise power. Gaussianity describes the amplitude probability distribution and is useful when many small noise contributions combine. These are separate properties: Gaussian noise can be colored, and white noise need not be Gaussian.

When the density represents available thermal-noise power,

$$N_0 = kT, \qquad S_W(f) = \frac{kT}{2}\,\mathrm{W/Hz},$$

where $k \approx 1.38 \times 10^{-23}\ \mathrm{J/K}$ and T is absolute temperature in kelvin. The negative and positive frequencies each contribute half the available power. Real receiver bandwidths are finite, so the infinite-band idealization does not imply infinite measured noise power. An ideal low-pass bandwidth B therefore admits kTB. This available-power convention does not by itself specify a resistor's voltage-noise density.

Filtering the spectrum

For white noise passing through a filter,

$$S_Y(f) = \frac{N_0}{2}|H(f)|^2,$$

$$P_Y = \frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 \, df.$$

The integral must be finite for finite output noise power. The filter shapes the flat input spectrum into colored noise: some frequency ranges now carry more noise power than others. Gaussian white input gives Gaussian output.

Example: Ideal low-pass filtering

Question

White Gaussian noise with PSD $N_0/2$ passes through a unity-gain ideal filter with passband $[-B, B]$, where $B > 0$. Find its output PSD, autocorrelation, and power.

Solution

The output PSD is

$$S_Y(f) = \begin{cases} N_0/2, & |f| \le B, \\ 0, & |f| > B. \end{cases}$$

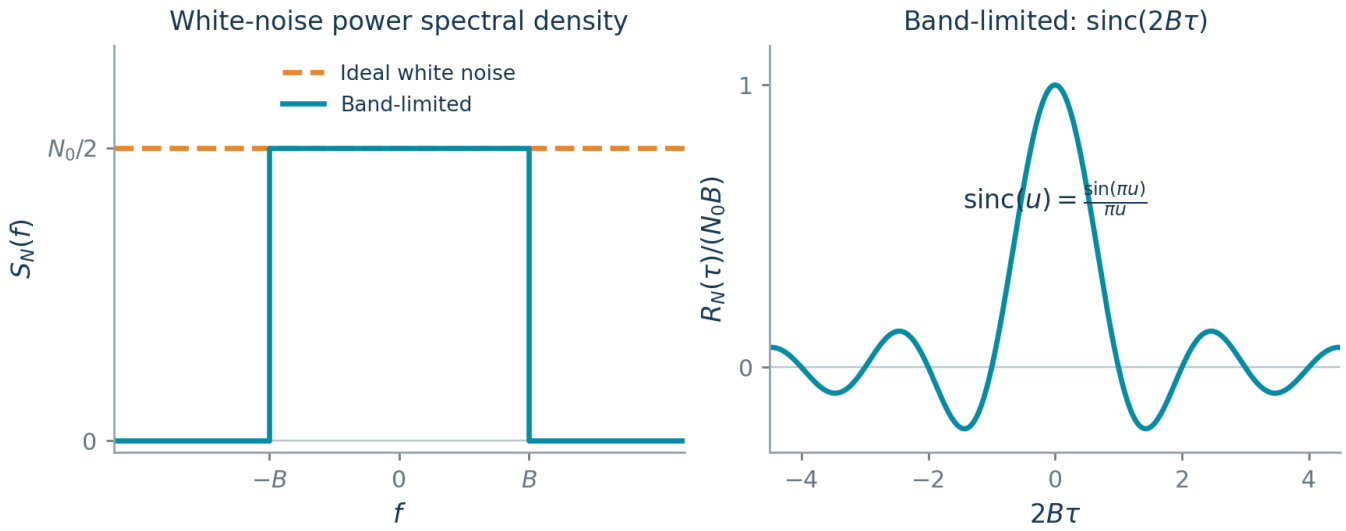
Using $\operatorname{sinc}(u) = \sin(\pi u)/(\pi u)$, with $\operatorname{sinc}(0) = 1$,

$$\begin{aligned} R_Y(\tau) &= \tfrac{N_0}{2} \textstyle\int_{-B}^{B} e^{j2\pi f\tau} \, df \\ &= N_0 B \operatorname{sinc}(2B\tau). \end{aligned}$$

Therefore,

$$P_Y = \sigma_Y^2 = N_0 B.$$

Bandwidth limitation also spreads the correlation over time: nearby output samples can resemble one another because the filter averages contributions from overlapping input intervals. Its correlation zeros occur at $\tau = m/(2B)$ for nonzero integers m. Distinct Gaussian samples on that grid are independent; at other separations, they are generally correlated.



White-noise PSD and the finite-bandwidth output correlation.

Example: RC filtering

Question

White Gaussian noise passes through an RC low-pass filter. Find the output spectrum, power, and autocorrelation.

Solution

For $RC > 0$**,**

$$H(f) = \frac{1}{1 + j2\pi RCf}, \qquad f_{3\mathrm{dB}} = \frac{1}{2\pi RC}.$$

Hence

$$S_Y(f) = \frac{N_0/2}{1 + (2\pi RCf)^2}.$$

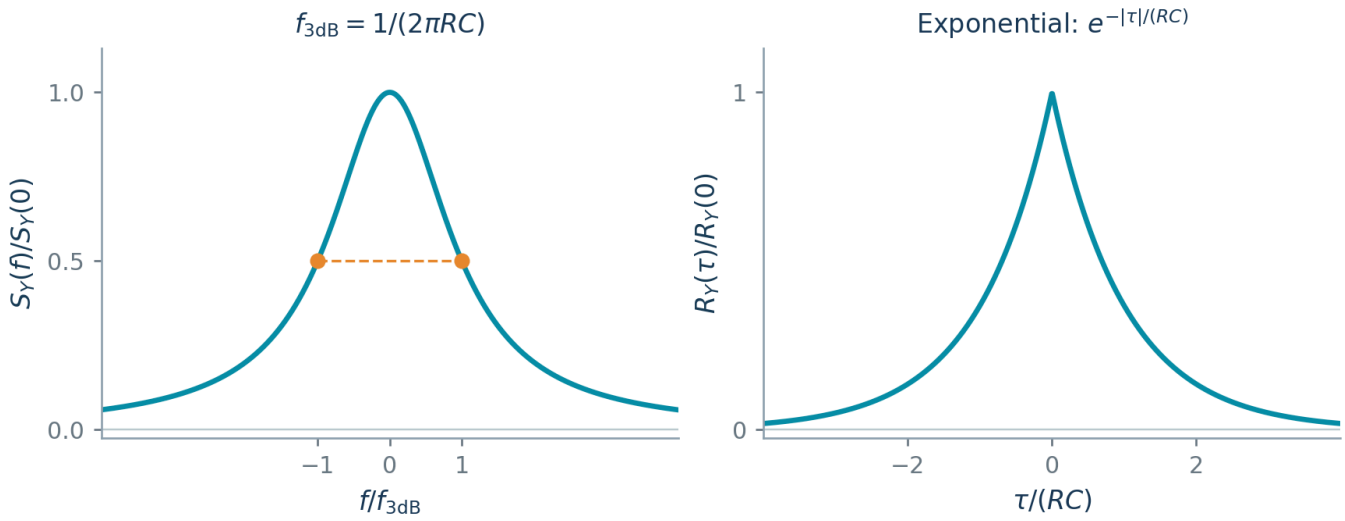
Substitute $u = 2\pi RCf$ **in the power integral:**

$$\begin{aligned} P_Y &= N_0 \int_0^\infty \frac{df}{1 + (2\pi RCf)^2} \\ &= \frac{N_0}{2\pi RC} \int_0^\infty \frac{du}{1 + u^2} \\ &= \frac{N_0}{4RC} = \frac{\pi}{2} N_0 f_{3\mathrm{dB}}. \end{aligned}$$

The corresponding correlation is

$$R_Y(\tau) = \frac{N_0}{4RC} e^{-|\tau|/(RC)}.$$

Its zero-lag value confirms the same power. Increasing RC lowers the noise power and broadens the correlation in time.



RC noise spectrum and exponential correlation.

Noise-equivalent bandwidth

Let $g = \max_f |H(f)|^2 > 0$ be the reference power gain of a real filter. Define its one-sided noise-equivalent bandwidth by

$$B_N = \frac{1}{g}\int_0^\infty |H(f)|^2\, df = \frac{1}{2g}\int_{-\infty}^\infty |H(f)|^2\, df.$$

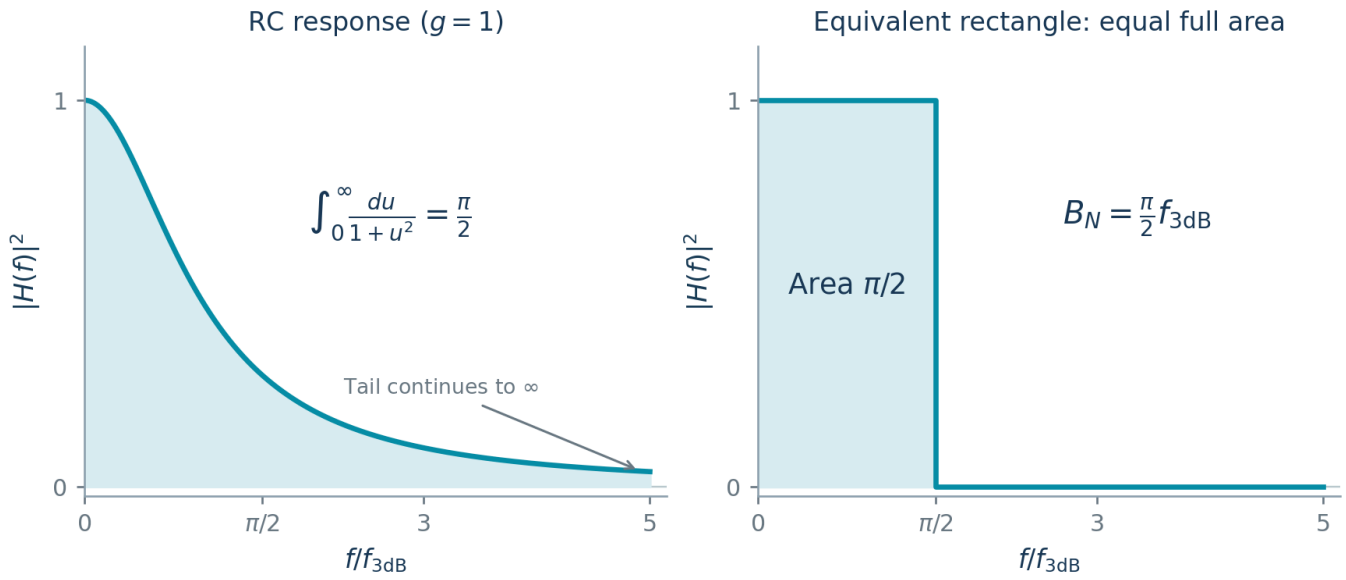
An ideal rectangle of height g and positive-frequency width B_N has the same area as the squared-gain curve. Therefore,

$$P_Y = N_0 g B_N.$$

For the RC filter, $g = 1$, giving

$$B_N = \int_0^\infty \frac{df}{1 + (2\pi RCf)^2} = \frac{1}{4RC} = \frac{\pi}{2} f_{3\text{dB}}.$$

The 3-dB frequency marks one point on the response; equivalent bandwidth measures its area. Filters with the same 3-dB frequency can therefore admit different noise powers. Equivalent bandwidth provides a direct comparison by replacing each actual response with an ideal rectangle that admits the same white-noise power.



Equal positive-frequency areas define the equivalent bandwidth.

Worked Problems

White noise through a first-order low-pass filter

Question

Zero-mean white noise $N_i(t)$ has

$$R_{N_i}(\tau) = \frac{N_0}{2}\delta(\tau).$$

It drives a filter with

$$H(\omega) = \frac{1}{1 + j\omega/\omega_0}, \qquad \omega_0 > 0, \qquad \omega = 2\pi f.$$

Determine:

- The impulse response.
- The output-noise autocorrelation.
- Both orders of input–output cross-correlation.
- The output noise power.

Solution

Impulse response

$$H(\omega) = \frac{\omega_0}{\omega_0 + j\omega} \quad \Rightarrow \quad h(t) = \omega_0 e^{-\omega_0 t} u(t).$$

Output spectrum and autocorrelation

$$\begin{aligned} S_{N_o}(f) &= \tfrac{N_0}{2}|H(f)|^2 \\ &= \tfrac{N_0\omega_0^2}{2\left[\omega_0^2 + (2\pi f)^2\right]}. \end{aligned}$$

Using $e^{-a|\tau|} \leftrightarrow 2a/\left[a^2 + (2\pi f)^2\right]$,

$$R_{N_o}(\tau) = \frac{N_0\omega_0}{4} e^{-\omega_0|\tau|}.$$

Input–output cross-correlations

With $R_{XY}(\tau) = E\{X(t+\tau)\,Y(t)\}$,

$$\begin{aligned} R_{N_iN_o}(\tau) &= \int_{-\infty}^{\infty} h(v)\,R_{N_i}(\tau+v)\,dv \\ &= \tfrac{N_0}{2} h(-\tau). \end{aligned}$$

Therefore,

$$R_{N_iN_o}(\tau) = \frac{N_0\omega_0}{2} e^{\omega_0\tau} u(-\tau),$$

$$R_{N_oN_i}(\tau) = \frac{N_0\omega_0}{2} e^{-\omega_0\tau} u(\tau).$$

Output noise power

The output mean is zero, so

$$P_{N_o} = R_{N_o}(0) = \frac{N_0\omega_0}{4}.$$

Autocorrelation of ideal low-pass filtered white noise

Question

White noise has two-sided PSD $N_0/2$ and passes through

$$H(f) = \begin{cases} 1, & |f| \le B, \\ 0, & \text{otherwise}, \end{cases} \qquad B > 0.$$

Determine the output-noise autocorrelation.

Solution

Output spectrum

$$S_{N_o}(f) = \frac{N_0}{2}\,\mathrm{rect}\left(\frac{f}{2B}\right).$$

Inverse Fourier transform

$$\begin{aligned} R_{N_o}(\tau) &= \tfrac{N_0}{2}\int_{-B}^{B} e^{j2\pi f\tau}\,df \\ &= \tfrac{N_0 \sin(2\pi B\tau)}{2\pi\tau}. \end{aligned}$$

With $\mathrm{sinc}(x) = \sin(\pi x)/(\pi x)$ **and its continuous value at zero,**

$$R_{N_o}(\tau) = N_0 B\,\mathrm{sinc}(2B\tau).$$

Zero-lag check

$$P_{N_o} = R_{N_o}(0) = N_0 B.$$

The first nonzero-lag zeros are $\tau = \pm 1/(2B)$ **.**

Band-pass filtering of triangular-spectrum noise

Question

Zero-mean noise has the two-sided PSD

$$S_{N_i}(f) = \begin{cases} 10^{-8}\left(1 - \dfrac{|f|}{10^8}\right), & |f| < 10^8\ \mathrm{Hz}, \\ 0, & \text{otherwise}. \end{cases}$$

A unity-gain ideal band-pass filter has a positive-frequency passband of width $2\ \mathrm{MHz}$ centered at $50\ \mathrm{MHz}$, and a mirrored negative-frequency passband.

Determine:

- The output PSD.
- The input and output noise powers.

Solution

Identify the passbands

The positive-frequency edges are $49\ \mathrm{MHz}$ and $51\ \mathrm{MHz}$. Therefore,

$$S_{N_o}(f) = \begin{cases} 10^{-8}\left(1 - \dfrac{|f|}{10^8}\right), & 49\times10^6 \le |f| \le 51\times10^6, \\ 0, & \text{otherwise.} \end{cases}$$

Input power

$$P_{N_i} = 2\times10^{-8}\int_0^{10^8}\left(1-\frac{f}{10^8}\right)df = 1.$$

Output power

$$\begin{aligned} P_{N_o} &= 2\times10^{-8}\int_{49\times10^6}^{51\times10^6}\left(1-\tfrac{f}{10^8}\right)df \\ &= 2\times10^{-8}\left[f-\tfrac{f^2}{2\times10^8}\right]_{49\times10^6}^{51\times10^6} \\ &= 0.02. \end{aligned}$$

Thus $100P_{N_o}/P_{N_i} = 2\%$. If the PSD is in $\mathrm{W/Hz}$, the powers are in watts.

Triangular-spectrum noise through a ramp-gain filter

Question

Zero-mean noise has the two-sided PSD

$$S_{N_i}(f) = \begin{cases} 10^{-8}\left(1 - \dfrac{|f|}{10^8}\right), & |f| < 10^8\ \mathrm{Hz}, \\ 0, & \text{otherwise.} \end{cases}$$

The filter has

$$|H(f)|^2 = \begin{cases} |f| - 49\times10^6, & 49\times10^6 \le |f| \le 51\times10^6, \\ 0, & \text{otherwise.} \end{cases}$$

Evaluate f numerically in hertz and use the specified gain without normalization.

Determine:

- The output PSD.
- The output noise power.
- The output-to-input power ratio.

Solution

Output spectrum

$$S_{N_o}(f) = \begin{cases} 10^{-8}\left(1 - \dfrac{|f|}{10^8}\right)\left(|f| - 49\times 10^6\right), & 49\times 10^6 \le |f| \le 51\times 10^6, \\ 0, & \text{otherwise.} \end{cases}$$

Integrate both passbands

Set $a = 49\times 10^6$, $L = 2\times 10^6$ and $u = f - a$. Then

$$\begin{aligned} P_{N_o} &= 2\times 10^{-8}\int_0^L \left(0.51 - \tfrac{u}{10^8}\right) u\, du \\ &= 2\times 10^{-8}\left[\tfrac{0.51L^2}{2} - \tfrac{L^3}{3\times 10^8}\right] \\ &= \tfrac{59600}{3} \approx 1.98667\times 10^4. \end{aligned}$$

Power ratio

The input triangular spectrum has area $P_{N_i} = 1$. Hence

$$\frac{P_{N_o}}{P_{N_i}} = \frac{59600}{3}, \qquad 100\frac{P_{N_o}}{P_{N_i}} \approx 1.98667\times 10^6\%.$$

The specified maximum squared gain is 2×10^6. Powers are in watts when the input PSD is in $\mathrm{W/Hz}$.

Autocorrelation of ideal band-pass filtered white noise

Question

White noise with two-sided PSD $N_0/2$ passes through an ideal band-pass filter. Its magnitude is K on each interval $|f - f_c| \le B$ and $|f + f_c| \le B$, and zero elsewhere, with $f_c > B > 0$.

Determine the output-noise autocorrelation.

Solution

Output spectrum

Each passband has width $2B$:

$$S_{N_o}(f) = \frac{K^2 N_0}{2}\left[\operatorname{rect}\left(\frac{f - f_c}{2B}\right) + \operatorname{rect}\left(\frac{f + f_c}{2B}\right)\right].$$

Inverse Fourier transform

Using the rectangular-spectrum transform and frequency translation,

$$R_{N_o}(\tau) = K^2 N_0 B \,\mathrm{sinc}\,(2B\tau)\left(e^{j2\pi f_c\tau} + e^{-j2\pi f_c\tau}\right)$$
$$= 2K^2 N_0 B \,\mathrm{sinc}\,(2B\tau)\,\cos\,(2\pi f_c\tau).$$

Therefore,

$$R_{N_o}(\tau) = 2K^2 N_0 B \,\mathrm{sinc}\,(2B\tau)\,\cos\,(2\pi f_c\tau).$$

Here $\mathrm{sinc}\,(x) = \sin\,(\pi x)/(\pi x)$**, with** $\mathrm{sinc}\,(0) = 1$**.**

Power check

$$P_{N_o} = R_{N_o}(0) = 2K^2 N_0 B.$$

First-order filtering with a 10 rad/s cutoff

Question

White noise has $R_X(\tau) = (N_0/2)\,\delta(\tau)$ **and drives**

$$H(\omega) = \frac{1}{1 + j\omega/10}, \qquad \omega = 2\pi f.$$

Determine:

- **The impulse response.**
- **The output-noise autocorrelation.**

Solution

Impulse response

$$H(\omega) = \frac{10}{10 + j\omega} \quad \Rightarrow \quad h(t) = 10e^{-10t}u(t).$$

Output spectrum

$$S_Y(f) = \frac{N_0}{2}|H(f)|^2 = \frac{N_0}{2}\frac{100}{100 + (2\pi f)^2}.$$

Output autocorrelation

Using $e^{-a|\tau|} \leftrightarrow 2a/\left[a^2 + (2\pi f)^2\right]$ **with** $a = 10$**,**

$$R_Y(\tau) = \frac{5N_0}{2}e^{-10|\tau|}.$$

The output power is $P_Y = 5N_0/2$, and the cutoff frequency is $10/(2\pi)$ Hz.

Finite-band powers after first-order noise filtering

Question

A zero-mean white Gaussian input has $S_X(f) = \eta/2$ and drives a filter with $h(t) = e^{-t}u(t)$.

Determine:

- **The output PSD, autocorrelation and mean-square value.**
- **The input and output powers in the band $|f| \leq B$, with positive low-pass edge $B = 10^3$ Hz.**
- **The percentage of total output power within that band and the output-to-input power percentage within the same band.**

Solution

Frequency response and output PSD

$$\begin{aligned} H(f) &= \int_0^\infty e^{-t}e^{-j2\pi ft}\,dt = \tfrac{1}{1+j2\pi f}, \\ S_Y(f) &= \tfrac{\eta}{2}|H(f)|^2. \end{aligned}$$

Therefore,

$$S_Y(f) = \frac{\eta}{2\left[1+(2\pi f)^2\right]}.$$

Autocorrelation and mean square

The exponential Fourier pair gives

$$R_Y(\tau) = \frac{\eta}{4}e^{-|\tau|}, \qquad E\{Y^2(t)\} = P_{Y,\text{total}} = \frac{\eta}{4}.$$

Powers in the measurement band

$$\begin{aligned} P_{X,B} &= \int_{-B}^{B}\tfrac{\eta}{2}\,df = \eta B = 1000\eta, \\ P_{Y,B} &= \tfrac{\eta}{2}\int_{-B}^{B}\frac{df}{1+(2\pi f)^2} = \tfrac{\eta}{2\pi}\tan^{-1}(2\pi B). \end{aligned}$$

Power percentages

At $B = 10^3$ Hz,

$$100\frac{P_{Y,\,B}}{P_{Y,\,\text{total}}} = 100\frac{2}{\pi}\tan^{-1}\ (\,2\pi B\,)\ \approx 99.9899\%,$$

$$100\frac{P_{Y,\,B}}{P_{X,\,B}} = 100\frac{\tan^{-1}\ (\,2\pi B\,)}{2\pi B} \approx 0.0249975\%.$$

The first ratio measures how much of the total filtered noise lies in the measurement band; the second compares output and input powers over that same band.

Random noise and deterministic interference

> ### Question
>
> **Is an unwanted noise waveform necessarily deterministic or random? Classify the main sources of unwanted electrical signals and explain which model applies to thermal noise.**

Solution

Waveform model. Thermal noise is random: its exact voltage cannot be predicted, so it is described by probability distributions, correlation, and a power spectrum. Some unwanted signals are deterministic or nearly periodic, such as power-supply hum. The word noise describes their unwanted effect; it does not force every disturbance to have the same statistical model.

Source classification.

- **Natural sources: atmospheric disturbances, extraterrestrial radiation, and internal circuit noise. Thermal noise is caused by the random motion of charge carriers in conducting materials.**
- **Man-made sources: pickup from other transmitting equipment, power-supply hum, and switching transients.**

A Gaussian model describes the amplitude distribution of thermal noise, while a white-noise model describes an approximately flat spectrum over the frequency range of interest. These are separate properties. A receiver filter selects a finite band and therefore limits the noise power reaching the next stage.

Equivalent bandwidth of a stepped low-pass gain

Question

A real low-pass filter has magnitude response

$$|H(f)| = \begin{cases} 2, & |f| \le 15\times 10^{3}\ \text{Hz}, \\ 1, & 15\times 10^{3} < |f| \le f_m, \\ 0, & |f| > f_m, \end{cases}$$

where $f_m > 15\times 10^{3}$ Hz. Determine f_m when the one-sided noise-equivalent bandwidth is $B_N = 20$ kHz.

Solution

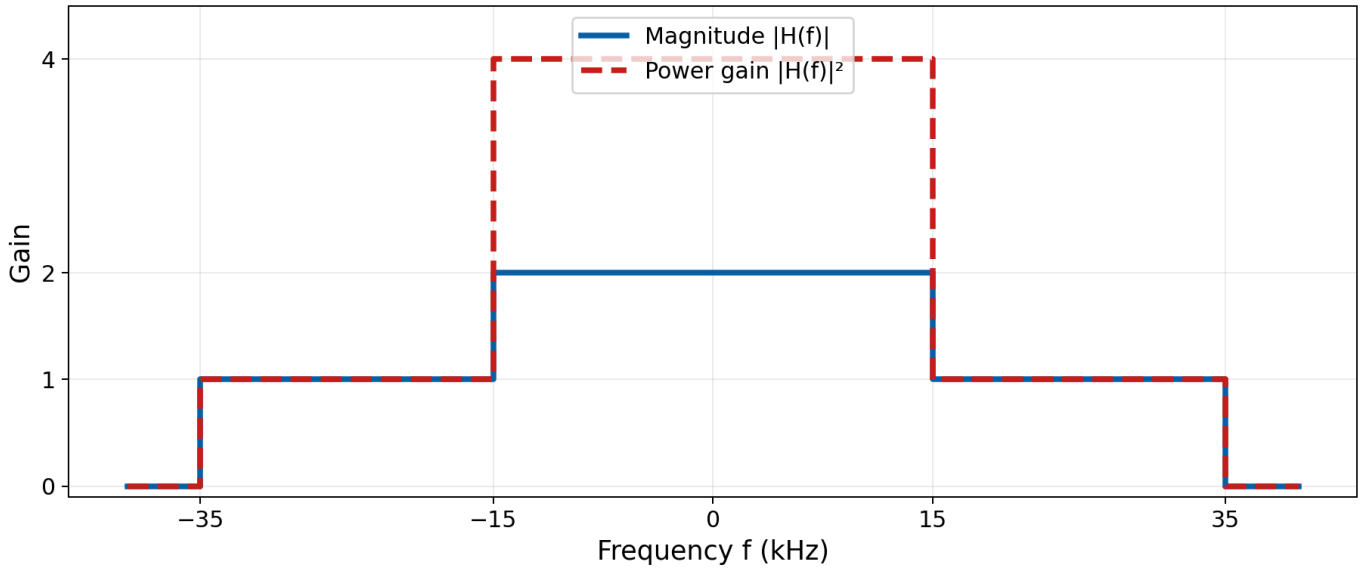
Use squared gain. The maximum power gain is $g = \max |H(f)|^2 = 4$. Thus

$$B_N = \frac{1}{g}\int_0^{\infty} |H(f)|^2\, df = \frac{1}{4}\left[4\left(15\times 10^{3}\right) + \left(f_m - 15\times 10^{3}\right)\right].$$

Equating this area to the specified bandwidth gives

$$20\times 10^{3} = \frac{f_m + 45\times 10^{3}}{4}, \qquad f_m = 35\times 10^{3}\ \text{Hz} = 35\ \text{kHz}.$$

The outer gain of 1 contributes one quarter as much noise power per hertz as the central gain of 2; noise power depends on $|H|^2$.



Amplitude gain and squared gain: the squared response determines the noise-equivalent bandwidth.

Comparing equivalent and half-power bandwidths

Question

A filter has frequency response

$$H(f) = \frac{4}{10 + j2\pi f}.$$

Find the ratio of its noise-equivalent bandwidth to its positive-frequency 3 dB bandwidth.

Solution

Half-power edge.

$$|H(f)|^2 = \frac{16}{100 + (2\pi f)^2}, \qquad |H(0)|^2 = \frac{16}{100}.$$

Half the peak occurs at $(2\pi f)^2 = 100$**:**

$$f_{3\text{dB}} = \frac{10}{2\pi}.$$

Normalize before integrating.

$$\begin{aligned} B_N &= \tfrac{1}{|H(0)|^2} \int_0^\infty |H(f)|^2 \, df \\ &= \int_0^\infty \tfrac{df}{1 + (2\pi f/10)^2} = \tfrac{10}{2\pi}\tfrac{\pi}{2} = \tfrac{10}{4}. \end{aligned}$$

Therefore

$$\frac{B_N}{f_{3\text{dB}}} = \frac{\pi}{2} \approx 1.57080.$$

The constant gain cancels in this bandwidth ratio.

Half-power and equivalent widths of a spectrum

Question

A random process has the two-sided PSD

$$S_X(f) = \frac{10}{\left[1 + \left(\frac{2\pi f}{3}\right)^2\right]^2}.$$

Find its positive-frequency half-power edge and its equivalent noise bandwidth, defined by

$$B_N = \frac{1}{S_X(0)} \int_0^\infty S_X(f)\, df.$$

Solution

Half-power edge. Set the spectrum equal to half its peak:

$$\left[1 + \left(\frac{2\pi f_{3\text{dB}}}{3}\right)^2\right]^2 = 2.$$

Thus

$$f_{3\text{dB}} = \frac{3}{2\pi}\sqrt{\sqrt{2} - 1} \approx 0.307294 \text{ Hz}.$$

The two-sided half-power width is $2f_{3\text{dB}}$. Equivalent area. Substitute $u = 2\pi f/3$:

$$\begin{aligned} B_N &= \tfrac{3}{2\pi} \int_0^\infty \frac{du}{(1+u^2)^2} \\ &= \tfrac{3}{2\pi}\left[\frac{u}{2(1+u^2)} + \tfrac{1}{2}\tan^{-1} u\right]_0^\infty = \tfrac{3}{8}\,\text{Hz}. \end{aligned}$$

The total power is $2S_X(0)\, B_N = 7.5$.

White-noise power in low-pass filters and a differentiator

Question

White noise has one-sided spectral level $N_0 = 0.001\ \mathrm{mW/Hz}$, so its two-sided PSD is $N_0/2$. Determine the output noise powers for an RC low-pass filter with $R = 1000\ \Omega$ and $C = 0.1\ \mathrm{mF}$, a unity-gain ideal low-pass filter with positive-frequency edge $B = 1000$ Hz, and an ideal differentiator with $H(f) = j2\pi f\tau$, $\tau = 0.01$ s. Explain how doubling a low-pass cutoff changes the output power. For the differentiator, distinguish ideal white input from input restricted to $|f| \le B$.

Solution

Units.

$$N_0 = 10^{-6}\ \mathrm{W/Hz}, \qquad C = 10^{-4}\ \mathrm{F}, \qquad RC = 0.1\ \mathrm{s}.$$

RC filter.

$$P_{\mathrm{RC}} = \frac{N_0}{4RC} = 2.5 \times 10^{-6}\ \mathrm{W}.$$

Its cutoff is $f_{3\mathrm{dB}} = 1/(2\pi RC) \approx 1.59155$ Hz. Since $P_{\mathrm{RC}} = (\pi/2) N_0 f_{3\mathrm{dB}}$, doubling the cutoff doubles the power to 5×10^{-6} W. Ideal low-pass filter.

$$P_{\mathrm{LP}} = N_0 B = 10^{-3}\ \mathrm{W}.$$

Doubling B doubles this to 2×10^{-3} W. Differentiator. Ideal white input gives

$$P_{\mathrm{diff}} = N_0 (2\pi\tau)^2 \int_0^\infty f^2\, df = \infty.$$

A differentiator alone therefore gives no finite noise power. With a finite input band $|f| \le B$,

$$P_{\mathrm{diff},\,B} = \frac{N_0 (2\pi\tau)^2 B^3}{3}.$$

Doubling that band edge multiplies the finite result by $2^3 = 8$; it does not make the ideal-white-input result finite.

Signal and white noise through a low-pass filter

Question

A real WSS input is $X(t) = s(t) + w(t)$. The desired signal and noise are uncorrelated. The signal has $R_s(\tau) = Ke^{a|\tau|}$, with $K > 0$ and $a < 0$; the white noise has two-sided PSD $N_0/2$. An ideal unity-gain low-pass filter passes $|f| \le W$. Find the output signal-to-noise ratio as a function of $W > 0$.

Solution

Find the signal spectrum. Write $\alpha = -a > 0$. Then

$$S_s\left(f\right) = \int_{-\infty}^{\infty} Ke^{-\alpha|\tau|}e^{-j2\pi f\tau}\,d\tau = \frac{2K\alpha}{\alpha^2 + \left(2\pi f\right)^2}.$$

The condition $a < 0$ makes the correlation decay; a positive exponent would violate $|R_s\left(\tau\right)| \le R_s\left(0\right)$.

Integrate the passed signal and noise.

$$\begin{aligned} P_{s,o} &= \int_{-W}^{W} \frac{2K\alpha}{\alpha^2 + \left(2\pi f\right)^2}\,df = \frac{2K}{\pi}\tan^{-1}\left(\frac{2\pi W}{\alpha}\right), \\ P_{n,o} &= \int_{-W}^{W} \frac{N_0}{2}\,df = N_0 W. \end{aligned}$$

Therefore

$$\mathrm{SNR}_o\left(W\right) = \frac{2K}{\pi N_0 W}\tan^{-1}\left(\frac{2\pi W}{\alpha}\right), \qquad \alpha = -a.$$

Widening the filter admits more signal power, approaching K, while the noise power continues to grow in proportion to W.

A random sinusoid multiplying white noise

Question

Let

$$X\left(t\right) = 2\cos\left(4400\pi t + \Phi\right) W\left(t\right),$$

where $\Phi \sim \mathcal{U}\left[-\pi, \pi\right]$ is independent of zero-mean white noise with two-sided PSD $S_W\left(f\right) = N_0/2 = 4$. The filter magnitude is

$$|H\left(f\right)| = \begin{cases} 5, & 2100 \le |f| \le 2400 \text{ Hz}, \\ 2, & 2000 \le |f| < 2100 \text{ Hz or } 2400 < |f| \le 2500 \text{ Hz}, \\ 0, & \text{otherwise}. \end{cases}$$

Determine the input and output spectra, the output power, and whether an output signal-to-noise ratio is defined by this model.

Solution

Input correlation. Averaging over the independent uniform phase gives

$$E\{\cos\left(4400\pi t + \Phi\right)\cos\left(4400\pi\left(t+\tau\right) + \Phi\right)\} = \frac{1}{2}\cos\left(4400\pi\tau\right).$$

Since $R_W\left(\tau\right) = 4\delta\left(\tau\right)$,

$$R_X(\tau) = 4\left[\frac{1}{2}\cos(4400\pi\tau)\right]4\delta(\tau) = 8\delta(\tau).$$

Therefore

$$S_X(f) = 8, \qquad S_Y(f) = 8|H(f)|^2.$$

Finite output power. Both mirrored bands contribute. Each has a 300 **Hz interval at spectral level** $8\left(5^2\right) = 200$ **and a total** 200 **Hz at level** $8\left(2^2\right) = 32$**:**

$$P_Y = 2\left[200(300) + 32(200)\right] = 132800.$$

Meaning of SNR. The specified input multiplies the noise by a random sinusoid. It supplies no separate additive desired signal and noise components, so a signal-to-noise ratio cannot be determined from this model.

Narrowband Noise and Its Quadrature Components

A radio receiver selects a narrow band around the carrier, so the noise reaching its detector is also concentrated in that band. Its waveform oscillates rapidly at approximately the carrier frequency while its amplitude and phase vary more slowly. Describing those slow variations makes the noise easier to analyze than following every carrier cycle. The quadrature representation also connects directly to the two carrier references used by an I/Q receiver.

A time-domain approximation

A real zero-mean stationary Gaussian narrowband process has mirrored spectral lobes near $\pm f_c$, with bandwidth small compared with f_c. Consider first a rectangular spectral pair of height S_0, each lobe having full width Δf and center $\pm f_c$. Then

$$R_Y(\tau) = 2S_0\Delta f \ \mathrm{sinc}\ (\Delta f\tau)\ \cos\ (2\pi f_c\tau).$$

For $|\tau| \ll 1/\Delta f$,

$$R_Y(\tau) \approx 2S_0\Delta f \cos\ (2\pi f_c\tau).$$

This correlation is reproduced by

$$\tilde{Y}(t) = A\cos\ (2\pi f_c t) + B\sin\ (2\pi f_c t),$$

where A, B are independent zero-mean Gaussian variables with

$$E\{A^2\} = E\{B^2\} = 2S_0\Delta f.$$

A general narrowband spectrum can be divided into thin spectral pairs. With pair centers $\pm f_m$ and width Δf,

$$Y(t) \approx \sum_m \left[A_m \cos\ (2\pi f_m t) + B_m \sin\ (2\pi f_m t)\right],$$

$$E\{A_m^2\} = E\{B_m^2\} = 2S_Y(f_m)\,\Delta f.$$

Coefficients from different pairs are uncorrelated and jointly Gaussian. The resulting mean square is the spectral-area approximation

$$E\{Y^2(t)\} \approx \sum_m 2S_Y(f_m)\,\Delta f.$$

In-phase and quadrature representation

The same noise can be written using slowly varying baseband components:

$$Y(t) = Y_c(t)\cos(2\pi f_c t) - Y_s(t)\sin(2\pi f_c t).$$

Y_c is the in-phase component and Y_s the quadrature component. They are the noise coordinates along cosine and sine carrier references, separating the fast carrier oscillation from the slower noise variations. For zero-mean stationary Gaussian bandpass noise, they are zero-mean, low-pass, jointly stationary Gaussian processes.

Translate the positive lobe left by f_c and the negative lobe right by f_c, then add:

$$S_{Y_c}(f) = S_{Y_s}(f) = S_Y(f+f_c) + S_Y(f-f_c),$$

within the retained baseband support. With

$$R_{Y_cY_s}(\tau) = E\{Y_c(t+\tau)\,Y_s(t)\},$$

the cross spectra are

$$S_{Y_cY_s}(f) = j[S_Y(f+f_c) - S_Y(f-f_c)],$$

$$S_{Y_sY_c}(f) = S^*_{Y_cY_s}(f).$$

Their imaginary, odd shape gives $R_{Y_cY_s}(0) = 0$. The component powers satisfy

$$P_{Y_c} = P_{Y_s} = P_Y = \int_{-\infty}^{\infty} S_Y(f)\,df.$$

For real components,

$$R_Y(\tau) = R_{Y_c}(\tau)\cos(2\pi f_c\tau) + R_{Y_cY_s}(\tau)\sin(2\pi f_c\tau).$$

Symmetry about the carrier means that equal offsets above and below it carry matching noise densities. If the lobes are symmetric about the chosen carrier, the cross spectrum vanishes at every frequency. The components are then independent Gaussian processes, and $R_Y = R_{Y_c}\cos(2\pi f_c\tau)$. Without that symmetry, the same-time pair remains independent, but different-time component values can be correlated.

Example: Symmetric bandpass white noise

Question

White Gaussian noise of PSD $N_0/2$ passes through a unity-gain filter with positive passband $[f_c - B, f_c + B]$ and its negative-frequency mirror, where $0 < B \ll f_c$. Find the noise and component spectra, correlations, and powers.

Solution

Each lobe has full width $2B$, so

$$P_Y = 2\left(2B\frac{N_0}{2}\right) = 2N_0B.$$

The translated lobes coincide at baseband:

$$S_{Y_c}(f) = S_{Y_s}(f) = \begin{cases} N_0, & |f| \le B, \\ 0, & |f| > B. \end{cases}$$

Thus

$$R_{Y_c}(\tau) = R_{Y_s}(\tau) = 2N_0B\,\text{sinc}\,(2B\tau),$$

$$R_Y(\tau) = 2N_0B\,\text{sinc}\,(2B\tau)\,\cos\,(2\pi f_c\tau),$$

$$P_{Y_c} = P_{Y_s} = P_Y = 2N_0B.$$

Each component has the same power as the bandpass process. Sinusoidal reconstruction contributes a factor one-half to each component's power.

Example: An asymmetric triangular spectrum

Question

A zero-mean Gaussian process has a positive triangular lobe with vertices $(4,0)$, $(5,1)$, $(7,0)$, with frequency in hertz, and a reflected negative lobe. Find the component spectra at reference carrier $f_c = 5$ Hz and the same-time joint density.

Solution

The total spectral area is

$$P_Y = 2\left[\frac{1}{2}(7-4)\right] = 3.$$

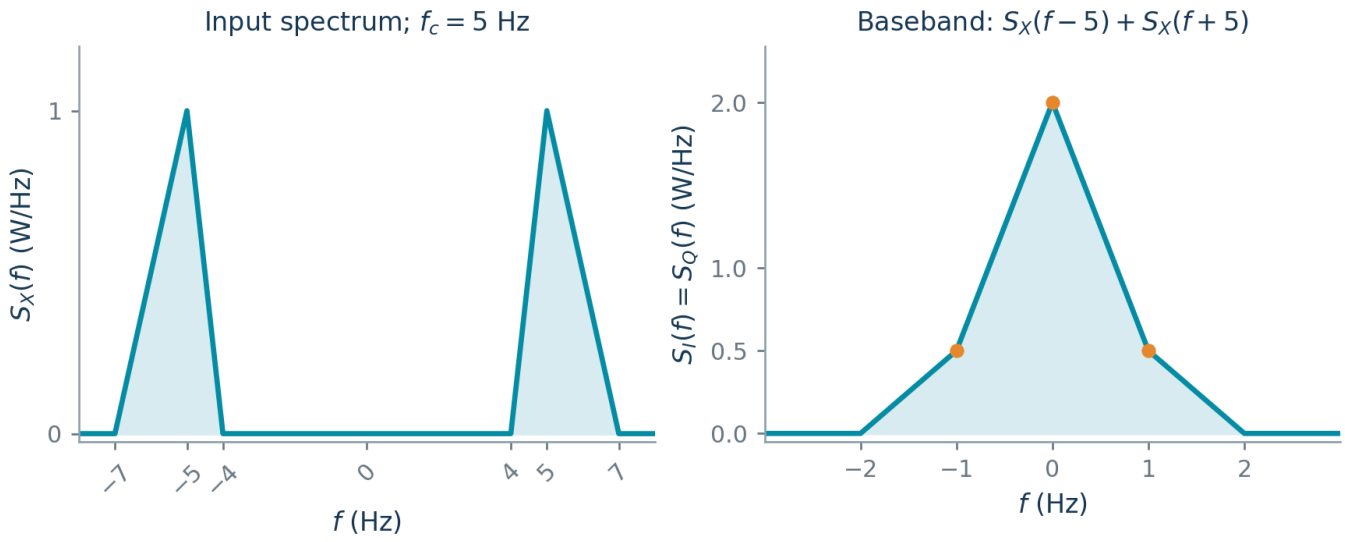
Adding the two translated shapes gives

$$S_{Y_c}(f) = S_{Y_s}(f) = \begin{cases} 2 - \frac{3}{2}|f|, & |f| \le 1, \\ 1 - \frac{1}{2}|f|, & 1 < |f| \le 2, \\ 0, & |f| > 2. \end{cases}$$

The peak is two, and each component spectrum has area three. At a common time, the Gaussian pair has zero covariance and variances three:

$$f_{Y_c, Y_s}(u, v) = \frac{1}{6\pi} \exp\left[-\frac{u^2 + v^2}{6}\right].$$

The spectrum's asymmetry affects unequal-time cross-correlation, while the same-time pair is independent.



A bandpass triangle and its shifted-and-added component spectrum.

Envelope and phase

The envelope measures the instantaneous strength of the narrowband noise, and the phase locates its oscillation relative to the carrier reference. When this noise is added to a received signal, its quadrature components perturb the signal's amplitude and phase, affecting the detector output.

Define

$$R(t) = \sqrt{Y_c^2(t) + Y_s^2(t)}, \qquad \Theta(t) = \operatorname{atan2}\left(Y_s(t), Y_c(t)\right).$$

Then

$$Y(t) = R(t) \cos\left[2\pi f_c t + \Theta(t)\right].$$

At a fixed time, write $\sigma^2 = P_Y > 0$. The independent Gaussian components have joint density

$$f_{Y_c, Y_s}\left(u, v\right) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{u^2 + v^2}{2\sigma^2}\right].$$

Substitute $u = r\cos\theta$, $v = r\sin\theta$ with Jacobian magnitude r:

$$f_{R,\Theta}\left(r, \theta\right) = \frac{r}{2\pi\sigma^2} e^{-r^2/\left(2\sigma^2\right)}, \qquad r \geq 0, \qquad -\pi < \theta \leq \pi.$$

It factors into the Rayleigh envelope density and uniform phase density:

$$f_R\left(r\right) = \frac{r}{\sigma^2} e^{-r^2/\left(2\sigma^2\right)}, \quad r \geq 0, \qquad f_\Theta\left(\theta\right) = \frac{1}{2\pi}.$$

The envelope and phase are independent at a fixed time. Their useful envelope moments are

$$E\{R\} = \sigma\sqrt{\frac{\pi}{2}}, \qquad E\{R^2\} = 2\sigma^2, \qquad \mathrm{Var}\left(R\right) = \left(2 - \frac{\pi}{2}\right)\sigma^2.$$

Worked Problems

Envelope and phase of a carrier plus narrowband noise

Question

A carrier $x\left(t\right) = A\cos\left(2\pi f_c t\right)$ is added to narrowband noise

$$n\left(t\right) = n_c\left(t\right)\cos\left(2\pi f_c t\right) - n_s\left(t\right)\sin\left(2\pi f_c t\right).$$

Express $y\left(t\right) = x\left(t\right) + n\left(t\right)$ in the form

$$y\left(t\right) = \rho\left(t\right)\cos\left[2\pi f_c t + \theta\left(t\right)\right].$$

Solution

Combine the quadrature components

$$y\left(t\right) = \left[A + n_c\left(t\right)\right]\cos\left(2\pi f_c t\right) - n_s\left(t\right)\sin\left(2\pi f_c t\right).$$

Match the envelope–phase coefficients

Expanding $\rho\left(t\right)\cos\left[2\pi f_c t + \theta\left(t\right)\right]$ gives

$$\begin{aligned}\rho\left(t\right)\cos\theta\left(t\right) &= A + n_c\left(t\right),\\ \rho\left(t\right)\sin\theta\left(t\right) &= n_s\left(t\right).\end{aligned}$$

Hence

$$\rho\,(\,t\,) \;=\sqrt{[\,A+n_c\,(\,t\,)\,]^{\,2}+n_s^2\,(\,t\,)}\,,$$

$$\theta\,(\,t\,) \;=\; \operatorname{atan2}\,(n_s\,(\,t\,)\,,A+n_c\,(\,t\,)\,).$$

The two-argument arctangent selects the correct quadrant. When $\rho\,(\,t\,)\;=0$, phase may be assigned arbitrarily.

I/Q spectra of asymmetric triangular-band noise

Question

The PSD $S_n\,(\,f\,)$ has a positive-frequency triangular lobe supported on $4\le f\le 7\ \mathrm{Hz}$, with height 1 at $f=5\ \mathrm{Hz}$. It rises linearly from zero at 4 Hz and falls linearly to zero at 7 Hz. The negative-frequency lobe is its mirror.

Use $f_c=5\ \mathrm{Hz}$ and

$$n\,(\,t\,) \;=n_c\,(\,t\,)\;\cos\;(\,2\pi f_c t\,)\;-n_s\,(\,t\,)\;\sin\;(\,2\pi f_c t\,)\,.$$

Determine:

- **The in-phase and quadrature PSDs.**
- **Both cross-spectral densities, using $R_{XY}\,(\,\tau\,)\;=E\{X\,(\,t+\tau\,)\,Y(\,t\,)\,\}$.**

Solution

Express the band-pass spectrum

$$T(\,f\,)\;=\begin{cases} f-4, & 4\le f\le 5,\\ \dfrac{7-f}{2}, & 5\le f\le 7,\\ 0, & \text{otherwise,}\end{cases}\qquad S_n\,(\,f\,)\;=T(\,f\,)\;+T(\,-f\,)\,.$$

Translate and add the lobes

$$S_{n_c}\,(\,f\,)\;=S_{n_s}\,(\,f\,)\;=S_n\,(\,f+5\,)\;+S_n\,(\,f-5\,)\,.$$

For $r=|f|$, with f in hertz,

$$S_{n_c}\,(\,f\,)\;=S_{n_s}\,(\,f\,)\;=\begin{cases} 2-\dfrac{3}{2}r, & 0\le r\le 1,\\ 1-\dfrac{1}{2}r, & 1<r\le 2,\\ 0, & r>2.\end{cases}$$

Cross spectra

$$\begin{aligned} S_{n_c n_s}(f) &= j\left[S_n(f+5) - S_n(f-5)\right], \\ S_{n_s n_c}(f) &= S^*_{n_c n_s}(f). \end{aligned}$$

Define

$$q(r) = \begin{cases} \dfrac{r}{2}, & 0 \le r \le 1, \\ 1 - \dfrac{r}{2}, & 1 < r \le 2, \\ 0, & r > 2. \end{cases}$$

The explicit answers are

$$S_{n_c n_s}(f) = j\,\mathrm{sgn}\,(f)\,q(|f|), \qquad S_{n_s n_c}(f) = -j\,\mathrm{sgn}\,(f)\,q(|f|).$$

Power and covariance checks

$$\begin{aligned} P_{n_c} = P_{n_s} &= 2\left[\int_0^1 \left(2 - \tfrac{3}{2}r\right) dr + \int_1^2 \left(1 - \tfrac{1}{2}r\right) dr\right] \\ &= 3. \end{aligned}$$

The imaginary odd cross spectrum integrates to zero, so $C_{n_c n_s}(0) = 0$. It is not identically zero, so unequal-time cross-correlation can remain.

I/Q spectra with three carrier choices

Question

Zero-mean white noise has two-sided PSD $N_0/2$ and passes through a filter with

$$|H(f)|^2 = \begin{cases} \frac{1}{2}, & 10 \le |f|/10^3 < 12, \\ 1, & 12 \le |f|/10^3 \le 14, \\ 0, & \text{otherwise}, \end{cases}$$

where f is in hertz.

Determine:

- The output PSD and total noise power.
- The I/Q PSDs and both cross spectra for $f_c = 10, 12, 14\,\text{kHz}$.

Use $n = n_c \cos(2\pi f_c t) - n_s \sin(2\pi f_c t)$ and $R_{XY}(\tau) = E\{X(t+\tau)\,Y(t)\}$.

Solution

Output PSD and power

$$S_n(f) = \begin{cases} \frac{N_0}{4}, & 10 \le |f|/10^3 < 12, \\ \frac{N_0}{2}, & 12 \le |f|/10^3 \le 14, \\ 0, & \text{otherwise}. \end{cases}$$

Both lobes contribute, with each subband width equal to $2000\ \text{Hz}$:

$$P_n = 2\left[\frac{N_0}{4}(2000) + \frac{N_0}{2}(2000)\right] = 3000N_0.$$

Shifted-spectrum formulas

Let $S_I(f) = S_{n_c}(f) = S_{n_s}(f)$ and $r = |f|/10^3$. Then

$$\begin{aligned} S_I(f) &= S_n(f+f_c) + S_n(f-f_c), \\ S_{n_c n_s}(f) &= j\,[S_n(f+f_c) - S_n(f-f_c)], \\ S_{n_s n_c}(f) &= S^*_{n_c n_s}(f). \end{aligned}$$

Carrier $f_c = 10\,\text{kHz}$

$$S_I(f) = \begin{cases} \frac{N_0}{4}, & 0 < r < 2, \\ \frac{N_0}{2}, & 2 < r < 4, \\ 0, & r > 4, \end{cases}$$

$$S_{n_c n_s}(f) = j \operatorname{sgn}(f) S_I(f).$$

Carrier $f_c = 12\,\mathrm{kHz}$

$$S_I(f) = \begin{cases} \frac{3N_0}{4}, & r < 2, \\ 0, & r > 2, \end{cases}$$

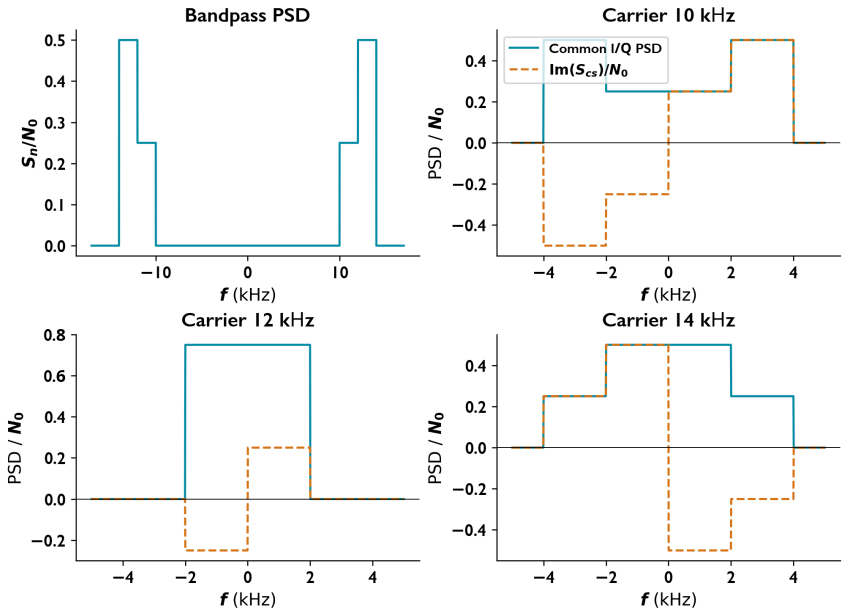
$$S_{n_c n_s}(f) = \begin{cases} j \operatorname{sgn}(f) \frac{N_0}{4}, & 0 < r < 2, \\ 0, & r > 2. \end{cases}$$

Carrier $f_c = 14\,\mathrm{kHz}$

$$S_I(f) = \begin{cases} \frac{N_0}{2}, & 0 < r < 2, \\ \frac{N_0}{4}, & 2 < r < 4, \\ 0, & r > 4, \end{cases}$$

$$S_{n_c n_s}(f) = -j \operatorname{sgn}(f) S_I(f).$$

For every carrier, $S_{n_s n_c} = S^*_{n_c n_s}$ **and each component power is** $3000 N_0$**. Values at isolated spectral breakpoints do not affect these powers.**



Frequency axes are in kHz. The dashed orange curve is the imaginary cross spectrum under the stated first-variable lag convention.

Band-pass noise translated to baseband

Question

Zero-mean white Gaussian noise with two-sided PSD $N_0/2$ passes through a unity-gain ideal band-pass filter with passbands $|f-f_c| \le B$ and $|f+f_c| \le B$. The result is multiplied by $\cos\,(\,2\pi f_c t\,)$ and passed through a unity-gain ideal low-pass filter with $|f| \le B$. Assume $f_c > 2B$. Find the final output PSD and autocorrelation, its mean and variance, and the greatest uniform sampling rate for which all distinct output samples are uncorrelated.

Solution

Translation and gain. Write the band-pass noise as

$$v\,(\,t\,) = n_c\,(\,t\,)\,\cos\,(\,2\pi f_c t\,) - n_s\,(\,t\,)\,\sin\,(\,2\pi f_c t\,)\,.$$

Multiplication by the cosine produces a baseband term $n_c\,(\,t\,)/2$ and terms near $2f_c$. The low-pass filter removes the latter. Since the in-phase component has PSD N_0 on $[-B, B]$,

$$S_Y\,(\,f\,) = \begin{cases} N_0/4, & |f| \le B, \\ 0, & \text{otherwise.} \end{cases}$$

Correlation and moments. Inverting the rectangular spectrum gives

$$R_Y\,(\,\tau\,) = \frac{N_0 B}{2}\,\mathrm{sinc}\,(\,2B\tau\,)\,.$$

The process remains zero mean, and

$$\mu_Y = 0, \qquad \sigma_Y^2 = R_Y(0) = \frac{N_0 B}{2}.$$

Uncorrelated samples. The first nonzero correlation zero is $\tau = 1/(2B)$. For samples spaced by $T_s = 1/(2B)$, every nonzero integer lag is also a zero:

$$R_Y(kT_s) = \frac{N_0 B}{2}\,\mathrm{sinc}\,(k) = 0, \qquad k \neq 0.$$

$$f_{s,\,\max} = 2B.$$

Because these samples are jointly Gaussian, they are also independent.

Narrowband Gaussian noise and the envelope of a noisy carrier

Question

Zero-mean white Gaussian noise with two-sided PSD $N_0/2$ passes through an ideal unity-gain band-pass filter. Each passband has width $2W$ and is centered at $\pm f_c$, with $f_c > W$. With the quadrature carrier chosen at f_c, find the power in each quadrature component and the density of the noise envelope. Then add $A\cos(2\pi f_c t)$ and find the density of the resulting envelope.

Solution

Translate the two noise bands to baseband. Write

$$n(t) = n_c(t)\cos(2\pi f_c t) - n_s(t)\sin(2\pi f_c t).$$

The component spectra add the two translated bands:

$$S_{n_c}(f) = S_{n_s}(f) = \begin{cases} N_0, & |f| \le W, \\ 0, & |f| > W. \end{cases}$$

Consequently

$$E\{n_c^2(t)\} = E\{n_s^2(t)\} = 2N_0 W \equiv \sigma_n^2.$$

This is also the band-pass noise power. At a fixed time, the components are jointly Gaussian, uncorrelated, and therefore independent.

Noise-only envelope. With $V = \sqrt{n_c^2 + n_s^2}$, the polar-coordinate Jacobian is v. Integrating over phase gives the Rayleigh density:

$$f_V(v) = \begin{cases} \dfrac{v}{\sigma_n^2} e^{-v^2/(2\sigma_n^2)}, & v \ge 0, \\ 0, & v < 0. \end{cases}$$

Add the carrier before taking the magnitude. The new envelope is

$$U = \sqrt{\left(A + n_c \right)^2 + n_s^2}.$$

For $u \geq 0$, the shifted Gaussian joint density and the polar Jacobian give

$$\begin{aligned} f_U\left(u \right) &= \frac{u}{2\pi\sigma_n^2} \int_0^{2\pi} \exp\left[-\frac{u^2 + A^2 - 2Au\cos\theta}{2\sigma_n^2} \right] d\theta \\ &= \frac{u}{\sigma_n^2} e^{-\left(u^2 + A^2 \right) / \left(2\sigma_n^2 \right)} I_0\left(\frac{|A| u}{\sigma_n^2} \right). \end{aligned}$$

It is zero for $u < 0$. Here $I_0\left(z \right) = \left(2\pi \right)^{-1} \int_0^{2\pi} e^{z\cos\theta}\, d\theta$ is the modified Bessel function. This is the Rice envelope law; it reduces to the Rayleigh law when $A = 0$. The carrier shifts one Gaussian component's mean, which changes the distribution of the magnitude.

Modulation and noise

A radio transmitter uses modulation to let a high-frequency carrier carry a lower-frequency message, such as speech. The receiver selects the carrier band and then recovers the message.

Sinusoidal power and AM sidebands For a unit resistance, a sinusoid of amplitude A has power $A^2/2$.

$$\begin{aligned} s_{\mathrm{AM}}\left(t\right) &= A_c\left[1+\mu\cos\left(2\pi f_m t\right)\right]\cos\left(2\pi f_c t\right), \\ s_{\mathrm{AM}}\left(t\right) &= A_c\cos\left(2\pi f_c t\right)+\frac{A_c\mu}{2}\cos\left[2\pi\left(f_c+f_m\right)t\right]+\frac{A_c\mu}{2}\cos\left[2\pi\left(f_c-f_m\right)t\right], \\ P_{\mathrm{AM}} &= \frac{A_c^2}{2}\left(1+\frac{\mu^2}{2}\right). \end{aligned}$$

The two sidebands place the message tone on either side of the carrier. After filtering, multiply each sinusoidal amplitude by its filter gain before calculating power. For DSB-SC,

$$s\left(t\right)=A_c m\left(t\right)\cos\left(2\pi f_c t\right), \qquad P_s=\frac{A_c^2 P_m}{2}.$$

Filtered and coherently translated noise A receiver filter admits both the desired signal and noise within its passband, so increasing the admitted bandwidth can increase noise power. Let $N_0/2$ be the two-sided white-noise PSD.

$$S_n\left(f\right)=\frac{N_0}{2}\left|H\left(f\right)\right|^2, \qquad P_n=\int_{-\infty}^{\infty}S_n\left(f\right)df.$$

For multiplication by $A_L\cos\left(2\pi f_c t\right)$ followed by a lowpass filter, let $\kappa=A_L/2$. For separated passbands,

$$S_{n,o}\left(f\right)=\kappa^2\left[S_n\left(f-f_c\right)+S_n\left(f+f_c\right)\right]$$

within the output passband. Ideal DSB-SC reception gives $m_o\left(t\right)=\kappa A_c m\left(t\right)$. For AM, remove carrier-generated DC when finding message SNR.

$$\mathrm{SNR}=\frac{P_s}{P_n}, \qquad \mathrm{SNR}_{\mathrm{dB}}=10\log_{10}\left(\mathrm{SNR}\right).$$

SNR compares useful signal power with the noise power that competes with it at the same point in the receiver.

FM bandwidth, power, and threshold Increasing the frequency deviation can improve the recovered message SNR above threshold, but it also requires a wider transmission band. Assume a unit-peak message, ideal above-threshold reception, and no pre-emphasis or de-emphasis.

$$\beta=\frac{\Delta f}{W}, \qquad B_T\simeq 2\left(\beta+1\right)W,$$

$$\mathrm{SNR}_o=\frac{3\beta^2 P_m P_r}{N_0 W}, \qquad P_r=P_T 10^{-L/10}.$$

Here W is the message bandwidth, P_m its mean-square value, and L the channel power loss in decibels. If the threshold ratio is ten,

$$P_{n,\mathrm{BP}} = N_0 B_T, \qquad \frac{P_r}{N_0 B_T} \geq 10.$$

Worked Problems

AM signal and noise through a nonuniform bandpass filter

Question

A carrier with $A_c = 10\,\mathrm{mV}$ and $f_c = 90\,\mathrm{kHz}$ is 50% amplitude modulated by a $750\,\mathrm{Hz}$ sinusoid. The two-sided noise PSD is $\eta/2 = 10^{-3}\,\mathrm{W/Hz}$.

The filter magnitude is even; on the positive-frequency lobe,

$$|H(f)| = \begin{cases} 1, & 89.5 \leq f \leq 90.5\,\mathrm{kHz}, \\ 0.5, & 89 \leq f < 89.5\,\mathrm{kHz} \text{ or } 90.5 < f \leq 91\,\mathrm{kHz}, \\ 0, & \text{otherwise}. \end{cases}$$

The output is multiplied by $2\cos(2\pi f_c t)$ and passed through an ideal LPF with bandwidth $1\,\mathrm{kHz}$.

Find the demodulator input SNR and the recovered-message output SNR.

Solution

Assume coherent phase at the carrier and sidebands, and a unit resistance.

Input signal and noise With $\mu = 0.5$, each sideband initially has amplitude $A_c\mu/2$. The carrier gain is one; the sideband gains are $1/2$.

$$\begin{aligned} P_{s,i} &= \tfrac{A_c^2}{2} + 2\tfrac{1}{2}\left(\tfrac{A_c\mu}{4}\right)^2 = 5.15625\times 10^{-5}\,\mathrm{W}, \\ \int |H(f)|^2 df &= 2\left[1000 + 2(500)(0.5)^2\right] = 2500\,\mathrm{Hz}, \\ P_{n,i} &= 10^{-3}(2500) = 2.5\,\mathrm{W}. \end{aligned}$$

$$\mathrm{SNR}_i = 2.0625\times 10^{-5} \approx -46.86\,\mathrm{dB}$$

Recovered message

$$s_o(t) = A_c + \frac{A_c\mu}{2}\cos(2\pi f_m t), \qquad P_{m,o} = \frac{1}{2}\left(\frac{A_c\mu}{2}\right)^2 = 3.125\times 10^{-6}\,\mathrm{W}.$$

Exclude the carrier-generated DC term from message power.

Output noise

$$S_{n,o}(f) = \frac{\eta}{2}\left[|H(f-f_c)|^2 + |H(f+f_c)|^2\right].$$

$$S_{n,o}(f) = \begin{cases} 2\times 10^{-3}, & |f| \le 500, \\ 0.5\times 10^{-3}, & 500 < |f| \le 1000, \\ 0, & \text{otherwise.} \end{cases}$$

$$P_{n,o} = \left(2\times 10^{-3}\right)(1000) + \left(0.5\times 10^{-3}\right)(1000) = 2.5\,\text{W}.$$

$$\text{SNR}_o = 1.25\times 10^{-6} \approx -59.03\,\text{dB}$$

If DC is included, $\left(A_c^2 + P_{m,o}\right)/P_{n,o} = 4.125\times 10^{-5}$.

FM power, Carson bandwidth, and threshold limitation

Question

An audio signal has $W = 15\,\text{kHz}$, average power $P_m = 1/6$, and peak magnitude one. FM transmission experiences 60 dB power loss and two-sided AWGN PSD $\eta/2 = 10^{-12}\,\text{W/Hz}$. The required output SNR is at least 40 dB.

- **Find the bandwidth and transmitter power for deviation ratio $\beta = 5$.**
- **Find the maximum deviation ratio and bandwidth when the carrier power must be at least ten times the channel noise power.**

Solution

Assume ideal above-threshold FM without pre-emphasis or de-emphasis. Let $N_0 = \eta = 2\times 10^{-12}$.

Bandwidth and power

$$B_T \simeq 2(\beta+1)W = 180\,\text{kHz}$$

$$\begin{aligned} \text{SNR}_o &= \frac{3\beta^2 P_m P_r}{N_0 W} = 10^4, \\ P_r &= \frac{10^4\left(2\times 10^{-12}\right)(15000)}{3(25)(1/6)} = 2.4\times 10^{-5}\,\text{W}, \\ P_T &= 10^6 P_r. \end{aligned}$$

$$P_T = 24\,\text{W}$$

Here $N_0 B_T = 3.6\times 10^{-7}\,\text{W}$ **and** $P_r/(N_0 B_T) = 66.67 > 10$.

Threshold at fixed transmitter power Keep $P_T = 24\,\text{W}$.

$$\begin{aligned} \frac{P_r}{N_0 B_T} &\geq 10, \\ B_T &\leq \frac{P_r}{10 N_0} = 1.2 \times 10^6 \text{ Hz}, \\ \beta &\leq \frac{1.2 \times 10^6}{2\,(\,15000\,)} - 1. \end{aligned}$$

$$\beta_{\max} = 39, \qquad B_{T,\,\max} = 1.2\,\text{MHz}$$

These are boundary values; strictly remaining above threshold requires a smaller bandwidth.

Power retuned for each deviation ratio If power is instead minimized anew at each β while maintaining 40 dB output SNR,

$$P_r = \frac{6 \times 10^{-4}}{\beta^2}, \qquad P_r \geq 6 \times 10^{-7}\,(\,\beta + 1\,)\,.$$

$$\beta^2\,(\,\beta + 1\,) \leq 1000.$$

$$\beta_{\max} = 9.67753, \qquad B_T = 320.326\,\text{kHz}$$

Minimum FM transmitter power under a channel bandwidth limit

Question

A normalized message has bandwidth $W = 5\,\text{kHz}$ and power $P_m = 0.5$. The channel has bandwidth $60\,\text{kHz}$, 40 dB power attenuation, and two-sided noise PSD $10^{-9}\,\text{W/Hz}$.

For FM without pre-emphasis or de-emphasis, find the minimum transmitter power and deviation ratio needed for at least 40 dB output SNR.

Solution

Assume unit peak message magnitude and ideal above-threshold FM.

Maximum deviation

$$2\,(\,\beta + 1\,)\,(\,5000\,) \leq 60000 \quad \Rightarrow \quad \beta \leq 5.$$

Since required power is proportional to $1/\beta^2$,

$$\beta = 5, \qquad \Delta f = 25\,\text{kHz}$$

Minimum power The two-sided density gives $N_0 = 2 \times 10^{-9}$.

$$
\begin{aligned}
\mathrm{SNR}_o &= \frac{3\beta^2 P_m P_r}{N_0 W} = 10^4, \\
P_{r,\ \min} &= \frac{10^4 \left(2 \times 10^{-9} \right) \left(5000 \right)}{3 \left(25 \right) \left(0.5 \right)} = 2.66667 \times 10^{-3}\ \mathrm{W}, \\
P_{T,\ \min} &= 10^4 P_{r,\ \min}\ .
\end{aligned}
$$

$$
P_{T,\ \min} = 26.667\ \mathrm{W}
$$

The predetection ratio is

$$
\frac{P_r}{N_0 B_T} = \frac{2.66667 \times 10^{-3}}{\left(2 \times 10^{-9} \right) \left(60000 \right)} = 22.22,
$$

above a ten-to-one threshold criterion.

DSB-SC filtering and coherent-detection SNR

Question

The transmitted signal is

$$
s\left(t \right) = 10 \left[3 \cos \left(2\pi 10^2 t \right) + 2 \cos \left(2\pi 10^4 t \right) \right] \cos \left(\pi 10^8 t \right).
$$

White noise has two-sided PSD $N_0/2 = 2 \times 10^{-6}\ \mathrm{W/Hz}$. A unity-gain ideal band-pass filter is centered at the carrier and has full positive-frequency passband width $10\ \mathrm{kHz}$, with the mirrored negative-frequency passband.

Determine:

- **The modulation type.**
- **The SNR at the band-pass filter output.**
- **The in-phase noise PSD and average power.**
- **The SNR after coherent detection with multiplier $2 \cos \left(2\pi f_c t \right)$ and low-pass edge $5\ \mathrm{kHz}$.**

Use a unit-resistance power normalization.

Solution

Carrier frequency and passed signal

The modulation is double-sideband suppressed carrier, with

$$
f_c = \frac{10^8}{2} = 5 \times 10^7\ \mathrm{Hz}.
$$

The positive passband edges are $f_c \pm B$, with $B = 5000\ \text{Hz}$. The 100 Hz tone passes; the 10 kHz tone is rejected. Therefore,

$$s_o\left(t\right) = 30\cos\left(2\pi 100t\right)\cos\left(2\pi f_c t\right).$$

Its mean-square power is

$$P_s = 30^2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 225\ \text{W}.$$

Band-pass noise and SNR

The combined width of both passbands is $4B$:

$$P_n = \frac{N_0}{2}\left(4B\right) = 0.04\ \text{W}.$$

Hence

$$\text{SNR}_{\text{BPF}} = \frac{225}{0.04} = 5625 \approx 37.50\ \text{dB}.$$

In-phase noise

For $n = n_c\cos\left(2\pi f_c t\right) - n_s\sin\left(2\pi f_c t\right)$,

$$S_{n_c}\left(f\right) = \begin{cases} 4\times 10^{-6}\ \text{W/Hz}, & |f| \le 5000\ \text{Hz}, \\ 0, & \text{otherwise}. \end{cases}$$

$$P_{n_c} = \int_{-5000}^{5000} 4\times 10^{-6}\,df = 0.04\ \text{W}.$$

Coherent detector output

Mixing by $2\cos\left(2\pi f_c t\right)$ and retaining $|f| \le 5000\ \text{Hz}$ gives

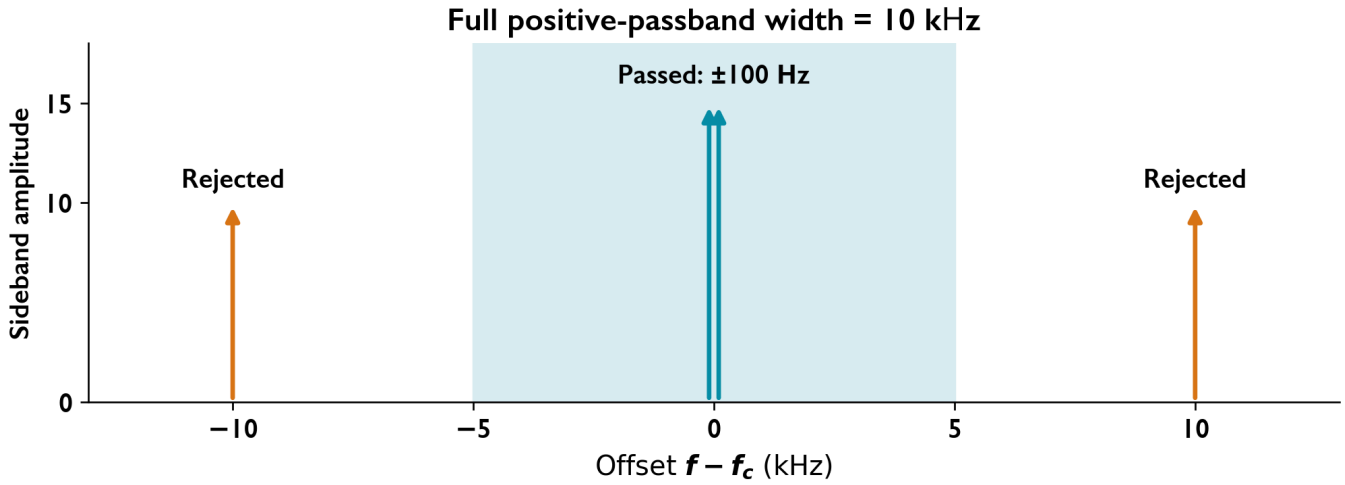
$$y_d\left(t\right) = 30\cos\left(2\pi 100t\right) + n_c\left(t\right).$$

Thus

$$P_{s,d} = \frac{30^2}{2} = 450\ \text{W}, \qquad P_{n,d} = 0.04\ \text{W},$$

$$\text{SNR}_d = \frac{450}{0.04} = 11250 \approx 40.51\ \text{dB}.$$

For a narrower detector edge B_d, with $100 \le B_d \le 5000\ \text{Hz}$, the signal power remains 450 W and the noise power becomes $2N_0B_d$.



Positive-frequency sidebands relative to the carrier. The reflected negative-frequency band also contributes noise power.

Sampling, quantization, and prediction

A digital receiver first measures an analogue voltage waveform at regularly spaced instants. These measurements form a sequence that can be stored, transmitted, or processed digitally.

Samples and correlation

$$X[n] = X(nT_s), \qquad T_s = \frac{1}{f_s}, \qquad r_X[k] = R_X(kT_s).$$

A PSD specifies sample statistics; it does not specify individual random sample amplitudes.

Exact quantization powers A quantizer replaces each sample by one of a finite set of levels that can be represented by bits. The difference between the original value and its chosen level is quantization error; its mean-square value measures the distortion. Let bin $\mathcal{B}_i$ have reconstruction level q_i.

$$\begin{aligned} p_i &= \int_{\mathcal{B}_i} f_X(x)\,dx, \\ P_X &= \int x^2 f_X(x)\,dx, \\ P_q &= \sum_i \int_{\mathcal{B}_i} \left(x - q_i\right)^2 f_X(x)\,dx, \\ P_{\text{out}} &= \sum_i q_i^2 p_i, \qquad \text{SQNR} = \frac{P_X}{P_q}. \end{aligned}$$

For a uniform input over the complete operating range and uniform bins without overload,

$$\Delta = \frac{2A}{Q}, \qquad P_q = \frac{\Delta^2}{12}.$$

For a nonuniform density, use the bin integrals rather than assuming a uniform error. A valid density must be nonnegative and integrate to one.

Minimum mean-square prediction When successive samples are correlated, past samples help estimate the next value. Quantizing the smaller prediction error instead of the full sample can reduce distortion for a given number of levels, or reduce the number of levels needed for a specified distortion. For a real zero-mean WSS sequence,

$$\widehat{X}[n] = aX[n-1], \qquad D[n] = X[n] - \widehat{X}[n].$$

Orthogonality gives

$$a = \frac{r_X[1]}{r_X[0]}, \qquad \sigma_D^2 = r_X[0] - \frac{r_X[1]^2}{r_X[0]}.$$

If $r_X[k] = \sigma_X^2 \rho^{|k|}$, a longer optimum predictor needs only its first tap:

$$a_1 = \rho, \quad a_2 = a_3 = 0, \qquad \sigma_D^2 = \sigma_X^2(1 - \rho^2).$$

For a Gaussian sequence, the innovations are independent Gaussian variables.

Differential quantization levels Under the specified Gaussian quantization model,

$$P_q = \frac{2\sigma_D^2}{Q^2}, \qquad Q_{\min} = \left\lceil \sqrt{\frac{2\sigma_D^2}{P_{q,\,\max}}} \right\rceil.$$

For binary coding, $b = \lceil \log_2 Q_{\min} \rceil$ and the implemented number of levels is 2^b.

Worked Problems

Prediction of a sampled Gaussian process and differential quantization

Question

A zero-mean Gaussian process has

$$S_X(f) = \frac{16}{1 + (16\pi f)^2}, \qquad f_s = 1\,\text{Hz}.$$

A differential quantizer subtracts the predictor

$$\widehat{X}[n] = a_1 X[n-1] + a_2 X[n-2] + a_3 X[n-3].$$

For a Gaussian input of variance σ^2 and Q quantization levels, the quantization-noise power is $2\sigma^2/Q^2$.

- Find the number of levels required for noise power at most 4.32×10^{-4}.
- Determine the sampled-process values at $T_s, 2T_s, 3T_s, 4T_s$.

Solution

Correlation and optimum prediction Assume minimum mean-square prediction. With $T_s = 1\,\text{s}$,

$$R_X(\tau) = e^{-|\tau|/8}, \qquad \rho = e^{-1/8} = 0.882496903, \qquad r_X[m] = \rho^{|m|}.$$

$$\begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} \rho \\ \rho^2 \\ \rho^3 \end{bmatrix} \quad \Rightarrow \quad a_1 = \rho, \quad a_2 = a_3 = 0.$$

$$X[n] = \rho X[n-1] + D[n], \qquad \sigma_D^2 = 1 - e^{-1/4} = 0.221199217.$$

Quantization levels

$$\begin{aligned} \frac{2\sigma_D^2}{Q^2} &\le 4.32 \times 10^{-4}, \\ Q &\ge \sqrt{\frac{2\left(1 - e^{-1/4}\right)}{4.32 \times 10^{-4}}} = 32.0011007. \end{aligned}$$

$$Q_{\min} = 33$$

For $Q = 32$, $P_q = 4.3202972 \times 10^{-4}$, just above the bound. If Q must be a power of two, $Q = 64,\ b = 6$.

Sample statistics A PSD does not determine individual sample amplitudes. The available results are

$$E\{X[i]\} = 0, \qquad \text{Var}\ (X[i]) = 1, \qquad \text{Cov}\ (X[i], X[j]) = \rho^{|i-j|}, \quad i, j \in \{1, 2, 3, 4\}.$$

Given $X[0]$ and independent Gaussian innovations $D[k]$, actual samples follow from

$$X[n] = \rho^n X[0] + \sum_{k=1}^{n} \rho^{n-k} D[k].$$

The corresponding autocorrelation samples are

$$\begin{aligned} R_X(T_s) &= 0.882497, & R_X(2T_s) &= 0.778801, \\ R_X(3T_s) &= 0.687289, & R_X(4T_s) &= 0.606531. \end{aligned}$$

Exact quantization powers for uniform and parabolic inputs

Question

A midrise quantizer has thresholds $-2, 0, 2$, reconstruction levels $-3, -1, 1, 3$, and input range $[-4, 4]$:

$$q(x) = \begin{cases} -3, & -4 \le x < -2, \\ -1, & -2 \le x < 0, \\ 1, & 0 \le x < 2, \\ 3, & 2 \le x \le 4. \end{cases}$$

- **For a uniform input on $[-4, 4]$, find quantization-noise power and SQNR.**
- **For $f_X(x) = A(1 - x^2/c)$ on $|x| \le 4$, zero otherwise, find the normalization, quantization-noise power, and quantized-signal power when the density is valid. The positive denominator is uncertain: test $c = 10$ and give the $c = 16$ results conditionally, together with the general valid model.**

Solution

Uniform input

$$\Delta = 2, \qquad P_X = \frac{4^2}{3} = \frac{16}{3}, \qquad P_q = \frac{\Delta^2}{12} = \frac{1}{3}.$$

$$\text{SQNR} = \frac{16/3}{1/3} = 16 \approx 12.04\,\text{dB}$$

Validity and normalization For $c = 10$, $1 - x^2/10$ changes sign on the stated support. No nonzero A makes it a density. For positive c, validity requires $c \geq 16$.

$$1 = A \int_{-4}^{4} \left(1 - \frac{x^2}{c}\right) dx = A\left(8 - \frac{128}{3c}\right), \qquad A = \frac{3c}{8\,(\,3c - 16\,)}.$$

Exact noise power

$$\begin{aligned} P_q = 2A[\quad & \int_0^2 (\,x-1\,)^{\,2} \left(1 - \tfrac{x^2}{c}\right) dx \\ & + \int_2^4 (\,x-3\,)^{\,2} \left(1 - \tfrac{x^2}{c}\right) dx]. \end{aligned}$$

$$P_q = \frac{5c - 28}{5\,(\,3c - 16\,)}$$

Quantized-signal power

$$\begin{aligned} P_1 &= P(\,0 < X < 2\,) = \tfrac{3c-4}{4\,(\,3c-16\,)}, \\ P_2 &= P(\,2 < X < 4\,) = \tfrac{3c-28}{4\,(\,3c-16\,)}. \end{aligned}$$

$$E\{q\,(\,X\,)^{\,2}\} = 2\,(\,P_1 + 9P_2\,) = \frac{15c - 128}{3c - 16}$$

Conditional model $c = 16$

$$A = \frac{3}{16}, \qquad P_q = \frac{13}{40} = 0.325, \qquad E\{q\,(\,X\,)^{\,2}\} = \frac{7}{2} = 3.5$$

Here $P_1 = 11/32$, $P_2 = 5/32$, and $E\{X^2\} = 16/5 = 3.2$. Exact quantization error can be correlated with X, so subtracting noise power from input power does not give output power.

Matched filters and binary detection

A receiver must often detect a known pulse hidden in noise. A matched filter aligns its response with the pulse shape so that the signal contributions combine at the chosen sampling time, giving the largest output SNR in white Gaussian noise.

Matched filter For a known pulse $s(t)$ in two-sided AWGN with PSD $N_0/2$,

$$E_s = \int |s(t)|^2 dt, \qquad h(t) = Ks^*(T-t).$$

For a real pulse,

$$y_s(t) = KR_s(t-T), \qquad y_s(T) = KE_s.$$

$$\sigma^2_{n,o} = \frac{N_0}{2}|K|^2 E_s, \qquad \mathrm{SNR}_{\max} = \frac{2E_s}{N_0}.$$

The gain $K \neq 0$ changes amplitude but not SNR.

Coherent binary error probabilities The receiver uses the noisy output to choose a transmitted bit, and the error probability tells us how often that choice will be wrong. Let $T_b = 1/R_b$ and $E_b \simeq A^2 T_b/2$ for a passband sinusoid of amplitude A.

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-u^2/2} du.$$

For equiprobable, undistorted signaling,

$$P_{e,\mathrm{BFSK}} = Q\left(\sqrt{\frac{E_b}{N_0}}\right), \qquad P_{e,\mathrm{BPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

The BFSK formula requires orthogonal signaling; $\Delta f T_b = 1$ is a suitable envelope-model spacing.

PSK bandwidth assumptions For binary PSK and positive-lobe channel width B,

$$B = (1+\alpha) R_b$$

with raised-cosine roll-off α. Rectangular pulses instead give the first-null approximation $B \simeq 2R_b$; their sidelobes extend beyond this interval.

MAP detection with Laplace noise A decision threshold separates observations assigned to the two possible signal levels, balancing the two kinds of wrong decision. If one level occurs more often, its larger prior probability shifts the threshold in its favour to minimize the average error probability. For polar levels $\pm A$, priors p_+, p_-, and $f_N(n) = \frac{\lambda}{2} e^{-\lambda|n|}$, decide $+A$ if

$$\lambda(|v+A| - |v-A|) > \ln \frac{p_-}{p_+}.$$

If $\left|\ln\left(p_-/p_+\right)\right| < 2\lambda A$,

$$\gamma = \frac{1}{2\lambda} \ln \frac{p_-}{p_+}, \qquad P_e = \sqrt{p_+ p_-}\, e^{-\lambda A}.$$

The average signal power is A^2 and the noise variance is $2/\lambda^2$, so

$$\mathrm{SNR} = \frac{(\lambda A)^2}{2}.$$

Worked Problems

Matched filter for a flat-topped, linearly falling pulse

Question

The pulse is

$$s(t) = \begin{cases} 1, & 0 \le t < 1, \\ 2-t, & 1 \le t < 2, \\ 0, & \text{otherwise.} \end{cases}$$

The two-sided AWGN PSD is $(2/3) \times 10^{-3}\,\mathrm{W/Hz}$.

- Find and sketch a matched-filter impulse response and its signal output.
- Find the time and magnitude of the maximum output.
- Determine the maximum output SNR.

Solution

Matched impulse response Choose unit gain and sampling time $T = 2$.

$$h(t) = s(2-t) = \begin{cases} t, & 0 < t \le 1, \\ 1, & 1 < t \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Signal output Let $a = |t-2|$. Then $y_s(t) = R_s(t-2)$. For $0 \le a \le 1$,

$$R_s(a) = \int_a^1 du + \int_1^{1+a} (2-u)\,du + \int_{1+a}^2 (2-u)(2-u+a)\,du.$$

For $1 < a \le 2$, $R_s(a) = \int_a^2 (2-u)\,du$. Hence

$$y_s(t) = \begin{cases} \frac{4}{3} - \frac{1}{2}a - \frac{1}{2}a^2 + \frac{1}{6}a^3, & 0 \le a \le 1, \\ \frac{1}{2}(2-a)^2, & 1 < a \le 2, \\ 0, & a > 2, \end{cases} \qquad a = |t-2|.$$

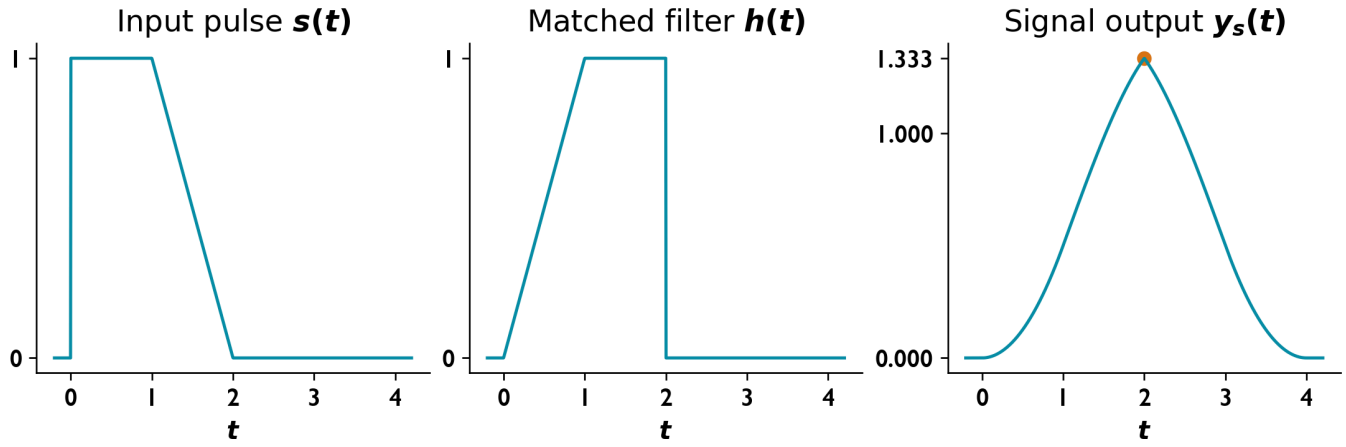
The output is symmetric about $t = 2$ and passes through

$$(t, y_s) = (0,0),\ \left(1, \tfrac{1}{2}\right),\ \left(2, \tfrac{4}{3}\right),\ \left(3, \tfrac{1}{2}\right),\ (4,0).$$

Peak and SNR

$$E_s = \int_0^1 dt + \int_1^2 (2-t)^2 dt = \frac{4}{3}, \qquad t_{\max} = 2, \quad |y_s(t_{\max})| = \frac{4}{3}.$$

$$\mathrm{SNR}_{\max} = \frac{E_s}{N_0/2} = \frac{4/3}{(2/3)\,10^{-3}} = 2000 \approx 33.01\,\mathrm{dB}$$



Input pulse, causal matched-filter response, and signal output with peak 4/3 at t = 2.

Coherent FSK and PSK over a finite bandpass channel

Question

A unity-gain channel passes $500 \le |f| \le 1500\,\mathrm{Hz}$ and rejects all other frequencies. The amplitude-to-noise ratio is $A^2/\eta = 8000$.

- **Find the coherent binary FSK error probability at $200\,\mathrm{bit/s}$ with tones at 1070 and 1270 Hz.**
- **Determine the fastest PSK rate and its coherent error probability.**

Solution

Assume equiprobable bits, two-sided AWGN PSD $\eta/2 = N_0/2$, sinusoid amplitude A, and negligible distortion of the shaped signals.

Coherent FSK

$$\begin{aligned} T_b &= 1/200 = 0.005\,\mathrm{s}, \\ (f_2 - f_1)\,T_b &= 200\,(0.005) = 1, \\ \frac{E_b}{N_0} &\simeq \frac{A^2 T_b}{2\eta} = 8000\frac{0.005}{2} = 20. \end{aligned}$$

The spacing satisfies envelope-model orthogonality.

$$P_{e,\mathrm{FSK}} = Q\left(\sqrt{20}\right) = 3.8721 \times 10^{-6}$$

Binary PSK with ideal Nyquist shaping A maximum rate requires a pulse-shaping and modulation-order assumption. For binary PSK with zero roll-off, the positive-lobe width $B = 1000\,\mathrm{Hz}$ equals R_b.

$$R_{b,\ \max} = 1000\,\mathrm{bit/s}$$

$$\frac{E_b}{N_0} = \frac{A^2}{2\eta R_b} = 4,$$

$$P_{e,\,\mathrm{BPSK}} = Q\left(\sqrt{8}\right) = 2.33887 \times 10^{-3}$$

For raised-cosine roll-off α, $R_{b,\ \max} = B/(1+\alpha)$.

Rectangular-pulse bandwidth approximation Using the first-null approximation $B \simeq 2R_b$,

$$R_b \simeq 500\,\mathrm{bit/s}, \qquad E_b/N_0 = 8,$$

$$P_e \simeq Q(4) = 3.16712 \times 10^{-5}$$

This assumes undistorted coherent detection. A brick-wall channel truncates rectangular-pulse sidelobes, so the approximation is not an exact distortion analysis. Higher-order PSK requires a specified order.

MAP detection of polar data in Laplace noise

Question

At a binary receiver's sampling instant,

$$V_0(T_b) = s_0(T_b) + N_0(T_b), \qquad f_N(n) = \frac{\lambda}{2} e^{-\lambda|n|}.$$

The polar levels are $+A$ for bit one and $-A$ for bit zero, with probabilities 0.25 and 0.75. Find the average SNR required for $P_e = 10^{-4}$ using the optimum threshold.

Solution

MAP threshold Decide $+A$ when

$$\frac{f_{V|+}(v)}{f_{V|-}(v)} = \exp\{\lambda(|v+A| - |v-A|)\} > 3.$$

Within $-A < v < A$, the log ratio is $2\lambda v$, giving

$$\gamma = \frac{\ln 3}{2\lambda}, \qquad +A \text{ if } v > \gamma.$$

The interior condition is $\lambda A > (\ln 3)/2$.

Error probability

$$\begin{aligned} P_e &= \tfrac{1}{8} e^{-\lambda\,(\,A-\gamma\,)} + \tfrac{3}{8} e^{-\lambda\,(\,A+\gamma\,)} \\ &= \tfrac{\sqrt{3}}{4} e^{-\lambda A}. \end{aligned}$$

$$\lambda A = \ln\left(\frac{\sqrt{3}}{4\times 10^{-4}}\right) = 8.37335216.$$

This satisfies the interior condition.

Average SNR

$$P_s = A^2, \qquad \sigma_N^2 = \frac{2}{\lambda^2},$$

$$\mathrm{SNR} = \frac{(\,\lambda A\,)^{\,2}}{2} = 35.0565 \approx 15.45\,\mathrm{dB}.$$